SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Engineer to Excel

SIMATS ENGINEERING

AICTE Sponsored International Conference

on

Artificial Intelligence-Driven Innovations in Computer, Electrical, Mechanical and Electronics Engineering

(ICAI-CEME 2026)

on

21-22 October 2026

PROCEEDINGS OF ICAI-CEME 2026

Organized By

Institute of Computer Science and Engineering

Institute of Computer Science and Engineering, SIMATS Engineering, Chennai.
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ABOUT SIMATS

Saveetha Medical and Educational Trust was established in 1986 mainly to provide medical relief, to promote Education and encourage Research. The Trust started a Saveetha Dental College and established an attached General Hospital in the year 1988. Subsequently the Trust established a Nursing College, Physiotherapy College, Occupational Therapy College and an Engineering College in order to comply with our commitment and to serve the people, especially those living in Rural Village in and around Chennai and adjoining districts of Tamil Nadu.

To fulfil the norms of the University Grants Commission, for the purpose of obtaining "Deemed to be University" status, Saveetha Medical and Educational Trust has sponsored another Trust (viz.) Saveetha Institute of Medical & Technical Sciences (SIMATS) in the year 2001. Subsequently, the said Trust (SIMATS) was declared as "Deemed to be University" with the following three Institutions under Section -3 of the UGC Act, 1956 as per Notification of the Govt. of India No.F.9- 3/2002-dated 18-03-2005, now known as Saveetha Institute of Medical and Technical Sciences (SIMATS) (Deemed to be University). https://www.saveetha.com/

1. Saveetha Dental College & Hospital
2. Saveetha College of Nursing
3. Saveetha College of Physiotherapy
4. Saveetha School of Engineering
5. Saveetha Medical College & Hospital
6. Saveetha School of Management
7. Saveetha School of Law
8. Saveetha College of Occupational Therapy
9. Saveetha School of Physical Education
10. Saveetha College of Allied Health Sciences
11. Saveetha College of Pharmacy

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12. Saveetha College of Architecture and Design
13. Saveetha College of Liberal Arts and Sciences

The SIMATS received Accreditation by the National Accreditation Board for Hospitals and Healthcare providers and the National Accreditation Board for Laboratories for RTPCR Testing for Covid-19.

The institution has a total student strength of 10000 and 1500 faculties with non-teaching staff around 2500.

SIMATS is ranked among the top 10 Indian Universities and claimed 395th rank in World in 2020 UI Green Metric World University Rankings on Sustainability by University of Indonesia, Jakarata.

SIMATS has been recognized as Social Entrepreneurship, Swachhta & Rural Engagement Cell (SES REC) Institution by Mahatma Gandhi National Council of Rural Education, Department of Higher Education, Ministry of Education, and Government of India.

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ABOUT SIMATS ENGINEERING

SIMATS Engineering, revolves around a highly dynamic and innovative curriculum, that puts down greater emphasis on hands-on training come market targeted value-added programs. The institute offers diverse undergraduate, post-graduate and doctoral programs, All UG programs comprise of enterprising platforms including interdisciplinary electives, choice-based credit system and a semester sandwich program option, in collaboration with renowned engineering institutions overseas. SIMATS Engineering is a NAAC A++, NBA Accredited and NIRF 11 ranking institute, offering idiosyncratic IET enabled choice-based credit courses. with a strong insistence on practical training.

SIMATS Engineering offers Under Graduate, Post Graduate and Doctoral programmes in the following Departments.

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- Department Of Automobile Engineering
- Department Of BioMedical Engineering
- Department Of Biotechnology
- Department Of Bioinformatics
- Department Of Computer Science Engineering
- Department Of Civil Engineering
- Department Of Electronics and Communication Engineering
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- Department Of Information Technology
- Department Of Mechanical Engineering
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- Artificial Intelligence and Data Science

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Doctorate-Ph.D.

- All disciplines of Engineering, Science & Humanities

Each department is enriched with experienced, dedicated and knowledgeable faculty members. With an extraordinary infrastructure facility and with innovative teaching methods like MILA, Flipped Classes, etc., SSE provides the best platform for the students to excel in their academics.

A faculty Member of SIMATS Engineering was included in top 2% Scientist in India from Stanford University, USA in the Building and construction category.

- The College has CLAB –Classroom Laboratories
- Saveetha Engineering Industries
- Study Flex @ Saveetha

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ABOUT THE ORGANIZING INSTITUTE

The Institute of Computer Science and Engineering (CSE) at SIMATS Engineering is dedicated to excellence in education, research, and technological innovation. The B.Tech Computer Science and Engineering programme is accredited by NBA, ABET, and IET (UK), reflecting globally recognized academic standards. The department combines strong theoretical foundations with practical applications in areas such as Artificial Intelligence, Data Science, High Performance Computing, Algorithm Design, Computational Modelling, and Software Engineering. Supported by experienced faculty, modern computing laboratories, industry collaborations, and interdisciplinary research, the department equips students with the technical expertise, analytical skills, and innovation mindset required to solve real-world challenges and excel in academia, research, and the global technology industry.

ABOUT THE CONFERENCE

The International Conference on AI-Driven Innovations in Computer, Electrical, Mechanical, and Electronics Engineering (ICAI-CEME 2026) is an AICTE-sponsored international conference that brings together academicians, researchers, industry professionals, and students to showcase and exchange innovative research in Artificial Intelligence and emerging engineering technologies. Centered on the theme, "Artificial Intelligence for Sustainable Engineering: Integrating Intelligent Technologies to Accelerate the Sustainable Development Goals (SDGs)," the conference promotes interdisciplinary collaboration across Computer Science, Electronics, Electrical, and Mechanical Engineering. Through keynote lectures, technical sessions, paper presentations, and industry interactions, ICAI-CEME 2026 fosters innovation, research collaboration, technology transfer, and the development of sustainable engineering solutions for global societal impact.

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CHANCELLOR'S DESK



It is admirable that the SIMATS has shaped a contemporary system to make it possible for staff and students to appraise their skills and talent in numerous areas like sciences, medical and technical disciplines because of quick evolving developments in modern era. Amongst the university, anyone can notice that the SIMATS Engineering has been fairly effective in sustaining stability accompanied by contemporary leanings and teaching modules. It gives me an immense pleasure to invite delegates to our serene campus to the AICTE Sponsored International Conference on AI-Driven Innovations in Computer, Electrical, Mechanical, and Electronics Engineering ICAI-CEME 2026 during 21-22 October 2026. Pupation is very important in the life of an educationist which is supported and encouraged through such events at SIMATS Engineering.

I take this opportunity to congratulate team ICAI-CEME 2026 for their efforts and initiative towards the success of the program.

Dr. N. M. VEERAIYAN
CHANCELLOR, SIMATS

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PRO CHANCELLOR'S DESK



It gives me immense pleasure and pride to be a part of the AICTE Sponsored International Conference on AI-Driven Innovations in Computer, Electrical, Mechanical, and Electronics Engineering ICAI-CEME 2026 during 21-22 October 2026. The motto of the Conference is to bring together experts from academic institutions, industries and researchers for sharing knowledge, expertise and experience in the emerging trends related to Computer Science Engineering and Information Technology. I am appreciative of the efforts undertaken by the SIMATS Engineering team in organizing the International Conference to create a platform for experts to share and learn.

I congratulate all concerned members and wish the conference a grand success.

Dr. DEEPAK NALLASWAMY
PRO CHANCELLOR , SIMATS

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DIRECTOR'S DESK



It’s a great honor to invite all the great scientists, academicians, young researchers, business delegates and students to attend two-day AICTE Sponsored International Conference on AI-Driven Innovations in Computer, Electrical, Mechanical, and Electronics Engineering ICAI-CEME 2026 during 21-22 October 2026 at SIMATS Engineering. I would like to congratulate the team ICAI-CEME 2026 for their long-sighted vision in coming up with this brilliant idea of giving a chance to students and scholars from other colleges and universities. At SIMATS Engineering, we discover new thrust areas for students who thirst after knowledge and excellence. I believe only through research and practice, we would be able to contribute to the nation and world at large with the advances and innovations in the field of science and technology. This conference will also act as a stepping stone for new researchers and students to attend and present papers in other conferences.I congratulate the team ICAI-CEME 2026 for their commitment and active participation. I wish you all the success.

Dr. RAMYA DEEPAK

DIRECTOR, SIMATS Engineering

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PRINCIPAL'S DESK



Warm and Happy greetings to all. I am immensely happy that SIMATS Engineering is organizing a two-day AICTE Sponsored International Conference on AI-Driven Innovations in Computer, Electrical, Mechanical, and Electronics Engineering ICAI-CEME 2026 during 21-22 October 2026. I hope that the theme of the conference about emerging technologies in the engineering field will provide a common platform for the academicians; researchers and industrialists to share their knowledge and experience about recent advancements in Engineering. I appreciate the active interest and participation shown by the faculty members involved in this conference and maintaining the research ambience in the SIMATS Engineering.

I wish the team ICAI-CEME 2026 all the very best in all their sustained pursuits for excellence and earnest efforts in making the conference a grand success.

Dr. B. RAMESH
PRINCIPAL
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CONFERENCE CHAIR'S DESK



On behalf of the Organizing Committee, it is My Distinct Honor and Privilege to Welcome you to the Official Proceedings of the AICTE Sponsored International Conference on AI-Driven Innovations in Computer, Electrical, Mechanical, and Electronics Engineering (ICAI-CEME 2026), held from 21–22 October 2026. Our technical program this year highlights the deep integration of AI across multi-disciplinary engineering domains. We received an exceptional response to our call for papers, attracting eight hundreds of high-quality submissions spanning diverse tracks in computer science, power systems, smart manufacturing, and advanced electronics. Based on the comprehensive evaluations, only the most innovative, technically sound, and impactful papers of around 500 were selected for inclusion in these proceedings, reflecting a highly competitive acceptance rate. Beyond the peer-reviewed technical tracks, the two-day program features visionary keynote addresses delivered by the experts and interactive technical sessions to spark meaningful dialogue on the real-world deployment and future trajectory of AI-driven engineering solutions.

The successful execution of an international conference of this magnitude is the result of immense collective dedication. First and foremost, we express our profound gratitude to the **All India Council for Technical Education (AICTE)** for their generous sponsorship and steadfast support, which made this benchmark event possible. We owe a special debt of gratitude to our respected **Director, Principal, and the Management** for their unwavering guidance, constant encouragement, and for providing the world-class infrastructure and resources that made this event possible. I extend my deepest appreciation to Technical Program Chairs, Track Chairs, and our esteemed reviewers for their meticulous efforts in upholding academic excellence. Special thanks are also due to our dedicated organizing committee and local arrangements team for their untiring work behind the scenes. Most importantly, we thank the authors for choosing **ICAI-CEME 2026** to showcase their valuable research.

We are confident that you will find this volume of proceedings to be a profound source of inspiration, offering critical insights that will stimulate your own academic inquiries and professional research endeavours.

Dr. N. Bharatha Devi
Professor
Institute of CSE
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An Optimized Feature Selection Model for Malware Detection

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Abstract:

Malware detection is becoming an important cybersecurity task as the number of malicious executable files is growing, and the identification of unknown or altered threats is challenging with traditional signature-based methods. Machine learning offers an efficient approach to learning patterns in the characteristics of the executable files. However, static malware analysis generates a number of features, which can lead to a high dimensionality of the data, and thus, include less relevant information in the classification process.This paper proposes an optimized feature selection model for malware detection task using the EMBER 2018 dataset. The proposed system extracts static features from Windows Portable Executable (PE) files and uses SelectKBest with Mutual Information to select the most relevant features for classification. The selected features are then passed to an XGBoost classifier to classify the executable files as malware or benign. To make the prediction more understandable, SHAP-based explainability is incorporated to identify the features contributing to each prediction. In addition, the proposed system provides prediction confidence, malware probability, risk assessment, security recommendations, and a PDF-based analysis report in a Streamlit web application. The overall objective is to combine feature selection, malware classification, and explainable security analysis in a single practical detection framework.

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CleanAlert: A Smart GPS-Enabled Real-Time Waste Collection Notification and Community Reward System for Efficient and Responsible Urban Waste Management

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Abstract:

In many Indian cities, household waste collection follows planned routes or fixed schedules, but residents may not know exactly when the collection vehicle will reach their street. This small timing gap can make people miss collection and leave waste outside longer than necessary. CleanAlert is proposed as an Android application that connects vehicle movement with timely information for residents. The system combines live location tracking, street-level notifications, and rewards for verified disposal. The collector location is captured using Android's FusedLocationProviderClient and displayed through Google Maps. Cloud Firestore supports real-time data synchronisation, while Firebase Cloud Messaging (FCM) delivers notifications. Residents can submit geo-tagged and timestamped photographs as disposal evidence. Verified submissions can contribute to points, daily streaks, and locality-level leaderboards. Prototype testing recorded 90% GPS street-detection accuracy (27/30 correct detections), an average notification latency of 2.82 seconds, an 88% disposal-verification rate (22/25 successful audits), and a 100% completion rate for the evaluated Login and Rewards tasks. These results provide preliminary evidence of technical feasibility under the tested conditions; broader field studies are still needed.

Keywords: Smart waste management; GPS tracking; geofencing; real-time notification; Firebase; gamification; mobile application; civic

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Design and Performance Analysis of a DC Motor Speed Control System Using Lag–Lead Compensator

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Abstract:

DC motors are widely used in industrial automation, robotics, electric vehicles, conveyor systems, and CNC machines due to their reliability and ease of speed control. However, maintaining constant motor speed under varying load conditions is a major challenge, as conventional open-loop control systems cannot automatically compensate for disturbances. This project presents the design and performance analysis of a DC motor speed control system using a Lag–Lead compensator to improve system stability, response speed, and steady-state accuracy. Initially, a mathematical model of the DC motor is developed, and its open-loop performance is analyzed using important parameters such as rise time, settling time, maximum overshoot, and steady-state error. A Lag–Lead compensator is then designed to enhance transient response and reduce steady-state error. The compensated closed-loop system is simulated using MATLAB/Simulink to evaluate its performance and compare it with the uncompensated system. Furthermore, an Arduino-based hardware prototype incorporating an Arduino Uno, L298N motor driver, DC geared motor, encoder, and potentiometer is used to demonstrate practical speed control. The proposed system aims to provide faster response, improved stability, reduced overshoot, and better speed regulation under varying operating conditions. This approach offers a cost-effective and scalable solution for accurate DC motor speed control in industrial and educational applications.

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BharatVoice AI: A Multilingual Voice Assistant for Grounded, Everyday Citizen Services in Five Indian Languages

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Abstract:

Many people in India would rather ask a question out loud in Tamil, Hindi, Telugu or Kannada than search an English government portal, and when they type, they often write in Roman letters with English words mixed in. BharatVoice AI is a web assistant built for these users. It covers a small, fixed set of everyday needs: identity documents and where to obtain them, emergency numbers, weather, mandi crop prices, reminders, arithmetic and unit conversion, and a question bank with cited answers on banking, online fraud, welfare schemes, travel, education, jobs and hospitals. No large language model is used. Speech is captured in the browser, and a FastAPI backend processes each request rules-first: a four-stage cascade identifies the language, emergency requests are matched before any classifier is consulted, and every factual reply either names its source or states that the assistant does not know. Two fine-tuned MuRIL classifiers, one for intent and one for language, extend the rules and are trusted only above a confidence floor. On a held-out test set written during development, intent accuracy rose from 67% with rules alone to 89%, and language identification from 60% to 95%; these figures are indicative, since the test set was not written by independent native speakers. The system also offers optional sign-in with per-user history, reminders delivered through Web Push after the tab is closed, and tap-to-call helpline numbers. All automated backend and frontend tests pass, and a short hands-on probe with romanized questions exposed real gaps that guide further work.

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LISA: An AI-Enabled Dual-QR System for Emergency Healthcare Information Access

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Abstract:

Timely access to essential medical information can be challenging during emergencies, particularly when patients are unable to communicate. This paper presents LISA (Life Saver), a prototype system designed to organize emergency healthcare information using a dual QR-code approach and artificial intelligence-assisted medical information summarization.

LISA incorporates two distinct QR-code workflows. The first provides access to a limited emergency profile and designated emergency contacts. The second is intended to facilitate access to additional medical information and is currently under development. Python supports the application logic, Firebase provides backend and database functionality, and QR-code libraries enable QR-code generation and scanning. An implemented artificial intelligence component supports medical information summarization; however, its underlying model or service requires further documentation.

Initial basic functionality testing has been performed. Systematic evaluation of information retrieval time, summary quality, reliability, and access-control behaviour remains future work. Therefore, no quantitative performance improvements, security guarantees, or clinical benefits are claimed at this stage.

LISA explores a practical approach to organizing emergency healthcare information through digital access workflows. Future development will focus on completing the second QR workflow, evaluating the AI-generated summaries, reviewing data-protection mechanisms, and conducting structured functional and usability testing. The prototype aims to support information accessibility during emergencies, while its clinical effectiveness remains to be established

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AI-Based Intelligent Crowd Monitoring and Real-Time Evacuation Assistance System

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Abstract:

Large public gatherings such as festivals, concerts, railway stations, shopping malls, and sports events often face challenges related to overcrowding, panic, and poor crowd management, which can lead to accidents or stampedes. The proposed AI-Based Intelligent Crowd Monitoring and Emergency Evacuation Assistance System uses Artificial Intelligence, Computer Vision, and real-time mapping to monitor crowd movement continuously through surveillance cameras. The system analyses crowd density and movement patterns to detect abnormal behaviour and predict potential risks before they become critical. When a dangerous situation is identified, it immediately alerts the concerned authorities and generates a live crowd map by dividing the area into multiple regions based on crowd density. It then recommends the safest evacuation routes and provides a region-wise movement sequence to ensure an organised evacuation, reducing congestion and preventing bottlenecks. The application also offers a real-time visual dashboard that helps authorities monitor crowd conditions and make faster, data-driven decisions during emergencies. By integrating intelligent monitoring, predictive analysis, automated alerts, and dynamic evacuation planning into a single platform, the proposed system aims to improve public safety, minimise response time, and significantly reduce the risk of crowd-related disasters in large public spaces.

Keywords: Real-Time Surveillance, Abnormal Behaviour Detection, Emergency Alert System, Smart Evacuation, Crowd Monitoring, Computer Vision, Path Planning, Region-Based Crowd Segmentation.

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A Robust Deep Learning Ensemble System for Social Media Rumour Classification

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Abstract:

Organizations need to monitor competitors, but relevant information is often scattered across public news, company websites, product announcements, hiring pages, and other permitted sources. Reviewing these signals separately can make it difficult to identify meaningful developments, understand their possible relevance to a particular organization, and keep earlier intelligence up to date. This paper presents the design of a Multi-Agent Corporate Intelligence and Competitive Monitoring Platform that coordinates specialized agents to collect and normalize public information, combine related signals, and produce evidence-linked intelligence summaries. The proposed workflow includes source tracking, entity resolution, event formation from related signals, mapping of detected events to defined internal business assets, and identification of previously generated conclusions that may require reassessment. Rather than regenerating every report after each new signal, the platform is designed to identify and re-evaluate only conclusions connected to the affected assets, while retaining source references and distinguishing observed facts from analytical interpretations. The system is intended to support human decision-making rather than make strategic decisions autonomously. The proposed design will be evaluated using test scenarios that assess event relevance, evidence traceability, update scope, processing effort, and the correctness of affected-conclusion identification. The platform uses publicly accessible and permitted data sources and is designed with source attribution and verification as core requirements.

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SMART GRAIN STORAGE CONDITION MONITORING AND SPOILAGE RISK PREDICTION FOR PUBLIC DISTRIBUTION SYSTEM (PDS) USING AI AND IoT

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Abstract:

Food grain spoilage during storage is a major challenge in the Public Distribution System (PDS) due to environmental factors such as high temperature, humidity, excess moisture, and poor air quality, which reduce grain quality and lead to food wastage and economic loss. This project proposes a Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System using Artificial Intelligence (AI) and the Internet of Things (IoT) to continuously monitor storage conditions and provide early warnings before spoilage occurs. The proposed system uses an ESP32 microcontroller integrated with a DHT22 temperature and humidity sensor, grain moisture sensor, and MQ135 gas sensor to collect real-time environmental data from grain storage facilities. The collected data is transmitted to an IoT cloud platform for continuous monitoring and analysis. A Machine Learning model processes the sensor data and predicts the spoilage risk by classifying the storage condition as Safe, Moderate Risk, or High Risk, enabling warehouse authorities to take timely preventive actions. The proposed solution aims to reduce food wastage, improve grain storage efficiency, minimize manual inspection, and support smart warehouse management in PDS godowns through a low-cost, scalable, and intelligent monitoring system.

Keywords: Artificial Intelligence, Internet of Things, Smart Grain Storage, Spoilage Risk Prediction, ESP32, Grain Moisture Monitoring, Public Distribution System

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AI-Based Electrical Fire Prevention System Using IoT Sensors and Machine Learning

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Abstract:

Electrical fires are often caused by electrical and environmental conditions and are not detected until the smoke and flames of the fire start. To address this issue, we have created a prototype that monitors the temperature, voltage, current and smoke of an electrical device using the ESP32 IoT platform. An machine learning model classifies the risk of the situation into four levels of risk safe,warning,high risk or critical.We created a data set of 6,400 samples to train the machine learning model. The random forest model that was trained with the data set achieved an accuracy of 99.06%. However, for the embedded device that we created, a decision tree model was used which achieved an accuracy of 98.12%. When the system detects a state that indicates a potential fire, it will trigger a relay switch that cuts the power to the electrical device. In addition, a buzzer will sound and an SMS message will be sent to a designated phone number via the SIM800L GSM module. All sensor readings and events that occur are also logged. We tested our prototype in a controlled environment. This is a student prototype and is not intended to be used for high current applications.

Keywords: Fire prevention, ESP32, Internet of Things, Machine learning, Decision Tree, Random Forest, Multi-sensor monitoring GSM alert

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An Agentic Framework Using Slime Mold Intelligence for Transportation Network Optimization

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Abstract:

Designing transportation networks that connect many stations with limited infrastructure, while staying reliable when individual links fail, is a central challenge in smart urban mobility. Biological systems offer an alternative to centralized optimization: the slime mold Physarum polycephalum self-organizes efficient, fault-tolerant transport networks between food sources. This paper presents an agentic framework that implements the agent-based, raster-based Jones model, in which a population of simple autonomous agents senses, deposits, and follows a diffusing chemoattractant trail, with stations acting as nutrient stimuli. Because the process is stochastic, the simulation is repeated over multiple independent runs. A station-to-station graph is extracted from each trail field through thresholding, skeletonization, and station association, and the graphs are merged into a consensus network in which every link carries an edge frequency, the fraction of successful runs in which it appears. Links above a user-defined frequency threshold are treated as stable connections, while less frequent links reveal alternative candidate corridors. Developed using Python, NumPy, SciPy, NetworkX, and Streamlit, the system provides an interactive map for station entry and CSV import, adjustable model parameters, and exports in CSV, GeoJSON, and reproducible project formats for GIS tools. In a Chennai Metro case study, high-frequency links connected all stations, indicating the framework's potential as a transparent decision-support tool for early-stage network planning. Generated links are conceptual candidate alignments, not surveyed routes.

Keywords: Slime Mold Intelligence, Physarum polycephalum, Agentic Framework, Agent-Based Simulation, Transportation Network Optimization, Jones Model, Consensus Network, Geospatial Analysis.

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EchoGenesis: A Longitudinal, Degradation-Robust Framework for Voice Preservation in Speech Loss

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Abstract:

EchoGenesis is a voice preservation solution for people whose speech changes because of conditions such, as amyotrophic lateral sclerosis ALS, Parkinson's disease, stroke cancer treatments, or laryngectomy. Current voice cloning solutions and text-to-speech systems are mostly built around healthy speech and do not sufficiently consider longitudinal changes in the voice of a person. EchoGenesis is capable of preserving the unique voice of a speaker even when his/her speech becomes worse further. EchoGenesis uses degradation-robust speaker encoder which detects the speaker from damaged speech samples, and it compares more recent samples with those recorded voice earlier, while the speaker had a healthy voice. Natural speech is then generated from a limited number of voice samples through few-shot text-to-speech model, and prosody aware chunked synthesis and is used for longer sentences. Evaluation of the system is done using metrics for measuring speaker similarity, word error rate, and degradation robustness curve which is a proof-of-concept embedding-drift study shows that the drift signal increases with the degradation level and proves the variability between different sessions.

Keywords: Voice Cloning, Artificial Intelligence, Deep Learning, Amyotrophic Lateral Sclerosis, Throat Cancer, Speech Preservation, Personalized Voice Model, Word Error Rate.

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An IoT-Enabled Intelligent Classroom Monitoring Framework for Real-Time Occupancy and Noise Detection

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Abstract:

Many educational institutions face challenges such as overcrowded classrooms, inefficient seating utilization and excessive noise levels, all of which can negatively impact teaching effectiveness and the learning environment. This paper presents the design of an IoT-based Smart Classroom Occupancy Monitoring System, currently under development, designed to automate classroom monitoring and improve resource utilization. The system is built around an ESP32 microcontroller, which serves as the central processing unit for data collection, processing, and wireless communication. A camera module is used to count students, identify available desks, and determine classroom occupancy in real time. To enhance reliability, a radar sensor is integrated as a backup presence-detection mechanism, ensuring accurate occupancy monitoring even under poor lighting conditions or when visual obstructions affect camera performance. In addition, a noise sensor continuously monitors classroom sound levels and detects excessive noise that may disrupt the learning process. By combining occupancy monitoring and noise detection into a single platform, the proposed system provides a cost effective and efficient solution for smart classroom management. The system aims to improve classroom discipline, optimize resource allocation, and support a more productive and technology-enabled educational environment. The hardware prototype is being built, and the evaluation plan and expected outcomes are described in this paper. The system is expected to achieve 96.5% desk accuracy and 94.0% noise agreement under normal lighting. Even in dark conditions, it is expected to maintain 92.6% noise agreement, supporting reliable classroom monitoring.

Keywords: IoT, Occupancy Monitoring, ESP32, Smart Classroom, Radar Sensor, Camera-Based Detection.

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FPGA-Based Real-Time FIR Filter Accelerator for Audio Noise Reduction

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Abstract:
Audio signals captured in practical environments are often contaminated by unwanted noise that degrades sound quality and speech intelligibility, making frequency-selective filtering an essential processing stage. Real-time operation requires dedicated digital hardware that can process each incoming sample within a fixed sampling interval. Finite impulse response (FIR) filters are widely used for this purpose because of their inherent stability and linear-phase capability, but their hardware implementation requires careful consideration of arithmetic precision, computational complexity, and FPGA resource usage. This work proposes a real-time FIR low-pass filtering accelerator for audio noise reduction, targeted at the Spartan-7 Boolean Board. The FPGA processes digital audio samples, which in the proposed complete system are obtained through an anti-aliasing filter and an external ADC, with filtered samples passed to a DAC or audio output interface; these interfaces are proposed and are not yet implemented. The filter computes $y[n] = \Sigma\, h[k]x[n-k]$ using delay-line registers, coefficient storage, multiplication units, an adder/accumulator, fixed-point scaling, and a registered output. Preliminary design targets are an 8-tap filter, a sampling frequency of approximately 16 kHz, a cutoff frequency of approximately 3.5 kHz, and Q1.15 fixed-point coefficients, all subject to refinement. MATLAB will be used for coefficient design, frequency-response analysis, floating-point reference filtering, and comparison with quantized coefficients, while the RTL will be developed in Verilog/SystemVerilog and implemented using Vivado. The proposed small contribution is the practical implementation and evaluation of a symmetry-aware architecture. For an 8-tap linear-phase filter with symmetric coefficients, paired samples can be added before multiplication, reducing coefficient multiplications from eight to four, provided adequate arithmetic width and scaling are preserved. This reduction is an architectural property; the resulting FPGA savings depend on pre-adders, bit widths, synthesis optimization, timing, and device architecture, and will be measured rather than assumed. Planned evaluation compares conventional and symmetry-aware designs through testbench simulation against MATLAB reference outputs, followed by examination of LUT, flip-flop, DSP, and timing reports and board testing when the interfaces are available. The work aims to demonstrate the practical value of resource-conscious FIR implementation for low-cost embedded audio processing.

Keywords: FPGA, FIR Filter, Audio Noise Reduction, Spartan-7, Fixed-Point Arithmetic, Symmetric FIR, Digital Signal Processing, Coefficient Quantization, Resource Optimization, Real-Time Filtering

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Automated Food Expiry Detection and Shelf-Life Prediction Using OCR and Barcode-Based Identification

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Abstract

Food waste is a significant global problem causing economic inefficiencies, pollution, and improper usage of resources. Poor inventory management and ineffective monitoring of expiry dates cause unnecessary food waste in houses and small enterprises. This research paper presents a food expiry detection and reminders system using artificial intelligence (AI) for effective food inventory management. The suggested system combines optical character recognition (OCR) for food expiry date detection, barcode scanning for product identification, and food shelf life estimation for managing foods. For packaged food products, OCR technique is employed to detect manufacturing and expiry dates from food labels, while barcode scanning allows to identify the food product and manage the inventory. For the fresh foods without explicit expiry dates, the remaining shelf life is estimated based on the predefined food type dataset and user-defined storage conditions. Moreover, the proposed system provides centralized inventory management, flexible settings for shelf life, expiry detection, and automated reminders before food expiry. The system's performance is planned to be assessed based on accuracy, precision, recall, and F1-score for expiry date detection and proper error metrics for shelf life estimation. Preliminary performance goals are set for expiry-date identification, such as accuracy of 92%, precision of 94%, recall of 90%, and an F1-score of about 92%; these goals need to be verified experimentally via the use of an annotated dataset. The proposed framework is expected to contribute to improving food stock management as well as reducing food waste due to the application of automated expiry date detection.

Keywords - Artificial intelligence, barcode recognition, food inventory management, food waste reduction, optical character recognition (OCR), shelf-life estimation, expiry-date detection.

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AI-Based Expense Tracker with Analytics and Investment Recommendations

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Abstract

Managing personal expenses effectively is important for maintaining financial stability and achieving future financial goals. However, many users find it difficult to track their daily spending, understand their spending patterns, and plan their savings. This project, AI-Based Expense Tracker with Analytics and Investment Recommendations, proposes a web-based application that helps users manage their income, expenses, budgets, and savings in a simple and organized manner. The system allows users to manually enter their income and expenses and categorize them into areas such as food, transportation, shopping, education, and bills. The collected data is analyzed to identify spending patterns, predict future expenses, and provide personalized suggestions for better financial management. Users can also set specific financial goals, such as buying a car or a laptop within a particular period. Based on their income, expenses, savings, goal amount, and risk preference, the system provides suitable investment suggestions such as Fixed Deposits, Mutual Funds, Gold, or Stocks. The application also provides graphical analytics, budget monitoring, overspending alerts, and savings recommendations to help users make better financial decisions. The system can be developed using Python, Flask, HTML, CSS, JavaScript, MySQL, and Scikit-learn for machine learning-based analysis. The proposed system aims to make personal finance management easier by combining expense tracking, data analytics, AI-based predictions, and goal-oriented financial recommendations in a single platform.

Keywords: Artificial Intelligence (AI), Expense Tracking, Machine Learning, Financial Analytics, Expense Prediction, Investment Recommendation, Budget Management, Goal-Based Saving

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MindScape: AI-Based Mental Wellness and Stress Relief System Using Natural Language Processing and Voice Assistance

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Abstract

MindScape is a privacy-first, web-based personal wellness and self-reflection platform that unifies dream and scenario journaling, mood tracking, structured self-understanding tools, productivity aids, guided healing exercises, and AI-assisted reflective storytelling grounded in secular and user-selected wisdom traditions. Unlike conventional mood-tracking or journaling applications, MindScape is built around three deliberate design constraints: first, it never attempts to clinically diagnose or assess the user — all reflective scoring is transparent and self-reported; second, all personal data remains under the user's direct control through browser-local storage with explicit export and import; and third, any AI-generated content is screened against crisis-indicating language before generation and is presented as a clearly separate, optional reflection rather than being merged into the user's authentic journal. The current prototype is implemented entirely as a client-side single-page application using HTML5, CSS3, and vanilla JavaScript (ES6+), integrating a generative AI language model API for reflective content and native browser Web APIs — the Web Speech API for voice journaling and the SpeechSynthesis API for multilingual audio playback of wisdom content. No backend server or database is used in the current build; all data persists locally in the user's browser via the localStorage API. This paper documents MindScape's system design, technology stack, module-wise architecture, implementation highlights, safety and ethical design considerations, testing strategy, and the roadmap required to evolve the prototype into a production-grade, multi-user application backed by a real server, database, and authentication system.

Keywords Mental Wellness, Natural Language Processing, Voice Assistance, Generative AI, Digital Journaling, Mood Tracking, Web Speech API, Privacy-First Design, Self-Reflection.

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NEXAM: Online Examination and Real-Time System for Assessment Management

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Abstract

Online exams are now common in colleges, but most systems either trust students too much or punish them unfairly for false alarms, and many are too costly for institutions. NEXAM is a web-based platform that unifies examination, identity verification, and continuous proctoring in one affordable system built on open web technologies. Registration is limited to students on a faculty-uploaded college list and verified through their official college email, which prevents fake and duplicate accounts. Face descriptors captured at registration are matched against live webcam frames before and throughout the exam. Computer vision and object detection identify absent faces, additional persons, mobile phones, and extra monitors, while live speech-to-text detects suspicious vocal activity. NEXAM's key idea is fair proctoring. A tiered policy of flags, warnings, and termination, combined with a grace period, reduces false alarms. Every flag is linked to a timestamp in the full session recording and to an evidence photo. A risk score helps faculty review the highest-risk candidates first, and faculty can confirm or dismiss each flag before it counts. Candidates terminated for malpractice are still graded on the answers they wrote. NEXAM also supports text, image, and coding questions checked against test cases, per-student question shuffling, extra time for individual students, a live faculty monitor, and automated email reminders and results. Students receive detailed answer reports, and faculty can export roll-number-wise PDF reports. NEXAM has limitations. Webcam and microphone use raise privacy and bandwidth concerns, accuracy depends on lighting and network quality, and browsers cannot detect background applications or Bluetooth devices. Large-scale concurrent testing remains future work.

Keywords: Online Examination, Face Recognition, Proctoring System, Object Detection, Academic Integrity, Speech Detection, Automated Evaluation

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NeuroCalc: An AI-Powered Mathematical Problem-Solving and Learning Assistant

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Abstract

Many learners now turn to conversational AI for mathematics help, but these tools can make arithmetic mistakes even when their explanations sound convincing, while conventional calculators give exact answers without explaining how they were reached. This paper presents NeuroCalc, an AI-powered mathematical problem-solving and learning assistant that combines accurate computation with clear explanations. When a user types a question in natural language, the system first validates and normalizes the input and then uses an intent classifier to decide how the question should be handled. There are three processing paths. A calculation can be sent to SymPy for symbolic computation, a conceptual question can be answered by a large language model, or a problem can be solved by SymPy and then explained step by step by the language model. For every operation it supports, including equation solving, simplification, differentiation and integration, SymPy is treated as the authoritative source of the final result, so the language model is used only for explanation and not for calculating the answer. The system is built as a web application with a Next.js and React frontend, a FastAPI backend, and an SQLite database accessed through SQLAlchemy. The database stores users, conversations and messages so that learners can return to their earlier work. This paper describes the architecture, the routing method and the current prototype. It also defines an evaluation plan based on mathematical accuracy, routing accuracy, response latency and explanation quality. Quantitative results have not yet been collected and will be reported after deployment testing. By keeping exact computation separate from language-based explanation, NeuroCalc aims to give students answers they can trust together with explanations they can understand, supporting more effective mathematics learning

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AI CHATBOT FOR GRIEVANCE REDRESSAL IN LOCAL BODY GOVERNANCE WITH TAMIL LANGUAGE SUPPORT

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Abstract

Local body governance receives a large number of public grievances related to essential civic services such as roads, water supply, street lights, drainage, and waste management. Traditional grievance redressal systems often depend on man-ual complaint registration and classification, resulting in delays, inefficient department allocation, and limited accessibility for citizens using regional languages. To address these challenges, this paper proposes an AI-based chatbot for grievance redressal in local body governance with Tamil language support. The proposed system uses Natural Language Processing and Machine Learning techniques to understand citizen complaints, automatically classify them into appropriate categories, determine priority, and route them to the concerned department. The system also supports Tamil and English interaction, complaint status tracking, and geotagged photo and location information for improved verification and resolution. The per-formance of the proposed system is evaluated using classification accuracy, precision, recall, F1-score, complaint registration time, response time, and routing accuracy. The proposed approach aims to improve the efficiency, accessibility, accuracy, and transparency of the existing grievance management process and provide faster and more citizen-friendly public service delivery.

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GeneGuide AI: AI-Based Prenatal Genetic Risk Assessment and Decision Support System

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Abstract

Understanding genetic risk usually requires a doctor to review family history, parent details and past pregnancy results one by one. This process is slow and difficult when counsellors are not nearby. This paper introduces GeneGuide AI, a web-based tool that helps estimate risk using information from parents, family and past pregnancies. We used a Random Forest classifier trained on a dataset of 1,250 records with 14 input features. GeneGuide AI places each case into one of three groups: Low, Moderate or High risk. The result is intended to be educational, not a diagnosis. Every result page clearly states this. The system is built with Python, Flask, SQLite, HTML, CSS and JavaScript. It includes user registration and login, a form with sections that check for correct input, and a history of past assessments for each user. On a test set of 250 records, the model achieved 85.2 percent accuracy. The F1-scores were 0.92 for Low, 0.87 for Moderate and 0.75 for High risk. Importantly, GeneGuide AI never mixed up Low with High risk. By using machine learning with an easy-to-use web interface, GeneGuide AI demonstrates how risk screening can become more accessible and easier to reach, while still encouraging users to talk to real doctors for final decisions. I hope GeneGuide AI will help families make decisions.

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AI-BASED SPOT WELD STRUCTURAL ANALYSIS

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Abstract

Resistance Spot Welding (RSW) is widely used in automotive Body-in-White structures, where individual welds contribute to load transfer and structural integrity. This project develops a physics-based computational framework for an idealized DP600 resistance spot-welded lap-shear joint and an interpretable machine-learning surrogate. The primary geometry consists of two 100 mm × 25 mm × 1.5 mm sheets with a 25 mm overlap. Five nugget diameters from 4 to 8 mm are studied at six axial loads from 500 to 3000 N, producing 30 computational cases. The geometry is inspected in CAD, discretized in Gmsh using second-order tetrahedral elements, and solved as a linear-elastic static structural problem in CalculiX. Maximum global von Mises stress, equivalent strain, maximum displacement, reaction force, weld-region response and a preliminary Factor of Safety are extracted for every case. A second-order polynomial regression uses applied load and nugget diameter as inputs and maximum global von Mises stress as the target. Twenty-four cases are used for training and six 7 mm cases are held out for validation. The model gives training $R^2 = 0.9903$, MAE = 6.161 MPa and RMSE = 7.314 MPa; validation gives $R^2 = 0.9567$, MAE = 10.737 MPa and RMSE = 10.773 MPa. At 3000 N, maximum global stress decreases from 305.5 MPa for a 4 mm nugget to 159.1 MPa for an 8 mm nugget, while the model-based FOS increases from 1.146 to 2.201. The held-out validation indicates useful in-domain interpolation, while the negative predicted-minus-actual residuals show systematic stress underprediction; therefore, the ML-derived safety margin is not conservative without further calibration. The study is computational only and does not claim physical weld inspection, fracture, fatigue, corrosion monitoring or certification. The resulting FE-to-ML workflow provides an auditable baseline for future nonlinear, experimental and degradation-aware weld-health studies.

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Rare Earth Free Motors

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Abstract

The rapid growth of electric mobility has created a need for electric motors that are energy-efficient, cost-effective, reliable, and environmentally sustainable. Conventional electric vehicle motors such as Permanent Magnet Synchronous Motors (PMSMs) often rely on rare-earth permanent magnets, which can lead to high material costs, supply-chain challenges, and environmental concerns. To address these issues, this project proposes a Ferrite-Assisted Synchronous Reluctance Motor (FA-SynRM) as a sustainable alternative for electric vehicle propulsion, particularly for two- and three-wheelers.The proposed FA-SynRM combines the reluctance torque principle of a Synchronous Reluctance Motor with strategically positioned ferrite permanent magnets embedded within the rotor flux barriers. The use of ferrite magnets reduces dependence on rare-earth materials while maintaining efficient torque production. An optimized asymmetric rotor structure is designed to improve flux distribution, increase saliency, enhance torque, and reduce torque ripple. Hairpin windings are incorporated to improve electrical and thermal performance, while Field-Oriented Control (FOC) enables precise and efficient speed and torque control under varying operating conditions.An AI-based thermal monitoring system is also proposed to monitor temperature, current, speed, and torque, enabling early detection of overheating and improving motor reliability and operating life. The motor is evaluated through electromagnetic and thermal simulations based on parameters such as torque, efficiency, torque ripple, power factor, flux distribution, and temperature rise. Simulation-based optimization also helps reduce material consumption and prototype iterations.Overall, the proposed FA-SynRM provides a sustainable motor solution by combining reduced rare-earth dependency, efficient energy utilization, optimized material usage, improved thermal management, and intelligent monitoring. The proposed technology has the potential to support cleaner, affordable, and resource-efficient electric mobility while contributing to the development of sustainable next-generation electric vehicle propulsion systems.

Keywords: Sustainable Electric Mobility, Rare-Earth Material Reduction, Ferrite Permanent Magnets, Energy Efficiency, Sustainable Motor Design, Asymmetric Rotor, Hairpin Winding, Field-Oriented Control (FOC), AI-Based Thermal Monitoring, Electric Vehicles, Thermal Management, Resource Efficiency, Torque Optimization, Two- and Three-Wheeler EVs

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AI-Driven Email Prioritization System Using Llama for Intelligent Email Management

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Abstract

There has been an increase in the use of email for communication. This has made it harder for people to handle and decide which messages are most important. Old email programs sort messages by when they were sent. This means people have to look through their inbox and decide which emails need attention. This can lead to replies missing deadlines, too much information and less work done. To fix these problems this paper introduces ADEPS, an AI-Driven Email Prioritization System. This system uses the Llama Large Language Model to understand what emails mean and decide their importance.The system works with Gmail by using the Gmail API and Google OAuth 2.0. This makes it safe for people to use. Each email is looked at by the Llama model. Put into one of four groups: Urgent Important, Normal and Spam. The system can also make summaries find important things to do spot deadlines and meetings and suggest good replies based on the email. The priority is shown in Gmail using labels. This helps people see which messages are important in their inbox.ADEPS is built with Python, Flask, SQLite, HTML, CSS and JavaScript. This creates an easy-, to-use way to manage, search and sort emails. By using the power of Large Language Models to understand emails and adding it to Gmail the system cuts down on the work people do by hand. It helps with organizing. Lets people focus on important and urgent messages. The system shows how using AI in email can help people work better and handle emails faster and with knowledge.

Keywords— Artificial Intelligence, Large Language Model (LLM), Llama, Email Prioritization, Gmail API, Flask, Natural Language Processing, Email Summarization, Intelligent Email Management.

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DemandIQ: AI-Driven Retail Demand Forecasting and Inventory Optimization System

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Abstract

Small retail stores often struggle to maintain optimal inventory levels, leading to stock shortages or excess unsold goods. This project, DemandIQ, presents an AI-driven retail demand forecasting and inventory optimization system that predicts future product demand using historical sales data. Existing inventory management systems mainly rely on manual estimation and traditional forecasting methods, which often lead to inaccurate demand predictions, stock shortages, overstocking, increased inventory costs, and poor business decision-making. By combining machine learning and time-series forecasting techniques, the system analyzes past sales patterns to improve demand prediction accuracy and provide optimal restocking recommendations. This helps retailers reduce stockouts, minimize overstocking, and manage inventory more efficiently. The proposed system demonstrates how artificial intelligence can be applied to real-world retail challenges, providing a practical, low-cost solution for small businesses to optimize inventory management and support better business decision-making.

Keywords: Inventory Management, Demand Forecasting, Machine Learning, Time-Series Forecasting, Retail Analytics, Predictive Modelling.

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FreshAir Desk: AI-Powered Smart Desk Environment Monitor

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Abstract

Indoor environmental quality strongly affects health, comfort, concentration and productivity, yet poor air, heat and humidity often go unnoticed by students and professionals who spend long hours indoors. This paper presents FreshAir Desk, a compact IoT-based desk environment monitor built on an ESP32 microcontroller. It measures temperature, humidity, light intensity and air quality (RS/R0), and streams the readings in real time to a live web dashboard that shows current values, status labels and a short history. A rule-based model classifies conditions as Good, Attention or Poor and suggests actions such as improving ventilation or lighting. To evaluate the system, a comparative analysis was carried out across small rooms, big rooms and classrooms, with and without air conditioning, at increasing occupancy. A comparative analysis under Chennai climatic conditions identified the ideal indoor condition as an air-conditioned, adequately sized space with low occupancy, giving about 24–26 °C, 48–55 %RH and good air quality (RS/R0 above 3.6). Non-air-conditioned, small and crowded spaces deviated from this ideal, reaching over 30 °C and 62 %RH, with poor air quality. The results show that cooling, room size and occupancy strongly influence indoor comfort, and that FreshAir Desk enables proactive, data-driven decisions for healthier study and work environments.

Keywords: Internet of Things, Artificial Intelligence, Environmental Monitoring, ESP32, Smart Desk, Air Quality, Occupancy Analysis.

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Comparative Machine Learning Approach for Hydraulic Pump Fault Classification Using Vibration and Temperature

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Abstract

Hydraulic pumps play an important role in many systems and can fail for various reasons, causing efficiency drops and unexpected downtime. This study presents an AI-driven condition monitoring system for hydraulic pumps using vibration and temperature sensor fusion. An MPU6050 accelerometer and DS18B20 temperature sensor are integrated with an ESP32 for real-time data collection and monitoring. A prototype hydraulic pump setup was developed to collect data under normal and fault conditions. The collected vibration and temperature data are analyzed using machine learning to classify the pump condition as normal, abnormal, or critical. A comparative study is conducted using vibration-based classification and combined vibration- and temperature-based classification to evaluate performance. The proposed system aims to provide a low-cost, real-time, and scalable solution for early pump fault detection.

Keywords: Hydraulic Pump, Condition Monitoring, Fault Classification, Sensor Fusion, Machine Learning

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An Intelligent Voice Assistant for Visually Impaired Navigation in Public Transport using Real-Time Object Detection

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ABSTRACT

Visually challenged people find very difficult for their independent mobility in public transport. Even public transport systems does not support with the needed features such as tactile paving, Braille signage, or clear audio announcements leading them to difficult situation for a safe navigation. So considering their difficulty, an intelligent voice assistant is proposed which integrates a real-time object detection along with scene text recognition to direct a visually impaired commuter at a bus stop and inside a public transport environment. Using a smartphone or a wearable camera video frames are captured which are then analysed by a YOLO (You Only Look Once) based detector for locating buses, doors, people, poles and other obstacles. Meanwhile a text recognition module reads route numbers and destination boards for analysis. A depth estimation module is used for finding the distance of the identified objects and then a decision layer transforms the findings into a precise prioritised spoken instruction given through a text to speech engine in the language chosen by the individual. This proposed tool supports the visually challenged person to recognize the correct bus, assess the boarding location and avoid hindrances

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Intelligent Online Payment Fraud Detection with LightGBM and Multi-Layer Security

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Abstract

The rapid growth of e-commerce, Unified Payments Interface (UPI), internet banking, and digital wallet services has significantly increased the occurrence of fraudulent online payment transactions.Traditional rule-based fraud detection methods rely on predefined patterns and may fail to identify evolving and complex fraud behaviors. Hence, intelligent Machine Learning (ML) approaches are required to improve fraud detection and strengthen the security of digital payment systems.This project proposes an intelligent framework for online payment fraud detection and transaction security with Light Gradient Boosting Machine (LightGBM) as the primary classification model. The performance of LightGBM is compared with Logistic Regression, Decision Tree, Random Forest, and XGBoost using Accuracy, Precision, Recall,F1-Score, and ROC-AUC. This comparative analysis helps identify an effective model for distinguishing legitimate and fraudulent transactions. After successful authentication, transaction details are analyzed by the trained LightGBM model. Multiple fraudulent transaction attempts or repeated OTP verification failures temporarily lock the account and restrict further transactions. An administrative dashboard provides centralized monitoring of transaction history, fraud predictions, high-value transactions, locked accounts, and security alerts.The proposed framework combines machine learning-based fraud classification with authentication and automated security mechanisms to provide a reliable and secure online payment environment. The approach aims to improve fraud detection performance, reduce unauthorized transactions, and enhance the overall security and trustworthiness of digital financial services

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SECURE INTELLIGENT PORTABLE NOHONKING SYSTEM WITH ADAPTIVE SILENT ZONES AND CENTRALIZED MANAGEMENT

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Abstract

Unnecessary vehicle honking in sensitive areas such as hospitals, schools, residential areas, and wildlife zones contributes to noise pollution and disturbance. Existing approaches mainly depend on driver awareness, GPS based geofencing, which may have limitations in privacy, coverage, and infrastructure requirements. This project proposes a Portable RF-Based Intelligent Horn Management System for Silent Zones in Smart Vehicles. The system establishes communication between roadside infrastructure and vehicles using RF signals to automatically manage horn operation within designated zones. An ESP32-based vehicle unit processes and validate the received command and controls the horn, while an emergency override allows necessary horn usage. The roadside unit can be remotely monitored and configured through a centralized platform using 4G/LTE, MQTT, and secure communication mechanisms. The proposed system provides a portable, scalable, and practical approach to support silent-zone management and improve the commuting environment.

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WEARABLE DEVICE FOR PHYSIOLOGICAL RECOVERY-BASED EXERCISE MONITORING

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Abstract

This paper presents an adaptive wearable fitness system for real-time workout monitoring and Rate of Perceived Exertion (RPE) estimation. The system uses motion and physiological measurements through an MPU6050 IMU sensor and a MAX30102 heart rate monitor, implemented in an ESP32-based environment. Real-time computations are achieved via a dual-core FreeRTOS system that uses concurrent sensing, inference, and user interaction. The system uses a self-calibration process for personalized motion thresholding to accurately detect repetitions and sets in several different exercises. A TinyML model predicts Rate of Perceived Exertion from motion and heart rate signals, while an adaptive rest time algorithm determines the optimal rest time from the heart rate signal recovery patterns. All computations are executed in a standalone wearable fashion without any dependence on external hardware or cloud computing resources.

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A Differentiable Computational Framework for Reconstructing Intentional Acoustic Excitation from Archaeoacoustic Chamber Geometry

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Abstract

Ancient stone chambers, temple interiors, and rock-cut caves and rooms sometimes exhibit geometries that could have been intentional in their design for influencing sound behaviour and the flow of sound. However, details about the sound used in these rooms and the principles behind their acoustic design are typically unknown. The aim of this research is to investigate if, from the remaining geometries of heritage buildings, enough physical evidence exists to fully characterize their acoustics and infer the likely excitation used with them. The research focuses on developing an integrated computational system comprising 3-D reconstruction of heritage, physics-based acoustics modeling, machine learning, and inverse modelling. The process will start by generating 3-D digital models of selected heritage chambers located in and around Chennai using photogrammetric and LiDAR data. The models will be then analyzed with FEM/FDTD simulations in order to create valid acoustic signatures comprising the main frequencies, resonant behaviour, signal amplification, and modal analysis for every given room. These signatures, used for modeling their acoustical properties by means of an artificial neural network, will allow predicting the response of these rooms. An inverse modeling technique then identifies the most plausible excitative (source type) signal based on simulated resonance profiles using gradient-based optimization and evolutionary search will validate possible recovered solutions.The main output of this project is a documented and reusable computational tool for which three dimensional scans of any architectural place can be uploaded producing it, its acoustic behavior including characteristic frequencies, amplifications, distribution, and excitation factors that would fit the acoustic observed properties. Instead of trying to retrieve the exact sound that might have been historically produced into these spaces, this methodology offers a rigorously validated technique to infer the acoustical intentions associated with heritage architectural design.

Keywords Archaeoacoustics; Heritage Architecture; Acoustic Intent; Differentiable Computing; Inverse Acoustic Modelling; Physics-Based Acoustic Simulation; FEM/FDTD; LiDAR

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Circuit-Level Design and Analysis of a Low-Power CMOS Phase-Locked Loop Using Cadence Virtuoso

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Abstract

A Phase Locked Loop or PLL is a mixed-signal feedback system that realizes clock generation, frequency synthesis, and phase and frequency synchronization. The project examines the design and transistor-level implementation of a low-power full-custom CMOS PLL that is developed in a gpdk45-nm design environment utilizing Cadence Virtuoso. The proposed circuit comprises a Phase-Frequency Detector (PFD), charge pump, loop filter, ring-oscillator-based Voltage-Controlled Oscillator (VCO), and a feedback frequency divider. The PFD compares the reference clock with the divided feedback signal and generates UP and DOWN pulses according to the phase and frequency error. These control pulses are utilized to drive the charge pump that sources or sinks current into the loop filter. The loop filter converts the charge pump current into a control voltage that regulates the frequency of the voltage-controlled oscillator. Meanwhile, the frequency divider scales the VCO output and provides the feedback necessary for the closed-loop operation. The proposed design takes into account the influence of loop-filter properties on bandwidth and stability as well as control-voltage ripple, transient response, and noise characteristics. The behaviour of the circuit at a transistor level is evaluated using Cadence simulations, while robustness is analysed using PVT and Monte Carlo simulations that take into account process, voltage, temperature, and device variations. The project takes into account PLL non- idealities that include current mismatch, timing skew, parasitic effects, phase noise, jitter, and spurs that could compromise the locking response and spectral performance of the circuit. Thus, the project realizes the circuit-level implementation and analysis of a low-power CMOS PLL that considers the trade-offs between loop dynamics, frequency control, stability, noise, and power consumption.

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Development and mechanical testing of jute-glass fiber hybrid composites for light-weight automotive panel

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Abstract

This study develops novel jute fiber and glass fiber epoxy hybrid composites to address the limited toughness and thermal conductivity of natural-fiber epoxies for structural applications. Research gaps include scarce data on optimized fibre loadings with aligned jute mats, unclear interfacial modification strategies for balancing stiffness and impact resistance, and limited multi-scale characterization linking microstructure to macroscopic performance. We used alkali-treated, aligned jute mats combined with silane-functionalized fibres dispersed in epoxy via three-roll milling; composites were fabricated by vacuum-assisted resin transfer molding with controlled fibre volume fractions and fiber contents (0–5 wt%). Mechanical (tensile, flexure, impact), thermal conductivity, dynamic mechanical analysis, and fractography (SEM) were performed alongside nanoindentation and electrical conductivity mapping to correlate phases across scales. Key outcomes show that 1–3 wt% silane- yields a 28% tensile strength increase, 42% improvement in fracture toughness, and thrice the value of thermal conductivity versus neat jute/epoxy; optimal fibers enhance stress transfer and crack-bridging without significant agglomeration. The composites retained more than 85% tensile performance after hygrothermal ageing, indicating improved durability. Future scope includes scale-up via continuous processing, fatigue and fire-resistance studies, life-cycle and cost-benefit analyses, and exploration of bio-based epoxy matrices to further improve sustainability and multifunctionality for automotive and lightweight structural uses.

Keywords: Jute fiber, vacuum-assisted resin transfer molding, dynamic mechanical analysis, bio-based epoxy matrix composites

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BLOCKCHAIN-BASED SECURE EHR SHARING ACROSS HOSPITALS USING HYPERLEDGER FABRIC

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ABSTRACT

Electronic Health Records (EHRs) facilitate the digital storage, management, and access of patient medical information. However, securely sharing medical records across hospitals remains a major challenge. Many hospitals use isolated healthcare systems, making it difficult for doctors to access prior medical records when patients receive care at different institutions. This can lead to repeated medical tests, higher healthcare costs, treatment delays, and inconvenience for patients. Protecting patient privacy and preventing unauthorized access are additional critical concerns. This research proposes a Blockchain-Based Secure Electronic Health Record Sharing Across Hospitals system using Hyperledger Fabric to enable secure and controlled exchange of medical records. Medical records are protected using AES-256 encryption and stored in secure off-chain storage, while their SHA-256 hashes are stored on the blockchain to verify data integrity and detect unauthorized modifications. Smart Contracts (Chaincode) manage access permissions by verifying patient consent, doctor authorization, and hospital identity before granting access to medical records. Access requests and key transactions are recorded on the blockchain to provide transparency and a tamper-resistant audit trail. The system also offers an Emergency Read-Only Access Mode for situations in which patients cannot provide consent. Authorized emergency physicians can view essential information such as blood group, allergies, current medications, previous diagnoses, and treatment history. Physicians cannot modify or delete records during emergency access, and all activities are recorded on the blockchain. The proposed system aims to enhance patient privacy, data integrity, secure healthcare information exchange, and continuity of care while reducing repeated medical tests, treatment delays, and unnecessary healthcare costs.

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A Differentiable Computational Framework for Reconstructing Intentional Acoustic Excitation from Archaeoacoustic Chamber Geometry

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Abstract

Ancient stone chambers, temple interiors, and rock-cut caves and rooms sometimes exhibit geometries that could have been intentional in their design for influencing sound behaviour and the flow of sound. However, details about the sound used in these rooms and the principles behind their acoustic design are typically unknown. The aim of this research is to investigate if, from the remaining geometries of heritage buildings, enough physical evidence exists to fully characterize their acoustics and infer the likely excitation used with them. The research focuses on developing an integrated computational system comprising 3-D reconstruction of heritage, physics-based acoustics modeling, machine learning, and inverse modelling. The process will start by generating 3-D digital models of selected heritage chambers located in and around Chennai using photogrammetric and LiDAR data. The models will be then analyzed with FEM/FDTD simulations in order to create valid acoustic signatures comprising the main frequencies, resonant behaviour, signal amplification, and modal analysis for every given room. These signatures, used for modeling their acoustical properties by means of an artificial neural network, will allow predicting the response of these rooms. An inverse modeling technique then identifies the most plausible excitative (source type) signal based on simulated resonance profiles using gradient-based optimization and evolutionary search will validate possible recovered solutions.

The main output of this project is a documented and reusable computational tool for which three dimensional scans of any architectural place can be uploaded producing it, its acoustic behavior including characteristic frequencies, amplifications, distribution, and excitation factors that would fit the acoustic observed properties. Instead of trying to retrieve the exact sound that might have been historically produced into these spaces, this methodology offers a rigorously validated technique to infer the acoustical intentions associated with heritage architectural design.

Keywords

Archaeoacoustics; Heritage Architecture; Acoustic Intent; Differentiable Computing; Inverse Acoustic Modelling; Physics-Based Acoustic Simulation; FEM/FDTD; LiDAR

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Emotional Granularity as a Measure of Affective Understanding in Large Language Models

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Abstract

Large Language Models are increasingly deployed in emotion sensitive applications, including mental health chatbots, crisis text lines, and journaling tools, yet their capacity for emotional granularity, the ability to distinguish closely related emotions such as guilt and shame, remains unmeasured. Existing NLP emotion benchmarks evaluate classification accuracy alone, treating all misclassifications as equivalent regardless of whether a model confuses psychologically similar emotions or entirely unrelated ones. This paper introduces the Emotional Granularity Index, a multi metric evaluation framework grounded in Barrett's (2004) emotion differentiation theory, comprising four components: Confusion Density Score, Valence Corrected Accuracy, Cluster Separation Index, and Human Parity Ratio. Using LLaMA 3.1 8B Instruct evaluated on the GoEmotions dataset (27 fine grained emotion categories, 58,000 annotated Reddit comments), we assess classification performance under zero shot and chain of thought prompting conditions. Beyond behavioural evaluation, we extract the model's internal hidden state representations and apply embedding geometry and Representational Similarity Analysis to determine whether its internal organisation of emotions mirrors a psychologically predicted structure derived from appraisal theory. We further test whether the model can detect emotional distress expressed without emotion vocabulary, through somatic, behavioural, and cognitive seeking language, conditions under which individuals in genuine psychological distress most often communicate. Findings are expected to reveal specific, structurally meaningful gaps in LLM emotional understanding that conventional accuracy metrics cannot detect, with the largest gaps anticipated where emotional content is least explicit. These results carry direct implications for the safe and responsible deployment of large language models in mental health and emotionally sensitive contexts.

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Cyber Threat Intelligence Framework For Gender-Based Digital Abuse Against Women

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ABSTRACT

The increasing use of digital platforms and social media has led to a rise in technology-assisted abuse against women, including cyberstalking, online harassment, impersonation, image-based abuse, doxxing, and digital coercion. These incidents can result in privacy violations, emotional distress, reputational damage, financial loss, and physical safety concerns. Existing Cyber Threat Intelligence (CTI) systems mainly focus on organizational cybersecurity threats and give limited attention to threats targeting individuals, particularly women. This study proposes a Cyber Threat Intelligence Framework for identifying, assessing, and responding to gender-based digital abuse against women. The framework collects threat information from user reports, digital evidence, publicly available threat intelligence, suspicious accounts, malicious links, and harmful communication patterns. Predefined rules and risk factors are used to identify threats and assess their severity based on threat type, frequency, impact, and potential harm. LLM-based assistance provides guidance on response actions, digital safety practices, evidence preservation, and reporting procedures. The proposed framework provides a structured and user-focused approach that integrates threat identification, risk assessment, evidence management, and response guidance to improve awareness, digital safety, and timely responses to gender-based digital abuse.

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Dark Pattern Detection in Digital Interfaces Using Semantic Text Analysis and Severity Scoring

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Abstract

Dark patterns are deceptive interface design practices that manipulate users into performing actions contrary to their original intentions, including unwanted subscriptions, unintended purchases, and acceptance of unnecessary permissions. Their increasing prevalence across websites and mobile applications has raised concerns regarding user autonomy, transparency, and consumer protection. Existing automated detection approaches predominantly rely on machine learning classifiers, which require substantial labelled datasets and offer limited interpretability. This study proposes a semantic text analysis framework for automated detection and severity assessment of dark patterns in digital interfaces. The framework analyses textual interface elements, including button labels, pop-up messages, countdown timers, and consent dialogs, to determine their semantic intent and manipulative characteristics. Natural Language Processing techniques using NLTK and spaCy are employed for text preprocessing, linguistic analysis, and semantic feature extraction. The extracted features are evaluated using a weighted scoring mechanism to assign each detected instance a severity score ranging from 1 to 10, classifying it as mild, moderate, or severe. Unlike machine learning-based approaches, the proposed framework requires no supervised model training and provides interpretable outcomes by identifying the textual characteristics contributing to each detection. The framework is expected to offer a transparent and auditable tool for systematic dark pattern assessment in digital interfaces.

Keywords: Dark Patterns, Semantic Text Analysis, Severity Scoring, Natural Language Processing, Digital Interfaces.

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Generative AI for Technical Knowledge Retrieval in Electric Vehicles

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Abstract

Electric vehicles are becoming increasingly common; however, finding technical information when a problem occurs can still be difficult. Warning messages, fault codes, battery concerns, and other vehicle issues may not be easy for owners to understand. Technicians also have to refer to large service manuals and technical documents while diagnosing faults. Searching these documents using exact keywords may not work when the words used in a user's question are different from those in the manual. General AI tools may provide useful explanations, but their responses are not always connected to EV manufacturer information. This study proposes an Intelligent EV Knowledge Assistant that uses retrieval-augmented generation (RAG) to answer technical questions from a collection of EV documents. The knowledge base will contain service manuals, troubleshooting guides, fault-code information, and battery management system (BMS) documents. A user or technician can ask a question without uploading a manual each time. The system will search the prepared document collection, identify relevant information, and pass that information to a large language model (LLM) to prepare a response. Therefore, the answer will be based on the retrieved technical content rather than only the model's general knowledge. This study will explore suitable approaches for preparing technical documents, retrieving relevant information, and generating context-based responses. The system will be assessed based on retrieval relevance, answer accuracy, response time, and user feedback. The intended outcome is a practical knowledge-support tool that can help EV users and technicians find and understand relevant technical information more easily during troubleshooting and maintenance.

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Security Misconfiguration Detection Framework

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Abstract

Security misconfiguration means wrong settings that create security risks in software, cloud systems, and company networks. When settings aren't set up properly, users get to many permissions, and the system becomes open to attack. Even though tools exist to catch these problems, they often miss complicated errors and need constant updates.What makes this important is that a single small misconfiguration is often not the real danger by itself — multiple small misconfigurations can combine together to create a much bigger security risk, which most existing tools fail to notice because they check each setting separately.

Keywords
Security Misconfiguration, Misconfiguration Detection, Cloud Security, Network Security, Configuration Errors, Excessive Permissions, Vulnerability Detection, Security Risk Analysis

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A Survey on Automating the Detection of Access Control Vulnerabilities in Web Applications

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Abstract

Access control vulnerabilities are among the most critical security threats in web applications, allowing unauthorized users to access or manipulate protected resources. This research examines automated approaches for detecting access control vulnerabilities by analysing user interactions and authorization mechanisms across different user roles. Existing detection techniques include black-box testing, dynamic analysis, response comparison, request replay, and rule-based methods, which have demonstrated the potential to improve the identification of unauthorized access while reducing the limitations of manual security assessment. However, current approaches continue to face challenges, including restricted coverage of certain vulnerability types, dependence on application-specific information, false positives and false negatives, and difficulties in reliably detecting unauthorized modification operations. In addition, variations in application behaviour, user privileges, and authorization mechanisms can affect the effectiveness and reliability of existing detection techniques. These challenges highlight the need for continued research into more comprehensive and reliable methods for analysing authorization behaviour and identifying access control violations across diverse web applications. Overall, existing research demonstrates the importance of automated access control vulnerability detection while indicating the need for further improvements in detection coverage, accuracy, and reliability.

Keywords: Access Control Vulnerabilities, Broken Access Control, Web Application Security, Automated Security Testing, Web Crawling, HTTP Request Analysis, Request Replay, Role-Based Access Control, Rule-Based Detection, Vulnerability Assessment.

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A STUDY ON PREDICTIVE EDGE WEIGHTS MODELING FOR DYNAMIC SHORTEST PATH COMPUTATION

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Abstract

In light of escalating traffic congestion, shifting road conditions, and the need for intelligent transportation systems, dynamic route planning in road networks has recently grown in relevance as a realm of expertise. The shortest path algorithms currently in use primarily concentrate on reducing travel time and distance, while many methods fail to sufficiently account for a variety of factors that affect real-world travel, such as road stereo type, slopes, favorable and unfavorable weather, and numerous other dynamic road characteristics. By estimating projected travel conditions rather than depending solely on current or past prior information, accurate edge weight prediction plays a significant and crucial role in improving route planning. This analysis describes how graph-based algorithms for routing evolved in order to forecast future edge value estimations and adaptive shortest route planning. The route algorithms presently in use, such as Dijkstra's Algorithm, A* Algorithm, Bellman and Ford Algorithm, Floyd-Warshall Algorithm, and Yen's K-Shortest Path Algorithm, together with statistical approaches involving ARIMA, SARIMA, Kalman Filter, Holt-Winters Method, and Exponential Smoothing, typically focus on preventing travel distance and time. These method’s pros, cons, and suitability for dynamic shortest path computing are analyzed.In order to improve routing accuracy and flexibility under all dynamic traffic circumstances, a significant emphasis is made on combining several road-related information into forecast future edge weight utilizing mathematical and graphical approaches. In order to create an integrated road type-aware predictive edge weight modeling framework for effective dynamic shortest path, this paper wraps up by identifying current research concerns and future research plans determining in advanced transportation networks.

Keywords: Dynamic Route Planning, Predictive Edge Weight Modeling, Dynamic Shortest Path, Graph Theory, Intelligent Transportation Systems (ITS).

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Enhancing Digital Image Forensics Through Blockchain-Based Evidence Provenance and Chain-of-Custody Management

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Abstract

With digital evidence having become a more important part of current forensic investigation and evidence analysis, there is an increasing demand for reliability of methods for evidencing integrity, authenticity and traceability of digital evidence. Several researches of using blockchain for digital evidence, authenticity and Chain-of-Custody verification were discussed recently. This survey explores five recent researches to discuss blockchain use to preserve digital evidence, verify forensics evidence, manage chain-of-custody record, legal evidence and digital image provenances. The researches are based on distributed ledger, smart contract, distributed storage, cryptographic validation, time stamping, access control mechanism for verifying the reliability and transparency of the digital evidences as well as managing accountability. Most current methods emphasis on general digital evidence preservation and immutable trail storage while there are only a few researches dedicated to the specific work on managing Digital image forensic evidence across the whole forensic life cycle. It appears that the blockchain based digital evidence preservation system provides the most suitable research base for the presented research, and it lays out a good foundation regarding how to keep evidence integrity, traceable, Chain-of-Custody verification. There is an absence in researches for specific domain and specifically image manipulation forensics procedures, fast evidence integrity checking and tracking as well as proper legal validation and chain-of-custody management for image forensic. Based on current review and analysis, research for an improved system on reliable digital image Chain-of-Custody management is introduced based on above research base.

Keywords: Digital Evidence, Digital Image Forensics, Blockchain, Chain of Custody, Evidence Integrity, Evidence Traceability, Digital Evidence Preservation, Image Provenance, Forensic Investigation.

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QR Code Encoding Modes and Error Correction Levels: A Comparative Study and Optimization Framework

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Abstract

Quick Response (QR) codes have become a fundamental component of modern information systems, widely deployed across payment platforms, healthcare, logistics, event management, and retail environments due to their ability to store and retrieve structured data rapidly using any standard smartphone camera. Despite their widespread adoption, a critical limitation persists in the way QR codes are generated - existing tools, both commercial and open-source, apply identical default parameters to every code they produce, without considering the nature of the input data or the physical environment in which the code will ultimately be scanned.The QR code standard, defined by ISO/IEC 18004, provides four distinct data encoding modes - Numeric, Alphanumeric, Byte, and Kanji each designed to handle specific categories of input data with varying levels of bit efficiency. The same standard defines four error correction levels - L, M, Q, and H , based on Reed-Solomon coding, allowing between 7% and 30% of a physically damaged code to be reconstructed during scanning. Together, these parameters offer significant flexibility in balancing code density, data capacity, and scan reliability. However, current generation tools universally default to Byte encoding mode and Level M error correction, leaving this flexibility entirely unexploited.This paper presents a systematic study of QR code encoding theory, Reed-Solomon error correction mechanisms, data encoding modes, and existing generation tools and related literature. Through this study, we formally identify the absence of a content-aware, environment-sensitive parameter selection framework as a significant and unaddressed research gap. We further propose a conceptual rule-based optimization framework that classifies input data type and considers target deployment environment to guide the selection of the most appropriate encoding mode and error correction level. This proposed framework lays the theoretical foundation for future implementation work aimed at improving QR code efficiency and scan reliability across diverse real-world deployment scenarios including payment systems, patient identification, package tracking, and outdoor advertising.

Keywords - Quick Response (QR) Code, Reed–Solomon Error Correction, Encoding Modes, Rule-Based Optimization, Content-Aware QR Code Generation, Scan Reliability, Data Encoding, QR Code Generation.

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A REVIEW ON DIGITAL AUDIO INTEGRITY VERIFICATION

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Abstract

Digital Audio Integrity Verification aims to ensuring the protection, authentication, and reliability of audio after it has been compressed. The existing literature shows that current techniques have limitations regarding chunk-level verification, tamper detection, effects of compression, and computational complexity. This study discusses what are the methods that involve dividing audio into chunks and applying compression to preserve the original audio information. Each audio chunk after compression is denoted with a hash and arranged in the form of a tree to ensure integrity. During the process of verification, the input audio is converted back into chunks, and the new hashes are compared against the existing ones to identify any part that is altered. The review outlines the benefits of chunk- level audio integrity verification, which include efficient verification and effective detection of altered audio parts. The drawbacks of current approaches and research gap in relation to audio compression and audio integrity verification are also discussed. This review studies efficient methods for chunk-level audio integrity verification based on lossless audio compression.

Keywords : Audio integrity verification, Chunked audio compression, Secure Audio Storage, Hash Tree Structure, Tamper detection

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A BLOCKCHAIN BASED FINANCIAL AUDITING PLATFORM

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Abstract

Financial auditing is important to ensure that the financial information of an organization is correct. The old way of auditing required a lot of work and checking records at certain times, which took a long time to find mistakes, fraud, or problems. The use of blockchain technology can keep transactions secure and create an unchangeable history. A blockchain-based financial auditing platform provides auditors with a way to view transaction details through a blockchain network. Smart contracts can be used to set rules for activities and transactions that require a person to check easily.The proposed platform uses SHA-256, Merkle Tree, and RAFT consensus algorithms. SHA-256 secures transaction data, Merkle Tree enables efficient verification, and RAFT maintain consistency among authorized nodes. Blockchain auditing can make the process faster make it easier to follow where money has gone keep the data safe and let auditors get the information they need quickly.Therefore, a blockchain-based financial auditing platform can provide continuous transaction monitoring and support real-time reporting through secure and traceable records.Also using blockchain can make things more open by letting approved auditors see the records.. There are some problems that could make auditing with blockchain harder, such as keeping things private,handling a lot of data, managing rules, and using existing financial systems. Blockchain technology can offer a way to fix the problems, with how financial auditing works now.

Keywords: Blockchain, Financial Auditing, Smart Contracts, Data Integrity, Transparency, Transaction Monitoring.

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A Lightweight Policy-Based File Access Control Framework with Policy Conflict Detection and Resolution

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Abstract

Access control systems allow the protection of computing resources through a series of policies which determine access permissions to certain computing resources. However, the introduction of policy- based and attribute-based access control systems is becoming more efficient in today's environment as the possibility of generating rules according to certain policies and attributes arises. At the same time, there are several disadvantages associated with the existing approaches such as inefficient processing of the rules, difficult rule creation, redundant rules, policy conflicts which influence negatively both security and availability. Moreover, policy conflicts lead to decision errors concerning authorization. This paper aims at presenting the design and implementation of the lightweight policy-based file access control system that was implemented using the Python programming language. File access rules are generated on the basis of attributes and represented as logical expressions in the Python programming language in order to enhance the processing of the rules. In order to improve the reliability of the designed solution, there are implemented mechanisms for detecting and resolving policy conflicts, overlapping of attributes, positive and negative authorization rules, and conflict resolution before policy evaluation. There is conducted policy analysis for detecting authorization, policy interactions, missing attributes, invalid attribute values, and conflict resolution algorithms.

Keywords: Access Control, Policy-Based Access Control, Attribute-Based Access Control, Policy Conflict Detection, Conflict Resolution, File Access Control, Authorization, Rule-Based Systems, Python.

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Multimodal Human- Computer Interaction: A Survey

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Abstract

Virtual reality interaction is increasingly moving beyond traditional handheld controllers toward methods that allow users to communicate with immersive systems in more natural ways. This survey reviews multimodal human–computer interaction techniques that use voice, eye gaze, head gaze, dwell selection, body leaning, blinking, and controller-based interaction for selection and system-control tasks, particularly when the user's hands are occupied or cannot be used. Within immersive VR research, eye gaze, head gaze, and voice interaction are among the most widely investigated approaches, while alternatives such as brain–computer interfaces and facial-expression-based interaction are emerging. The reviewed studies use different evaluation strategies, including systematic reviews, controlled Fitts' law experiments, usability testing, and questionnaires examining presence, workload, cybersickness, and user satisfaction. Several interaction combinations have been explored, such as gaze for target pointing followed by dwell or voice for confirmation, as well as head gaze with leaning, eye gaze with blinking, and voice-only selection. Performance is commonly evaluated using measures such as target-selection time, Fitts' law metrics, selections per minute, and subjective usability. The findings indicate that gaze interaction combined with dwell-based confirmation is particularly promising for hands-free use, with performance in several studies coming close to that of conventional controllers. At the same
time, incorporating voice into multimodal interaction does not always produce better results; although it can improve task completion in some situations, it may introduce delays or additional effort that affects performance, comfort, and satisfaction. Overall, the reviewed evidence suggests that hands-free interaction can offer experiences comparable to controller- based techniques across a range of immersive tasks. This indicates that appropriately designed multimodal interfaces have considerable potential for practical VR applications, especially in situations where conventional controller use is inconvenient or impossible.

Keywords: Multimodal Human–Computer Interaction, Virtual Reality, Eye Gaze, Head Gaze, Voice Interaction, Hands-Free Interaction, Dwell-Based Selection, Fitts' Law.

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Digital Chain Framework for Secure Forensic Evidence Management

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Abstract

Digital evidence is an important part of forensic investigations. Digital evidence must be protected from the moment it is collected until the investigation is finished. Traditional ways of managing evidence can create problems such as access, changes to digital evidence missing records and difficulty in tracking digital evidence activities. These problems can affect the reliability and proper handling of evidence. I present a chain framework for secure forensic evidence management to support the chain of custody and provide a traceable way to manage digital forensic evidence throughout its lifecycle.The framework starts with the collection and registration of evidence. Each evidence item receives an evidence ID and relevant metadata. A SHA-256 hash is generated for each evidence file. Stored as a reference for integrity verification. Digital signatures verify the identity of users performing actions on evidence. The hash values of multiple evidence records are grouped using a Merkle tree, which allows the integrity of the evidence records to be checked using the Merkle root. AES-256 encryption protects evidence during storage. Role-Based Access Control (RBAC) limits access to evidence according to the users role and permissions. Activities such as viewing, transferring and examining evidence are recorded with timestamps in an audit log making unauthorized changes to the activity history detectable.At the end of the investigation the integrity of the stored evidence is checked again. The SHA-256 hash is. Compared with the hash recorded during registration. The Merkle root is also. Compared with the stored value. If the values differ this indicates a modification or integrity issue, with evidence. This framework offers an approach for recording, protecting, verifying and tracking forensic evidence throughout its lifecycle without relying on blockchain technology.

Keywords: Digital Forensics, Chain of Custody SHA-256 Digital Signature, AES-256 Merkle Tree, RBAC, Evidence Integrity, Audit Log.

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Post-Quantum Cryptography: Preparing Secure Systems for the Quantum Era

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Abstract

The public-key cryptographic systems currently in common use will be impacted as quantum computers become more widespread. The algorithms employed in practice today are Rivest–Shamir–Adleman (RSA), Diffie The security of these encryption techniques rests on mathematical functions which are difficult to solve computationally. However, quantum computers will very likely be able to solve these functions within practical time frames as technological progress is made. It follows that the encryption standards currently in use are not secure against future threats and must therefore be replaced. To maintain confidentiality, integrity, authentication, and secure digital communication in the face of both quantum and classical computers, the National Institute of Standards and Technology has recently developed standardized Post Cryptography algorithms. This paper looks at the standardized Post algorithms (Module- Lattice-Based Key Encapsulation Mechanism (ML Lattice-Based Digital Signature Algorithm (ML Signature Algorithm (SLH-DSA) and Hamming Quasi aspects of the security principles, implementation features, and applications of each algorithm. The paper also compares the standardized algorithms on the basis of factors such as security, performance, key length, ciphertext length, etc., according to current research standard documents. Furthermore, it reviews the most important difficulties involved in
moving from conventional cryptography to standard quantum proposes a migration plan. Thus, the purpose of this study is to give guidance of standardized algorithms should be used in order to ensure confidentiality, integrity, authentication, and secure digital communication against future attacks.

Keywords: Post-Quantum Cryptography, Quantum Computing, Public ML-KEM, ML-DSA, SLH-DSA, HQC, NIST, Cryptographic Migration.

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Vehicle License Plate Detection and Recognition Using Traditional Image Processing Techniques

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Abstract

Vehicle license plate recognition is an important application of image processing used in traffic monitoring, parking management and vehicle identification. However, variations in lighting, noise, blur and plate orientation can affect recognition accuracy. This research proposes a traditional image processing-based system for vehicle license plate detection and recognition without using AI or ML. The proposed approach uses preprocessing, edge detection, morphological operations, contour detection, plate extraction, character segmentation and OCR or template matching. The system will be evaluated using accuracy, precision, recall and processing time. The main aim is to develop a simple, lightweight and efficient license plate recognition system.

Keywords: License Plate Recognition, Image Processing, Edge Detection, Contour Detection, Character Segmentation, OCR, Template Matching, Traffic Monitoring.

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FINGERPRINT BIOMETRICS FOR SECURE AUTHENTICATION :SURVEY

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Abstract

Fingerprint-based authentication is one of the most widely used methods for identifying a person because every individual has a unique fingerprint pattern. The system works by capturing the fingerprint through a sensor, extracting important ridge and minutiae details, and comparing them with stored templates to verify identity. It is commonly used in access control, attendance systems, and device security because it is easy to use and gives quick results. At the same time, this technology still faces several practical problems such as poor fingerprint quality, sensor errors, spoofing attempts, and mismatches caused by finger position or skin condition. This paper reviews the architecture of fingerprint-based authentication systems, their working process, and the main challenges that affect performance and security. It also explains the use of fingerprint matching algorithms—such as minutiae-based methods (Bazen–Gerez, Ratha–Bolle, Jain's minutiae matcher, Minutiae Cylinder-Code), correlation-based methods using normalized cross-correlation and phase- only correlation, and pattern-based methods using orientation field comparison and Gabor- filter–based texture features—which help in comparing fingerprint features accurately. The
purpose of this study is to understand how fingerprint authentication works and to identify the areas where improvement is needed for better reliability and security in real-world applications.

Keywords: Fingerprint-based authentication, biometric identification, minutiae extraction, fingerprint matching algorithms, spoofing attacks, access control, reliability and security.

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Critical Node Identification in Dynamic Networks: A Systematic Literature Review

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Abstract

Critical nodes are those points in a network where failure or their removal results in substantial damage to the overall network's connectivity and performance. It is important to identify them accurately especially in transportation systems, communication networks, and critical infrastructure, since a single weak critical node has the potential to bring the whole system to a halt. Most of the current research in the identification of critical nodes assumes that the network is static. However, static networks are not common in real-time systems. The network in real-time is dynamic and nodes and links are constantly being added, deleted, or modified over time based on the necessity. This paper reviews the existing studies on critical node identification and looks at them from three point of view: the development of underlying methods, whether they consider static networks or dynamic network environments, and whether centralized or decentralized systems are involved. An analysis of the literatures shows two separate lines of research. One is concerned with developing robust methods that combine several importance measures into one comprehensive score for static networks. The other focusses on building methods that handle dynamic networks efficiently but rely on one importance measure at a time. The review of literatures identifies a major gap in the current research that does not combine the advantages of different measures in dynamic environments. This survey outlines the key requirements for creating an approach that can identify critical nodes in dynamic networks by using multiple criteria, without needing a central computation authority to overcome the gap identified.

Keywords: Critical Node Identification, Dynamic Networks, Temporal Network, Multi- Criteria Analysis, Decentralized Computing, Network Robustness

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COMPARATIVE SURVEY ON DIGITAL PAYMENT SYSTEMS: SECURITY VULNERABILITIES, CYBER THREATS, AND MITIGATION STRATEGIES

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Abstract

Digital payment systems are now widely used for everyday financial transactions because They make payments faster,easier,and more convenient.At the same time, the increasing use of digital payments has created several security concerns.Users and payments system may face threats such as phishing, malware, identify theft, payment fraud and unauthorized access These threats can affect confidentiality , integrity and authenticity of financial transactions and may lead to financial loss and lack of trust among users .This study reviews digital payment systems and focus on the major security vulnerabilities and common Cyber Threats and methods used to reduce these risks .A qualitative research approach is followed by studying relevant journal articles , conference papers,industry reports and cybersecurity .The study examines security techniques such as Advanced Encryption Standard (AES-256),Secure Hash Algorithm (SHA-256), and Multi Factor Authentication(MFA).The findings highlights importance of cryptographic and authentication techniques in enhancing transaction security and protecting the digital payment system from evolving cyber threats.

Keywords: Digital Payment Systems, cybersecurity, cryptography, Cyber Threats, security vulnerabilities, Mitigation Strategies.

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Design and Development of a Secure Software Supply Chain Integrity Verification Framework

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Abstract

Software supply chain security is becoming a part of cybersecurity. More and more developers rely on third-party software, open-source libraries and external packages to speed up development. These components save time. Reduce effort. They can also bring risks. If attackers tamper with or compromise these components serious problems can arise. Software supply chain attacks can lead to malware spreading, data breaches, financial damage and system failures. That's why checking the authenticity and integrity of software packages is so important for software distribution.This survey looks at research in software supply chain security. The focus is on signature-based methods for verifying software. It explores how digital signatures help confirm software sources detect changes made without permission and build trust in software distribution. A major part of the study is on the use of Ed25519, a signature algorithm. Ed25519 offers security and fast performance. It is well- suited, for checking the integrity and authenticity of software packages. The survey also compares the strengths and weaknesses of approaches. It identifies challenges that still need to be addressed in software supply chain security. Finally it outlines research directions. These could lead to software verification techniques and more secure reliable ways to distribute software.

Keywords
Software Supply Chain Security, Software Package Integrity, Software Authenticity, Digital Signature, Ed25519 Digital Signature Algorithm, Package Verification, Integrity Verification, Unauthorized Modification Detection, Trusted Publisher Verification, Audit Logging.

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Design and Development of an Enterprise Energy Monitoring and Management System for Sustainable IT Infrastructure

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Abstract

The rapid growth of Information Technology organizations has led to increased energy consumption due to the continuous operation of computers, servers, networking devices, and data centers. As organizations seek to improve operational efficiency while reducing their environmental impact, Green Computing has emerged as an effective approach to promoting sustainable and energy-efficient IT practices. However, many organizations continue to face challenges in monitoring and managing energy consumption across their IT infrastructure. This research aims to design and develop an Energy Monitoring and Management System for Large-Scale IT Organizations to support efficient energy utilization and sustainable resource management. The study focuses on monitoring energy consumption, identifying areas of excessive power usage, and supporting organizations in implementing effective energy management practices. The expected outcome is an efficient energy monitoring solution that helps reduce energy wastage, improve operational efficiency, and encourage the adoption of Green Computing principles for sustainable IT infrastructure.

Keywords: Green Computing, Energy Monitoring, Energy Management, Sustainable IT, Energy Efficiency.

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DNS Tunnelling Detection Using Anomaly-Based Methods: A Review

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Abstract

Domain Name System (DNS) tunnelling has become a significant threat vector in modern network security, exploited by attackers to infiltrate data, establish covert command-and-control channels, and bypass traditional firewall inspection, all while disguising malicious traffic as ordinary DNS activity. A critical challenge persists in detecting this activity because DNS queries are essential for hostname resolution, they are almost universally permitted to pass through network perimeters with minimal inspection, creating an exploitable blind spot that signature-based detection tools, which rely solely on known malicious domains and predefined patterns, are structurally unable to close. Anomaly-based detection addresses this gap by analysing the statistical and structural properties of domain names themselves, rather than depending on prior knowledge of specific threats. This paper presents a systematic review of anomaly-based DNS tunnelling detection techniques, examining domain-level features including entropy, chi-square distribution, character frequency, and domain name length, each of which reflects measurable deviations between tunnelled traffic and legitimate DNS queries. Entropy analysis captures the randomness introduced when data is encoded into subdomains; chi-square distribution testing compares observed character patterns against expected linguistic norms; character frequency analysis identifies irregular proportions of alphanumeric and symbolic characters; and domain length analysis reflects the tendency of tunnelling payloads to require longer, denser subdomains than typical hostnames. Through this study, we conduct a comparative analysis of existing anomaly-based techniques across the literature, evaluating each according to detection accuracy, false-positive rate, computational overhead, and the ability to detect zero-day tunnelling tools that signature-based methods cannot recognize. We further identify a recurring limitation across the reviewed literature: legitimate domains exhibiting naturally irregular structure, such as automatically generated or content-delivery-network subdomains, frequently trigger false positives, indicating an unaddressed gap in threshold calibration and feature selection methodology. This review consolidates current research trends in anomaly-based DNS tunnelling detection and lays a foundation for future work aimed at improving detection reliability and reducing false-positive rates across diverse network deployment environments.

Keywords: DNS Tunnelling, Anomaly-Based Detection, Network Security, Covert Channels, Entropy Analysis, Chi-Square Analysis, Character Frequency, Domain Name Length, Zero-Day Detection, False Positive Reduction, Threshold Calibration, Feature Selection

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Survey On Secure DataTransmission Using Image Steganography Based on RLSB

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Abstract

The rapid growth of digital communication has increased the need for secure transmission of confidential information due to cyber threats and unauthorized access. Image steganography is a data hiding technique that conceals secret information within digital images while preserving their visual quality. The Randomized Least Significant Bit (RLSB) algorithm is employed to embed secret data into randomly selected pixel locations of a cover image based on a secret key. Unlike the conventional Least Significant Bit (LSB) technique, random pixel selection enhances security by reducing the possibility of detecting hidden information through steganalysis. The embedded information can be extracted accurately using the same secret key during the extraction process. The technique ensures data confidentiality, integrity, and high image quality, making it suitable for secure communication in military networks, healthcare information systems, banking services, government organizations, and secure online data sharing. Keywords: Secure Data Transmission, Image Steganography, Randomized Least Significant Bit (RLSB), Da t a Hiding, St ego Image , Information Security.

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A Survey of Student Marksheet File Integrity Using BLAKE3 Hashing and Metadata Analysis

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Abstract

Academic certificates are now always being shared as digital documents. These digital files are uploaded online for easy access across the globe . Because of this some digital certificates may be accessed by unauthorized people and may be misused or modified without the administrator's permission .This can lead to malpractice and lack of reliability . This literature survey focuses on validating the integrity of academic certificates using digital forensics techniques. This survey is mainly based on the BLAKE3 hashing algorithm and metadata analysis. BLAKE3 is a cryptographic hashing algorithm that mainly produces a unique hash value for every file. In this literature study the original certificate hash is compared with the hash of the uploaded certificates to check whether the file has been changed. Metadata analysis such as file size, owner, creation time and modification time. These details can help us identify whether an unauthorized person has accessed or modified the change . This literature survey reviews existing research papers that discuss file integrity, hashing, metadata analysis and digital forensics. These papers are compared to understand the methods they use and the limitations that they face. Based on this literature review study identified a research gap. This shows the need for a framework that applies the BLAKE3 hashing algorithm and metadata analysis to verify the integrity of digital academic certificates.

Keywords: Digital Forensics, File Integrity Verification, BLAKE3, Metadata Analysis, Cryptographic Hash Function, Tamper Detection.

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A SURVEY ON SECURE GENOME MUTATION ANALYSIS USING BLOCKCHAIN WITH PRIVACY PRESERVATION AND ACCESS CONTROL

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Abstract

Genome mutation data are highly sensitive and require secure storage with strict prevention of unauthorized access. Traditional systems are centralized systems with a single point of failure, making the data vulnerable to tampering. To overcome this, Blockchain- based system uses a decentralized system where the data are immutable, securely store and tamper resistance storage for genome mutation data. Role-based access control (RBAC) is integrated to grant permission based on the role to prevent unauthorized individuals from accessing and altering the genome mutation data. An immutable audit log records all access and modifications of the genome mutation data, ensuring traceability and accountability. This approach improves data security and integrity compared to traditional systems, offering a reliable foundation for genome data protection.

Keywords — Blockchain, Genome mutation analysis, Privacy preservation, Access control, Genomic security.

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A Differentiable Computational Framework for Reconstructing Intentional Acoustic Excitation from Archaeoacoustic Chamber Geometry

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Abstract

Ancient stone chambers, temple interiors, and rock-cut caves and rooms sometimes exhibit geometries that could have been intentional in their design for influencing sound behaviour and the flow of sound. However, details about the sound used in these rooms and the principles behind their acoustic design are typically unknown. The aim of this research is to investigate if, from the remaining geometries of heritage buildings, enough physical evidence exists to fully characterize their acoustics and infer the likely excitation used with them. The research focuses on developing an integrated computational system comprising 3-D reconstruction of heritage, physics-based acoustics modeling, machine learning, and inverse modelling. The process will start by generating 3-D digital models of selected heritage chambers located in and around Chennai using photogrammetric and LiDAR data. The models will be then analyzed with FEM/FDTD simulations in order to create valid acoustic signatures comprising the main frequencies, resonant behaviour, signal amplification, and modal analysis for every given room. These signatures, used for modeling their acoustical properties by means of an artificial neural network, will allow predicting the response of these rooms. An inverse modeling technique then identifies the most plausible excitative (source type) signal based on simulated resonance profiles using gradient-based optimization and evolutionary search will validate possible recovered solutions. The main output of this project is a documented and reusable computational tool for which three dimensional scans of any architectural place can be uploaded producing it, its acoustic behavior including characteristic frequencies, amplifications, distribution, and excitation factors that would fit the acoustic observed properties. Instead of trying to retrieve the exact sound that might have been historically produced into these spaces, this methodology offers a rigorously validated technique to infer the acoustical intentions associated with heritage architectural design.

Keywords
Archaeoacoustics; Heritage Architecture; Acoustic Intent; Differentiable Computing; Inverse Acoustic Modelling; Physics-Based Acoustic Simulation; FEM/FDTD; LiDAR

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A Survey on Secure File Sharing Using Hybrid AES-256 and Elliptic Curve Cryptography (ECC)

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Abstract

Cloud computing is widely used for storing and sharing files because it provides flexible storage and easy access to information. However, storing important files in the cloud can create security problems such as unauthorized access, data leakage, modification of files, and misuse of encryption keys. Several existing studies have used hybrid encryption methods, including AES with RSA and AES with ECC, to protect data stored in cloud environments. These methods mainly focus on encryption, but secure file sharing also requires proper control over users, access duration, encryption keys, and file activities. This survey reviews the existing security techniques used for cloud data protection and considers these requirements in the design of a secure file-sharing framework. The proposed framework uses AES-256 to encrypt the file before it is stored or shared. ECC is used for protecting and exchanging the AES encryption key, while SHA-256 is used to verify the integrity of the file. In addition to these cryptographic techniques, Role-Based Access Control (RBAC) is used to provide permissions based on the user's role. Time-Limited Access allows a file to be available only for a specified period. Key Rotation is included to replace encryption keys periodically and reduce the risk of key exposure. Audit Logging keeps a record of activities such as file uploads, downloads, sharing, and access attempts. By combining encryption, integrity verification, access control, key management, and activity monitoring, the proposed framework provides multiple levels of protection for cloud-based file sharing. The survey suggests that such an integrated approach can address security requirements that cannot be handled by file encryption alone. The framework can be further implemented and evaluated to study its security,
performance, and suitability for practical cloud file-sharing applications.

Keywords: Cloud Computing, Cloud Security, Secure File Sharing, Hybrid Cryptography, AES-256, Elliptic Curve Cryptography (ECC), SHA-256, Role-Based Access Control (RBAC), Time-Limited Access, Key Rotation, Audit Logging, Data Integrity.

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A Survey: Energy Hole Mitigation in Wireless Sensor Networks

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Abstract

In Multi-hop wireless sensor networks have a clearly identified structural flaw in that the nodes which are closer to the base station have a greater communication load than those that are further away, as these near-the-base-station nodes not only have to transmit their own data but also have to relay data on behalf of the whole network. This imbalance, usually referred to as the energy hole or hotspot problem, leads to the early exhaustion of the nodes near the sink and as a result isolates the rest of the network even though a large amount of energy is still unused in other areas. Over the last twenty years, a lot of research has been carried out to deal with this problem by means of clustering, chain-based routing, and hybrid topological approaches. Two main design approaches have come about. The first one divides the network in an uneven way from the beginning, giving smaller cluster or sector sizes close to the sink according to factors such as distance, residual energy, and local nodedensity, in order that the nodes near the sink may still have the capacity to cope with the extra relaying duty. The second approach keeps an eye on the energy levels as the network is operating and only changes the topology when an imbalance is detected, generally by merging or reallocating areas that have depleted their energy. This paper looks at example protocols from each of these categories, following their development and comparing their basic assumptions, the methods they use to select a head node or cluster head, and the performance they have reported in terms of network lifetime and energy distribution. Special attention is paid to how each method deals with the compromise between predicting the imbalance and correcting it afterwards, as well as to the extent to which the control-message overhead a cost that is often neglected in evaluations that focus on network lifetime is taken into account. The review shows that strategies involving static and dynamic partitioning have mostly developed independently of each other, with little research having been done to see if combining them could provide advantages over what either one achieves alone. The paper also discusses open questions concerning base station placement, network scalability, and the measurement of overhead within these protocol families, indicating possible areas for future research in this field.

Keywords: Wireless Sensor Networks, Energy Hole Problem, Unequal Clustering, Network Lifetime, Control Overhead

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Survey Paper On Secure Electronic Health Record Management Using Cryptography

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Abstract

The use of Electronic Health Records (EHRs) has made it easier for healthcare organizations to maintain and access patient information. At the same time, keeping medical data safe from unauthorized users, data tampering and privacy breaches remains a major challenge. This project presents an EHR security framework that combines AES encryption, ECC, and Role- Based Access Control (RBAC). Patient information is encrypted with AES before being stored, while ECC is used to securely handle the encryption keys. The system also verifies users and provides access according to their assigned roles. As a result, sensitive records can be accessed only by permitted users, helping to protect patient information and improve the security of the EHR system. The aims to provide a reliable and efficient solution for protecting the EHR system.

Keywords
Electronic Health Records (EHR), AES Encryption, Elliptic Curve Cryptography (ECC), Role Based Access Control (RBAC), Healthcare Data Security.

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INFORMATION SECURITY IN PROCTORING SYSTEMS

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Abstract

Online Examinations has now transformed the way how institutions assess the students.However, there are also security challenges related to protecting examination data, student information, and examination systems. Ensuring the security of these systems is essential to maintain the confidentiality, integrity, fairness, reliability, and credibility of the examination process.Online exam platforms can be vulnerable to security threats like hacking, cheating, data leaks and cyberattacks. This paper presents an approach to designing and validating an Information Security Framework to address these challenges. Our goal is to provide a scalable security model that protects online exams from vulnerability and improves stakeholder confidence, in digital assessment environments. The online examination systems security is critical. This paper ensure the security of online examination systems.

Keywords: Information Security Framework, Cybersecurity, Authentication, Data Privacy, Digital Assessment, Secure Education Systems.

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SURVEY ON BLOCKCHAIN - ENABLED FINANCIAL FRAUD INVESTIGATION

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Abstract

Financial fraud has grown alongside digital banking, and investigative tools haven't kept pace. Two problems keep surfacing: records that can be altered after the fact, and evidence whose provenance nobody can fully guarantee. Centralized databases make this worse — when everything lives in one system and that system is breached, there's often no reliable way to confirm what the records said beforehand. This paper proposes a blockchain-based framework built for financial fraud investigations. Rather than treating data integrity as an afterthought, it writes every piece of evidence to the chain as it's generated, timestamped and traceable from the start. Because the ledger is decentralized and immutable, once something is recorded it can't be quietly altered — critical when evidence needs to hold up in court or survive a regulatory audit. The framework also maintains a continuous chain of custody, automatically logging every access and modification so the trail stays verifiable end to end. Access is role-based: banks, regulators, and law enforcement each get permissions suited to their function, which speeds up multi-agency investigations and reduces the coordination burden that typically slows them down.

Keywords: Blockchain, Financial Fraud Investigation, Digital Evidence, Evidence Integrity, Chain of Custody, Immutable Ledger, Digital Forensics, Financial Transactions, Role-Based Access Control.

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A Web-Based Clinical Support System for Screening Iron Deficiency Anemia and Thalassemia Trait Using Machine Learning

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Abstract

Iron deficiency anemia (IDA) and thalassemia trait are common blood disorders that can show similar hematological patterns, making their early differentiation difficult using routine blood tests alone. This study proposes an explainable machine learning framework to support the screening of IDA and thalassemia trait using routinely available hematological and laboratory parameters. The proposed framework compares five modern tabular learning approaches, namely XGBoost, LightGBM, CatBoost, TabNet, and TabPFN, along with Logistic Regression. The models are trained to identify patterns in parameters such as hemoglobin, RBC count, MCV, MCH, RDW, and relevant iron-study measurements. Their performance is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and other appropriate evaluation measures. To make the predictions easier to understand, SHAP-based explainable artificial intelligence is incorporated to identify the features that most strongly influence both overall and individual patient predictions. The proposed framework aims not only to determine whether a patient's profile is more consistent with IDA or thalassemia trait, but also to provide an understandable explanation for the prediction. This approach could support early screening and clinical decision-making while reducing reliance on complex diagnostic procedures in initial assessment.

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Pseudo-Morphology-Guided Learning for Cotton Leaf Curl Disease Classification

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Abstract

Cotton Leaf Curl Disease (CLCuD) causes noticeable changes in cotton leaves, including altered venation, leaf curling, deformation, and changes in leaf boundaries. Although deep learning models can achieve high classification accuracy, their performance alone does not confirm whether predictions are based on disease-relevant visual characteristics. This study presents MorphoReli- CLCuD, a pseudo-morphology-guided deep learning framework developed for reliable classification of healthy and Cotton Leaf Curl Disease images. The framework combines adaptive disease-preserving image enhancement, vein-aware attention, and reliability-based fusion of local texture, pseudo-venation, leaf-boundary, and global geometric information. The study used a canonical dataset containing 472 healthy-labelled and 2,990 CLCuD-labelled images. Perceptual and acquisition-level grouping was performed before dividing the data into fixed 70%, 15%, and 15% training, validation, and test subsets, respectively. A common test set of 520 images and identical training procedures were maintained across all experimental configurations. Eight controlled model configurations were evaluated using three predefined random seeds for repeated assessment. The correctly supervised gated-fusion configuration (C2) obtained the highest mean macro-F1 of 0.916 with a standard deviation of 0.006. It improved macro-F1 by 0.0217 over the parameter-matched diagnosis-only control and by 0.0209 over the mismatched-target control. The adaptive enhancement configuration achieved a mean AUROC of 0.974 and a mean Brier score of 0.032. Grad-CAM++ and Integrated Gradients were further used with foreground, background, and structural-region interventions to examine model behaviour. The findings demonstrate the potential of morphology-guided supervision for improving reliable CLCuD classification while supporting interpretable AI-based disease assessment.

Keywords: Cotton Leaf Curl Disease, Deep Learning, Convolutional Neural Network, Pseudo-Morphology Supervision, Computer Vision, Explainable Artificial Intelligence

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A CONTINUOUS MULTIMODAL FRAMEWORK FOR DETECTING, EXPLAINING AND MITIGATING DARK PATTERNS IN AI-BASED INTERFACES

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Abstract

As AI becomes more closely intertwined with interactive systems, novel types of dark patterns appear that are not restricted to visual and textual interface features but can also be instantiated through interaction and behaviour generated by AI. Most previous work focuses either on specific features of interface interfaces or on model-generated output, leading to a somewhat fragmented picture of dark pattern detection and judgement in AI-driven user interfaces. This research proposes a continuous multimodal framework for detecting, explaining, mitigating and re-evaluating dark patterns in AI-supported UI/UX. The framework leverages four types of evidence: visual and textual cues, interaction trajectories and interface state changes, AI-generated behaviour, and contextual information. Computer Vision and Natural Language Processing (NLP) techniques are used to analyse visual and textual cues, while interaction sequences and interface state changes are examined to capture user experience. AI-generated behaviour is systematically evaluated using controlled prompts and contextual scenarios. These evidence sources are integrated through an evidence-fusion mechanism to establish a dark-pattern fingerprint and generate an explainable evidence trail for each detected pattern. Unlike detection-driven methods, the proposed framework incorporates a continuous cycle of assessment involving context analysis, human validation, mitigation, and re-assessment after mitigation to determine whether the detected dark pattern persists. For uncertain cases where contextual interpretation or user impact cannot be formalized, a human-in-the-loop approach is adopted. Controlled interaction test scenarios will be used to evaluate the framework and its ability to detect different types of dark patterns. The framework will be evaluated based on detection effectiveness, multimodal detection, dynamic behaviour detection, contextual detection, explainability, robustness, dark-pattern mitigation capability, re-evaluation, and human oversight, where applicable. The proposed research aims to take dark-pattern detection one step further from an isolated identification task to an ongoing, evidence-based, explainable and human-centred process for evaluating AI-based user interfaces.

Keywords: Dark Patterns, Artificial Intelligence, AI-Based Interfaces, Multimodal Detection, UI/UX, Computer Vision, Natural Language Processing, Evidence Fusion, Explainability, Human-in-the-Loop, Dark-Pattern Mitigation.

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A Survey On Behavioural Biometrics in Authentication Systems

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Abstract

Behavioural biometrics has emerged as a promising approach for continuous authentication by verifying users based on their unique behavioural characteristics such as mouse movements, touch gestures, and login habits. Unlike static authentication methods that rely on one-time verification, behavioural biometrics enables continuous monitoring of user activity to ensure user identity, thereby enhancing system security. However, the widespread adoption of behavioural biometric systems is challenged by concerns regarding the poor performance of authentication caused by behavioural drift over time. This critical analysis examines recent research on Behavioural Biometric Continuous Authentication (BBCA) focusing on approaches for adapting to behavioural drift. The paper includes recent literature, discusses commonly adopted methods such as Adaptive Machine Learning Models like Long Short Term Memory (LSTM) and Auto-Encoder models and evaluates their strengths and limitations. The analysis indicates that existing studies have made considerable progress in improving authentication accuracy but few have addressed the challenge of using ever-changing human behaviour as an authentication metric in long-term systems. The paper concludes by highlighting future directions for developing behavioural biometric systems that can adapt to behavioural drift successfully.

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AgriCare: An AI-Enabled Precision Agriculture and Smart Farming Platform for Real-Time Crop Monitoring and Digital Farm Advisory.

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Abstract

Agriculture faces challenges such as crop diseases, inefficient irrigation, nutrient imbalance, limited access to timely advisory services, and difficulties in field-level monitoring. This work presents AgriCare, an AI-enabled precision agriculture and smart farming platform designed to integrate real- time sensing, automation, autonomous field inspection, crop disease detection, and digital advisory services into a unified system. The proposed platform consists of a solar-powered real-time monitoring system using master and slave kits. The slave kit integrates soil moisture, temperature and humidity, and 8-in-1 soil sensing for monitoring key field conditions, while ESP32-based communication with LoRa transfers sensor data to the master kit. Based on soil moisture conditions, the system can automatically control irrigation and periodically evaluate the field condition. Soil nutrient information, including nitrogen, phosphorus, and potassium, is used to support nutrient management and automated fertilizer application. In addition, an autonomous solar-powered field robot travels through predefined paths and captures crop images for detecting abnormal crop conditions, pests, and diseases. Image data are processed using a Raspberry Pi and AI-based analysis, and identified abnormalities are communicated to the AgriCare web dashboard for farmer alerts. The platform also provides real-time sensor visualization through MQTT and JSON-based data communication, crop and soil-based digital recommendations, market price information, yield-oriented decision support, multilingual farmer guidance, and farm investment and stock monitoring. The current implementation supports Tamil, English, and Marathi, with further language expansion planned. AgriCare aims to provide a scalable and integrated solution that combines IoT, AI, robotics, and digital agriculture services to support timely decision-making, resource-efficient farming, and improved farm management.

Keywords: Precision Agriculture, Smart Farming, Artificial Intelligence, Internet of Things, ESP32, LoRa, Crop Disease Detection, Soil Monitoring, Autonomous Solar Robot, Digital Farmer Advisory

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SAFE-SHIELD: An Intelligent IoT-Enabled Safety System for School Students

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Abstract

In While going to school, coming back from school, attending various extracurricular activities, or simply wandering around in unknown places, kids may encounter a variety of unforeseen circumstances. In these situations, being able to immediately notify the right person about the emergency, and then sharing pertinent information can be a life-saver. This paper proposes SAFE-SHIELD - An IoT based, personal security solution, tailored for school students of classes 5th to 12th. The detailed system design involves a connected emergency-response system comprising of a device (implemented using a Raspberry Pi 4, SOS button, GPS module, camera, 4G LTE communication module, LCD display and buzzer) and connected user. Pressing the SOS button will trigger a message sent via 4G communication from the device to a predefined trusted contact containing the location and other related picture information of the student. The system provides local audio-visual feedback through LCD and buzzer that the contact has received the emergency. The upgraded version overcomes the deficiencies of the previous version of the design like 2G communication reliance, shared power source, breadboard connections, no support for alert acknowledgment and retry after failure. The upgraded version comprises of new features like 4G, segregated power rails, hardware reliability, GPS fallback support, long press and confirmation-based alert delivery. SAFE-SHIELD is developing a straightforward emergency button into a student- centered connected safety system that gives a child as little as one press of a button to request help and trusted adults instant visual and location-based info. The concept shows how IoT can be used to deploy cost effective and readily available solutions that are easy to use in schools.

Keywords: SAFE-SHIELD, Internet of Things (IoT), Personal Safety System, Emergency Alert System, 4G LTE, GPS Tracking, Emergency Response, Trusted Contact, Raspberry Pi 4

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AI-ENABLED IOT SMART BLDC MOTOR PERFORMANCE AND FAULT DIAGNOSTIC SYSTEM

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Abstract

Brushless DC (BLDC) motors are widely used in drones, electric vehicles, robotics, and other electric propulsion applications, where accurate measurement of thrust, rotational speed, electrical power, temperature, and operating conditions is essential. This work presents an AI-enabled IoT-based smart BLDC motor performance and fault diagnostic system designed to integrate real-time propulsion testing, multi-parameter monitoring, performance analysis, and fault identification into a unified platform. The proposed system uses an ESP32 as the main controller and integrates an 80 kg load cell with an HX711 amplifier for thrust measurement, an RPM sensor for rotational-speed measurement, an ACS712 current sensor for current monitoring, a DC voltage sensor for voltage measurement, and temperature and vibration sensors for motor condition monitoring. The measured parameters are processed to determine thrust, RPM, voltage, current, electrical power, thrust-to-power performance, temperature, and vibration characteristics. The system performs controlled motor testing at different operating conditions and displays real-time measurements through a local display and an IoT-based web dashboard using wireless communication. The collected data are stored for performance comparison and condition analysis. In addition, a rule-based and AI-assisted diagnostic approach is proposed to identify abnormal operating conditions such as motor overload, propeller imbalance or damage, excessive vibration, overheating, and low-battery operation. The system provides a motor health status based on the combined sensor parameters and performance characteristics. The proposed platform aims to provide a low-cost, scalable, and intelligent solution for BLDC motor testing, propulsion performance evaluation, real-time condition monitoring, and fault diagnosis, with potential applications in drone propulsion systems, robotics, electric mobility, and laboratory motor testing.

Keywords: BLDC Motor, Thrust Measurement, Smart Motor Testing, ESP32, Internet of Things, Fault Diagnosis, Load Cell, RPM Measurement, Vibration Monitoring, Motor Health Monitoring, Artificial Intelligence, Propulsion Performance.

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An AI-Powered Student Assistance Chatbot for the Department of Technical Education, Government of Rajasthan

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Abstract

The increasing digitization of educational services has created a need for efficient, reliable, and user-friendly systems that can provide students with timely information about technical education. Students seeking diploma and polytechnic education in Rajasthan often need to access information from multiple sources regarding government polytechnic colleges, diploma courses, eligibility criteria, admission procedures, fees, examinations, scholarships, and student services. To address this challenge, this paper presents "An AI-Powered Student Assistance Chatbot for the Department of Technical Education, Government of Rajasthan." The proposed system provides a conversational platform through which students can obtain relevant technical-education information using natural-language queries. The system applies Natural Language Processing (NLP) techniques, including text preprocessing, tokenization, language detection, intent classification, entity extraction, and context understanding, to interpret student queries. A structured and verified Technical Education Knowledge Base is used to retrieve relevant information covering colleges, courses, admissions, eligibility, fees, examinations, scholarships, and student services. The retrieved information is processed through a response-generation module to provide concise, relevant, and student-friendly answers. The system supports English and Hindi, enabling bilingual interaction and improving accessibility for a wider range of students. A web-based chatbot interface is integrated with a Python/FastAPI backend for query processing, information retrieval, and communication with the knowledge base. The proposed system aims to reduce the effort required to search through multiple information sources and provide students with a convenient and centralized information-access mechanism. The framework can be further enhanced through additional regional languages, voice- based interaction, advanced retrieval techniques, and large language model integration.

Keywords: Artificial Intelligence, Chatbot, Natural Language Processing, Student Assistance, Technical Education, Information Retrieval, Knowledge Base, Bilingual Chatbot, Rajasthan.

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SMART POWER CONSUMPTION MONITORING AND AUTOMATION USING IOT

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Abstract

The increasing deployment of electrical and electronic appliances in residential, educational, laboratory, and small-scale commercial environments has resulted in higher energy consumption and the need for effective real-time monitoring and automated load management. This work presents an IoT-based Smart Power Consumption Monitoring and Automation System designed to integrate electrical parameter sensing, wireless data communication, remote monitoring, and automated load control into a unified platform. The proposed system utilizes a PZEM-004T energy monitoring module to measure key electrical parameters, including voltage, current, instantaneous power, and accumulated energy consumption. An ESP32 microcontroller acquires the measured parameters through serial communication, processes the data, and transmits the information over Wi-Fi to a web- based monitoring platform for real-time visualization and analysis.The system incorporates a relay- based load switching mechanism to enable automated control of connected electrical appliances. A threshold-based control algorithm continuously evaluates the measured power consumption against a predefined power limit. When the power demand exceeds the specified threshold, the ESP32 activates the relay control mechanism to disconnect the associated load, thereby minimizing excessive or unnecessary power consumption. The threshold can be configured according to the power rating and operational requirements of the connected appliance. The system also enables users to remotely monitor the operating status and electrical consumption of connected loads through the IoT interface.The proposed architecture combines electrical sensing, embedded processing, Wi-Fi communication, cloud/web-based monitoring, and automated load management within a low-cost and scalable framework. Real-time acquisition of electrical parameters provides users with improved visibility of consumption patterns and supports informed energy-management decisions. The automation mechanism further reduces dependence on manual switching and provides a basic protective strategy against prolonged operation under high-power conditions.The developed system is intended for applications such as smart homes, classrooms, laboratories, offices, and small commercial buildings, where continuous monitoring and basic automated energy management are required. The modular architecture allows additional sensing and control units to be incorporated for multi-load monitoring and advanced energy analytics. The proposed system demonstrates the feasibility of integrating IoT-enabled electrical measurement with automated load control to achieve efficient and accessible power management.

Keywords: Internet of Things (IoT), ESP32, PZEM-004T, Power Monitoring, Energy Management, Smart Energy, Load Automation, Relay Control, Wireless Monitoring.

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Design and Development of a Portable Gas Sensor-Based Food Freshness Monitoring System for Quality and Spoilage Assessment

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Abstract

Food spoilage, driven by microbial proliferation and enzymatic degradation of proteins and carbohydrates, causes substantial food loss, economic wastage, and foodborne health risks, whereas conventional assessment relies on subjective organoleptic evaluation or laboratory-based microbiological and physicochemical analysis that is destructive, time-consuming, and unsuitable for in-situ deployment. This work presents the design and development of a portable, IoT-enabled food freshness monitoring system based on a chemiresistive gas sensor array for real-time, non-destructive spoilage assessment. The system employs tin dioxide (SnO2) based metal-oxide semiconductor (MOS) sensors, whose resistance varies with the adsorption and desorption of oxygen species at the sensing surface in the presence of volatile organic compounds (VOCs) and spoilage biomarkers such as ammonia (NH3), hydrogen sulfide (H2S), and methane (CH4). Heater-regulated sensors are stabilized through a preheating period, interfaced via load-resistor voltage dividers, conditioned by low-pass filtering, and digitized by an analog-to-digital converter (ADC), with the sensor resistance ratio (Rs/R0) serving as the freshness indicator. Moving-average filtering and oversampling suppress measurement noise, while real-time temperature and humidity compensation mitigates cross-sensitivity and baseline drift. The sensor response is calibrated against the logarithmic Rs/R0 sensitivity characteristic to estimate relative gas concentration in parts per million (ppm). A microcontroller then performs feature extraction and threshold-based multi-class classification to categorize the food as fresh, moderately fresh, or spoiled. The classified status is displayed locally with LED and buzzer alerts, and is transmitted over a Wi-Fi link using lightweight MQTT/JSON messaging for remote monitoring and time-series data logging. The battery-powered prototype was evaluated on common perishable items by analysing response magnitude and temporal gas profiles, and the results indicate a strong correlation between rising VOC concentration and spoilage progression, enabling early detection prior to visible deterioration. The proposed system offers a low-cost, scalable, and energy-efficient solution for household, retail, and cold-chain applications.

Keywords: Food Freshness Monitoring, Spoilage Detection, Chemiresistive Gas Sensor Array, Metal- Oxide Semiconductor (MOS), Volatile Organic Compounds (VOCs), Sensor Calibration, Temperature- Humidity Compensation, Internet of Things (IoT), MQTT, Non-Destructive Sensing

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Campus Explorer : Smart Campus Discovery and Navigation Platform

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Abstract:

Finding the right place or getting information about a college campus can be difficult, especially for new students, visitors, and parents. Campus information is often available in different places, and users may have to depend on faculty or senior students for simple queries. To address this problem, this work presents Campus Explorer, an AI-powered smart campus discovery and navigation platform that provides campus information and navigation support through a single system. The proposed system allows users to interact with the platform using natural-language queries and obtain information about departments, faculty, laboratories, facilities, services, and campus events. The system uses React and Tailwind CSS for the frontend and Python with Flask for the backend. A SQLite database is used to store campus-related information. The Google Gemini API is integrated to provide AI-based responses, while Retrieval-Augmented Generation (RAG) helps the system retrieve relevant campus information before generating a response. The platform also uses Leaflet and OpenStreetMap to provide an interactive campus map and route guidance between selected locations. When a user asks a question, the system processes the query, retrieves the relevant information from the campus knowledge base, and provides a clear response through the AI assistant. Users can also search for campus locations and obtain route guidance, making it easier to reach departments, facilities, and other important places. By bringing campus information, AI assistance, and navigation together, the system can reduce repetitive queries to faculty and senior students and make campus exploration easier for new users. The modular design of Campus Explorer also allows future improvements such as voice-based assistance, IoT integration, and augmented reality (AR) navigation. Overall, the project aims to provide a simple, accessible, and intelligent platform that helps users find information, discover campus facilities, and navigate the campus more easily.

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AI-Based Mobile Application for Career Guidance and Placement Support

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Abstract

Engineering students often face difficulties in identifying suitable career paths and preparing effectively for campus placements. Generic career advice, limited awareness of job- specific skills, ATS-based resume screening, and insufficient interview practice affect their placement readiness. To overcome these challenges, this paper proposes an AI-Based Mobile Application for Career Guidance and Placement Support that provides personalized career guidance and placement preparation through a single platform. The application creates a student profile containing academic details, technical skills, interests, certifications, and career goals. Based on this profile, it recommends suitable career paths, identifies skill gaps, and generates personalized learning roadmaps. The system also provides an AI-powered resume analyzer and builder, job-role matching, company-specific placement preparation, certification recommendations, mock interviews, skill assessments, and coding practice. Gamification features such as points, streaks, and leaderboards encourage students to practise regularly and monitor their progress. The application is developed using Flutter for the mobile interface, Python with FastAPI for backend services, and Firebase for authentication and data management. Google Gemini API and Natural Language Processing (NLP) techniques support career analysis, resume evaluation, and interview assessment. By integrating career guidance, skill development, resume improvement, and interview preparation, the proposed application aims to enhance students' confidence, employability, and overall placement readiness.

Keywords: Artificial Intelligence, Career Guidance, Placement Support, Resume Analysis, Job Matching, Mock Interviews.

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GROUNDED RADIOLOGY GENERATION WITH FEATURE-VERIFIED FINDING FOR CHEST X-RAY INTERPRETATION

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Abstract

Automated radiology report generation from chest X-rays (CXR) can reduce reporting workload and turnaround time. However, current vision-language models often produce clinically unsupported statements, including hallucinated findings, omitted abnormalities, and incorrect laterality or severity. This limits their reliability in safety-critical clinical use. This work presents a grounded radiology report generation framework in which every generated finding is verified against image-derived evidence before it enters the final report. The proposed architecture has three stages. First, a vision encoder extracts global and region-level features from the CXR, and a multi-label classification head predicts the presence probabilities of thoracic pathologies such as cardiomegaly, pleural effusion, consolidation, atelectasis, pneumothorax, and edema. Second, a language decoder, conditioned on the visual features and on similar reports retrieved from a reference corpus, generates a draft report in which findings are explicitly separated into individual claims. Third, a feature- verification module checks each claim against the visual evidence. It does this by combining classifier confidence, attention- and Grad-CAM-based spatial localization, and anatomical consistency checks on region and laterality. Claims that are unsupported or contradicted are removed or revised, and the remaining findings are linked to image regions to give interpretable, traceable output. The framework is evaluated on public benchmarks such as MIMIC-CXR and IU X- Ray, using both natural language generation metrics (BLEU, ROUGE-L, METEOR) and clinical efficacy metrics (CheXbert F1, RadGraph F1). The proposed method achieves [X]% CheXbert F1 and [X]% RadGraph F1, and reduces hallucinated findings by [X]% compared with baseline report generation models. These results indicate that feature- verified grounding improves factual consistency, clinical reliability, and transparency, supporting its use as a trustworthy decision-support tool for radiologists in chest X-ray interpretation.

Keywords: Radiology Report Generation, Chest X-Ray, Visual Grounding, Feature Verification, Hallucination Mitigation, Vision-Language Models, Retrieval-Augmented Generation, Explainable AI, Medical Image Analysis

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MINDUNCLUTTER: AI-BASED MOBILE APPLICATION FOR MENTAL HEALTH ASSESSMENT AND COUNSELING SUPPORT-2026MIN393

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Abstract

Mental health challenges among students and young adults include academic stress, anxiety, overthinking, low mood, and difficulty maintaining awareness of emotional patterns. This work presents MindUnclutter, an AI-based mobile application designed for mental health self-monitoring, standardized screening, mood journaling, and personalized self-help support. The proposed system integrates PHQ-9 and GAD-7 screening with minimal daily mood check-ins, longitudinal mood tracking, personal baseline creation, context and trigger pattern detection, and an adaptive Mood Reset module. The system records mood before and after coping activities to estimate intervention effectiveness and uses previous responses to personalize future recommendations. Optional Gemini API analysis is used for journal entries to identify emotional themes and possible thought patterns, with user consent, while core data is maintained using a local-first storage approach. The application provides visual mood trends, mood calendar, weekly reflections, personalized insights, privacy controls, and support resources. The proposed closed-loop approach follows Track, Understand, Reset, Measure, Learn, and Personalize, enabling the system to learn which self-help activities are more useful for an individual user over time. MindUnclutter is intended for self-monitoring and emotional well-being support and does not provide medical diagnosis or replace professional care.

Keywords: Mental Health, Mood Tracking, PHQ-9, GAD-7, Artificial Intelligence, Gemini API, Personalized Recommendation, Mood Reset, Longitudinal Analysis, Mobile Application, Privacy.

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Automated Hostel Utility Optimization and Predictive Resource Management Dashboard for High-Density Institutions

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Abstract:

Managing utilities in high-density hostels is challenging because electricity, water, room usage, and other shared resources can vary significantly throughout the day. Manual monitoring can make it difficult to identify abnormal consumption, predict future demand, and respond to wastage quickly. This work presents an Automated Hostel Utility Optimization and Predictive Resource Management Dashboard designed to provide centralized monitoring and intelligent resource management for hostel environments. The proposed system collects utility and occupancy-related
data, processes the data through a centralized backend, and presents real-time information through a web dashboard. Analytical and predictive modules can be used to identify consumption patterns, detect unusual usage, estimate future resource requirements, and support timely decision-making. The dashboard brings together utility monitoring, resource analytics, alerts, historical information, and predictive insights in a single interface. By combining automated monitoring with predictive resource management, the system aims to reduce avoidable wastage, improve operational visibility, and support efficient management of resources in high-density institutional hostels.

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AI-Powered Adaptive Career Path Mapping and Personalized Skill-Gap Analysis System

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Abstract:

Engineering students often face difficulty in selecting suitable career paths due to rapidly changing industry requirements, diverse job roles, and a lack of personalized guidance. Existing career guidance platforms generally provide generic recommendations and do not continuously evaluate an individual's skills against the requirements of their desired career. This paper proposes an AI-powered adaptive career path mapping and personalized skill- gap analysis system that acts as an intelligent career companion for students. The system analyzes a student's academic background, technical and soft skills, interests, career goals, and current proficiency levels to identify suitable career paths. It compares the student's existing skill set with the skills required for targeted job roles and identifies specific skill gaps. Based on this analysis, the system generates a personalized and adaptive learning roadmap containing recommended skills, courses, certifications, projects, and learning activities. The career path is dynamically updated based on the student's learning progress, assessment performance, and changes in industry skill requirements. AI and natural language processing techniques are used to analyze career requirements, map skills to job roles, and provide personalized recommendations. The proposed system aims to provide continuous, data-driven career guidance and help students systematically develop the skills required to achieve their desired career goals.

Keywords: Artificial Intelligence, Career Path Mapping, Personalized Career Guidance, Skill Gap Analysis, Adaptive Learning, Skill Recommendation, Career Recommendation, Machine Learning, Natural Language Processing, Learning Roadmap.

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AUTOMATED STUDENT PROJECT EVALUATION SYSTEM WITH PLAGIARISM DETECTION AND CODE EXECUTION VALIDATION

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Abstract:

Evaluating student software projects manually is a time-consuming and challenging process for faculty members, particularly when assessing code quality, plagiarism, and program correctness. Existing project evaluation approaches often rely on manual inspection and separate tools, which may result in inconsistent evaluation, delayed feedback, and difficulty in identifying copied code. This paper proposes an Automated Student Project Evaluation System with Plagiarism Detection and Code Execution Validation that provides an integrated platform for systematic and efficient project assessment. The proposed system allows students to submit their projects through a web-based interface, where the submitted source code is analyzed automatically. The system detects potential plagiarism by comparing submitted code with existing student submissions and identifying similarities in source code. It also validates program functionality by compiling and executing the submitted code against predefined test cases and analyzing the execution results. Based on plagiarism analysis, code execution, test-case results, and evaluation criteria, the system generates an automated evaluation report that assists faculty members in assessing student projects. The platform aims to reduce manual effort, improve evaluation consistency, provide faster feedback to students, and promote originality in software development. The proposed system provides a centralized and transparent approach for managing student project submissions, plagiarism detection, code execution validation, and automated evaluation.

Keywords: Automated Project Evaluation, Student Project Assessment, Plagiarism Detection, Code Execution, Code Validation, Source Code Analysis, Test Case Evaluation, Software Project Evaluation, Web-Based System, Academic Evaluation.

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RIVER WATER LEVEL MONITORING SYSTEM

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ABSTRACT

River systems are inherently dynamic, with water levels capable of changing rapidly due to heavy rainfall, upstream discharge, tidal influence, and seasonal variation. Sudden, unmonitored rises in river water level are among the leading causes of flash floods, posing a severe threat to human life, agricultural land, property, and critical infrastructure in riverside settlements. Conventional monitoring relies on manual staff-gauge observations, which are labour-intensive, infrequent, and prone to delay, making them inadequate for the early warning that fast-onset floods demand. There is a clear need for an automated, low-cost, self-sustained, and remotely accessible solution capable of continuously observing river conditions and instantly alerting stakeholders without constant human presence at the site This paper presents the design and implementation of a River Water Level Monitoring System, an IoT- based floating prototype deployed directly on the river surface. The system integrates three complementary sensors: turbidity, accelerometer, and barometric pressure to build a richer picture of river condition than a single sensor alone. The barometric pressure sensor, positioned to sense the water column, estimates the actual water level through the hydrostatic pressure principle, where pressure increases proportionally with depth, providing a direct, low-cost measure of rising or falling water. The accelerometer monitors the tilt, motion, and stability of the floating unit, reflecting turbulence and abnormal displacement caused by rising currents. A GPS module continuously tracks the real-time geographic position of the free-floating unit as it moves with the water. All sensor and location data is acquired and processed by an ESP8266 Wi-Fi microcontroller, which continuously compares each reading against predefined safety thresholds. All sensor readings and GPS coordinates are transmitted through the GSM module to a centralized web-based dashboard for continuous visual monitoring and trend analysis, as well as to a designated remote office ensuring complete independence from Wi-Fi or internet connectivity at the deployment site. When any threshold is crossed, the system additionally triggers an on-site warning alarm, allowing both continuous monitoring and instant alerting to function reliably even in network-limited riverside regions. To operate in remote, off-grid environments, the system is powered entirely by a solar panel connected to a rechargeable battery, which supplies the ESP8266, sensors, GPS, and GSM module continuously, enabling fully autonomous operation without manual recharging. The complete assembly is housed in a lightweight, weatherproof, buoyant enclosure designed to float stably on the river surface. The proposed system demonstrates a practical, low-cost, and scalable approach to early flood detection, combining direct water-level sensing via hydrostatic pressure, multi-sensor fusion, GSM-based dual-channel communication, threshold-based automated alerting, and solar-powered self-sufficiency into a single deployable unit — offering a meaningful reduction in response time for flood-prone river communities.

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Performance Comparison of CNN Architectures and Gentle MRI Preprocessing for Brain Tumor Classification

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Abstract

Automated brain tumor classification using magnetic resonance imaging (MRI) is a clinically relevant medical image analysis task that mandates both robust feature representation and validation. This work gives a comparative analysis based on three models of convolutional neural network (CNN): ResNet50, EfficientNetB0, and DenseNet121 under two preprocessing settings. . A fixed, leakage-controlled dataset comprising 5,560 cleaned training images and 1,430 independent test images was used across four diagnostic categories: glioma, meningioma, no tumor, and pituitary tumor. The experimental framework consists of two preprocessing pipelines, which are a baseline that only performs image resizing and data augmentation in conjunction with ImageNet normalization, and the proposed gentle B2 pipeline that additionally uses Non-Local Means denoising followed by contrast-limited adaptive histogram equalization (CLAHE) to perform preprocessing before non-rigid registration, augmentation, and normalization (resize, augm. normal). For a fair ablation study, all architectures were trained under the same data split and experimental conditions. Among the evaluated configurations, ResNet50 with the baseline pipeline achieved the highest test accuracy of 98.95%, whereas EfficientNet-B0 with the gentle B2 pipeline attained the highest macro-averaged ROC-AUC of 0.999725. The inclusion of the B2 preprocessing pipeline resulted in a reduction in top-1 classification accuracy across all three architectures, while producing marginal improvements in macro ROC-AUC for ResNet50 and EfficientNet-B0. Pairwise McNemar tests further indicated that none of the observed differences in classification accuracy were statistically significant at the 0.05 significance level. In summary, these results prove that the efficacy of image preprocessing is architecture- and evaluation metric-
dependent. These findings also indicate that state of the art preprocessing does not always enhance CNN based brain tumor classification performance, reinforcing the need to carefully validate any general preprocessing with empirical analysis within a controlled and leakage free evaluation framework.

Keywords: Brain tumor classification; Magnetic resonance imaging; Deep learning; Convolutional neural networks; ResNet50; EfficientNet-B0; DenseNet121; Image preprocessing; CLAHE; Non-Local Means denoising; ROC-AUC; Medical image analysis.

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Automated Subject-Specific Tutoring Assistant Utilizing Generative AI and Interactive Concept Maps.

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Abstract

Conventional AI chatbots used for tutoring often generate hallucinated or curriculum-misaligned answers and present knowledge as linear text that hides the conceptual structure of a subject. This
work presents an automated subject-specific tutoring assistant that integrates Retrieval-Augmented Generation (RAG) with dynamically generated interactive concept maps. Curriculum material such as textbooks, lecture notes, and syllabi is segmented, converted into vector embeddings, and stored in a vector database, which grounds a Large Language Model (LLM) so that responses remain domain- specific and traceable to source content. For every learner query, the LLM extracts key concepts and their typed relations (prerequisite, part-of, related-to) as schema-validated JSON, which is converted into a knowledge graph and rendered as an interactive, expandable concept map. Selecting any node provides an explanation, worked examples, or a practice question. A learner-modelling module tracks concept-wise mastery using knowledge tracing and adapts explanation depth and quiz difficulty, with questions generated according to Bloom's taxonomy. The platform is deployed as a web application, supports multiple subjects through interchangeable knowledge bases, and can be extended to additional languages. The proposed system aims to improve answer reliability, conceptual understanding, and learner engagement compared with text-only AI tutors. Its performance is evaluated using retrieval accuracy (Precision@k), answer faithfulness, concept-map quality (node and edge F1 against expert maps), and pre-test to post-test learning gain.

Keywords: Generative AI, Large Language Models, Retrieval-Augmented Generation, Concept Maps, Knowledge Graph, Intelligent Tutoring System, Adaptive Learning, Personalized Education

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Smart Tender Fraud Detection System Using Graph Neural Networks for E-Procurement Platform Monitoring.

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Abstract

Public e-procurement platforms process large volumes of tenders, bids and contracts, making them vulnerable to fraudulent practices such as bid rigging, collusive and cover bidding, shell-company
participation, and contract splitting. Conventional rule-based audits and tabular machine learning models evaluate each bid in isolation and therefore fail to capture the relational structure through which such fraud is organised. This paper proposes a Smart Tender Fraud Detection System that models e- procurement data as a heterogeneous graph, in which bidders, procuring departments, officials, tenders and contracts are nodes, and bidding, awarding, ownership and co-participation relationships are edges. Portal data are collected, cleaned and converted into node and edge features, including bid-to-estimate price deviation, bidder win rate, co-bidding frequency, bid-timing patterns and shared contact or address attributes. Graph Neural Network models, namely GraphSAGE and Graph Attention Networks (GAT), learn neighbourhood-aware embeddings for node classification and link-level anomaly scoring of suspicious bidders and tenders, with class imbalance handled through weighted loss and sampling. Performance is evaluated using precision, recall, F1-score and AUC-ROC against baseline models such as logistic regression, random forest and XGBoost. GNNExplainer is integrated to highlight the suspicious sub-graphs and features behind each prediction, supporting auditor trust. A web dashboard provides continuous risk scoring and alerts for newly published tenders, enabling vigilance officers to prioritise investigations. The proposed system aims to offer a scalable, explainable and proactive framework for improving transparency, accountability and integrity in public procurement.

Keywords: Tender Fraud Detection, E-Procurement, Graph Neural Networks, Bid Rigging, Collusion Detection, Anomaly Detection, GraphSAGE, Graph Attention Network, Explainable AI, Public Procurement Monitoring

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Machine Learning-Based Electricity Usage Forecasting for the Campus

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Abstract

Accurate electricity demand forecasting is essential for efficient energy management, cost control, and sustainable operation of large educational campuses. University and institutional campuses exhibit highly dynamic and non-stationary load behaviour arising from academic schedules, hostel and laboratory operations, seasonal weather variation, and event-driven occupancy, which limits the effectiveness of conventional statistical and manual estimation methods. This work presents a machine learning-based framework for short-term electricity usage forecasting of a campus using historical smart-meter data. The proposed methodology comprises data acquisition from campus energy meters, preprocessing through missing-value imputation, outlier removal, and normalization, and feature engineering that incorporates temporal attributes (hour, day, month, and season), lag and rolling- window statistics, meteorological parameters such as temperature and humidity, and academic- calendar indicators such as holidays, examinations, and vacation periods. Multiple forecasting models, including SARIMA as a statistical baseline, Support Vector Regression, Random Forest, XGBoost, and deep learning models based on Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, are trained and tuned using time-series cross-validation to avoid temporal data leakage. Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2), and the best-performing model is deployed for hourly, day-ahead, and week-ahead load forecasting and peak- demand prediction. The forecasting results are presented through a visualization dashboard that supports block-wise consumption analysis, anomaly detection, and demand-side management. The proposed approach is expected to enable proactive load scheduling, reduced peak demand and energy cost, improved integration of renewable energy sources, and lower carbon emissions, thereby providing a scalable and data-driven solution for smart campus energy management.

Keywords: Electricity Load Forecasting, Machine Learning, Time-Series Analysis, LSTM, XGBoost, Smart Campus, Energy Management, Demand-Side Management, Peak Demand Prediction, Smart Meter Data

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CNN-Based Real-Time Crowd Monitoring and Counting System

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Abstract

Crowd monitoring has become important in places like bus stands, railway stations, shopping malls, colleges, and public events where large crowds are formed. This poses a challenge for people attempting to count manually as the process is time-consuming, complicated, and prone to errors. This paper proposes a CNN-Based Real-Time Crowd Monitoring and Counting System, that automatically detects and counts the number of people present in an image/video. The webcam or CCTV camera captures the video stream of the target area and is broken into individual frames. These frames are subjected to preprocessing techniques like resizing, normalizing, and removing noise. The processed frames are then fed into the CNN model in order to detect people and calculate the density of crowds. This model keeps a track of the number of people present in the target area and updates the count in real-time. Furthermore, the system maintains the history of crowd count, and the crowd count trends over a specific period of time. The proposed system facilitates a simplified approach to real-time crowd counting and monitoring.

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AI Bug Fixer for Small Programs: An Agentic Generative AI Approach to Automated Debugging and Code Repair.

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Abstract

Debugging small programs is time-consuming and difficult, especially for beginners, because compiler or runtime messages often indicate what went wrong but not why it happened or how it should be corrected. Different languages add further complexity through varied syntax rules, compilers, and runtime errors such as off-by-one errors, ZeroDivisionError, NameError, NullPointerException, and ReferenceError. This work presents AI Bug Fixer, an agentic generative AI-based multi-language platform for automated debugging and code repair, providing an end-to-end Run → Detect → Analyze Cause → Fix → Re-run → Verify workflow. The user selects a language such as JavaScript, Python, Java, C, or C++, enters the code, and executes it through a common web interface. A Node.js backend handles web requests and communicates with the language runtimes, while language-specific execution, bug-detection, and repair logic is maintained separately in languages.cjs. Programs run through Node.js, the Python interpreter, javac and java, and GCC and G++, and the captured output, compilation errors, and runtime errors are combined with source-code patterns to identify supported bugs. After detection, cause analysis explains the reason behind the problem rather than only displaying the error message. The detected bug type or bug ID is then used by the generateFix() function to generate a targeted fix that modifies only the responsible portion of the code instead of rewriting the entire program; for example, range(len(numbers) + 1) in a Python loop is corrected to range(len(numbers)). The corrected program is executed again, and the verification stage checks syntax or compilation success, successful execution, removal of the original defect, and regression conditions, reporting whether the fix is verified or has failed. The platform thus automates debugging and code repair across multiple programming languages.

Keywords: Automated Debugging, Code Repair, Agentic Generative AI, Bug Detection, Cause Analysis, Multi- Language Support, Node.js, Fix Verification

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AI-BASED SPOT WELD STRUCTURAL ANALYSIS

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Abstract

Resistance Spot Welding (RSW) is widely used in automotive Body-in-White structures, where individual welds contribute to load transfer and structural integrity. This project develops a physics-based computational framework for an idealized DP600 resistance spot-welded lap-shear joint and an interpretable machine-learning surrogate. The primary geometry consists of two 100 mm × 25 mm × 1.5 mm sheets with a 25 mm overlap. Five nugget diameters from 4 to 8 mm are studied at six axial loads from 500 to 3000 N, producing 30 computational cases. The geometry is inspected in CAD, discretized in Gmsh using second-order tetrahedral elements, and solved as a linear-elastic static structural problem in CalculiX. Maximum global von Mises stress, equivalent strain, maximum displacement, reaction force, weld-region response and a preliminary Factor of Safety are extracted for every case. A second-order polynomial regression uses applied load and nugget diameter as inputs and maximum global von Mises stress as the target. Twenty-four cases are used for training and six 7 mm cases are held out for validation. The model gives training $R^2 = 0.9903$, MAE = 6.161 MPa and RMSE = 7.314 MPa; validation gives $R^2 = 0.9567$, MAE = 10.737 MPa and RMSE = 10.773 MPa. At 3000 N, maximum global stress decreases from 305.5 MPa for a 4 mm nugget to 159.1 MPa for an 8 mm nugget, while the model-based FOS increases from 1.146 to 2.201. The held-out validation indicates useful in-domain interpolation, while the negative predicted-minus-actual residuals show systematic stress underprediction; therefore, the ML-derived safety margin is not conservative without further calibration. The study is computational only and does not claim physical weld inspection, fracture, fatigue, corrosion monitoring or certification. The resulting FE-to-ML workflow provides an auditable baseline for future nonlinear, experimental and degradation- aware weld-health studies.

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Smart Energy Monitoring System

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Abstract

Electrical energy monitoring plays an important role in understanding, analysing, and improving energy consumption in residential, commercial, and small-scale electrical systems. Conventional energy measurement methods may provide only basic consumption information and may not offer continuous visibility of important electrical parameters. In this project, a smart energy monitoring system is proposed for the real-time measurement and monitoring of electrical parameters such as voltage, current, power, power factor, and energy consumption. The proposed system uses suitable voltage and current sensing modules connected to a microcontroller for data acquisition and processing. The acquired sensor data is processed to determine the required electrical parameters and to continuously update the monitoring interface. The mathematical relationships between voltage, current, power, power factor, and electrical energy are considered for accurate parameter calculation and energy accumulation. The system is intended to provide a cost- effective and practical approach for monitoring electrical energy usage in real time. By presenting instantaneous electrical parameters together with accumulated energy consumption, the system can help users understand their consumption patterns and identify opportunities for more efficient energy utilization. The proposed approach also considers suitable communication methods for transmitting or displaying the measured information, making the system adaptable for future IoT-based monitoring. The preliminary hardware architecture, component requirements, data acquisition process, mathematical model, and algorithmic flow have been analysed as the foundation for future hardware implementation, testing, and performance evaluation.

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Vision-Based Autonomous RC Car Using Raspberry Pi and YOLO

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Abstract

Low-cost autonomous mobile robots are increasingly important for applications such as material handling, inspection, surveillance, and education. Many commercial autonomous platforms depend on expensive sensors such as LiDAR and high-resolution encoders, which limits their use in low-budget academic and small-scale applications. To address this, this project proposes a Vision-Based Autonomous RC Car built around a Raspberry Pi 5 running Debian 13, which combines embedded motor control with real-time computer vision.The robot uses a USB camera and a custom-trained YOLO object detection model to detect a red target box. A finite state machine (IDLE, SEARCH, ALIGN, MOVE, STOP, WAIT, ROTATE, RETURN, COMPLETE) governs the autonomous mission. When the target is not visible, the robot rotates slowly to search for it; once detected, the horizontal offset of the bounding- box centre from the frame centre is used as a steering error to align the robot, and the bounding-box area is used as an approximate indicator of distance. The robot stops when the area reaches a calibrated threshold, waits for 20 seconds, performs an approximately 180° timed rotation, and returns toward its starting point by replaying its recorded movement commands in reverse.The drive system consists of two DC geared motors in a differential-drive configuration, controlled through an L298N motor driver using direction signals and PWM speed control generated from the Raspberry Pi GPIO with the lgpio library. A Flask-based Wi-Fi web interface with a virtual joystick, speed slider, and emergency stop provides manual control from a mobile phone or laptop; the joystick X/Y values are converted into left and right wheel commands using differential-drive mixing. Safety features stop the motors on emergency-stop, page close, or loss of page visibility.Since the prototype has no wheel encoders or IMU, distance estimation, rotation, and return are approximate dead-reckoning methods and are affected by battery voltage, floor friction, and motor mismatch; these limitations are evaluated experimentally through calibration and repeated trials of detection accuracy, stopping distance, rotation angle, and return error.

Keywords: Autonomous Robot, Raspberry Pi 5, Computer Vision, YOLO Object Detection, OpenCV, Embedded Linux, DC Motor Control, PWM, L298N Motor Driver, Differential Drive, State-Machine Control, Flask Web Interface, Wi-Fi Remote Control, Real-Time Decision Making, Dead Reckoning

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Automatic Solar Tracking System

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Abstract:

The efficiency of photovoltaic systems is strongly influenced by the orientation of the solar panel with respect to incident sunlight. Conventional automatic solar tracking systems commonly employ LDR sensors, microcontrollers and software-based control, which can increase system complexity and introduce additional control and power requirements. This work proposes a simplified microcontroller-free single-axis automatic solar tracking system based on differential photodiode sensing and analog control. The proposed system employs two BPW34 photodiodes to detect the difference in incident light intensity from two directions. The corresponding sensor signals are conditioned using an LM358 operational amplifier, while an LM393 comparator determines the direction of required panel movement. An NE555 timer-based control stage generates the required control pulses for a 180° servo motor, enabling controlled east-to-west movement of the photovoltaic panel. A key feature of the proposed approach is the adjustable gain and response tuning of the sensor-conditioning stage, allowing the system sensitivity and actuator response to be optimized according to operating conditions. The developed system is intended to reduce unnecessary panel movement while maintaining effective sunlight alignment. Its performance will be evaluated by comparing the electrical output of the tracking panel with that of a fixed panel using voltage, current and power measurements. The proposed architecture provides a low-complexity alternative to microcontroller-based trackers and demonstrates the potential of analog differential sensing for practical solar-energy harvesting applications.

Keywords: Automatic Solar Tracking, BPW34 Photodiode, Differential Sensing, Analog Control, Gain Tuning, LM393 Comparator, NE555 Timer, Servo Motor, Photovoltaic System.

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ONLINE PRODUCTIVITY AND GOAL TRACKING TOOL FOR CORPORATE OFFICES

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Abstract

Organizations often manage goals, projects and tasks in separate tools, rely on manual progress updates, and lack a reliable way to verify completed work. An employee may report a task as largely complete, yet the manager has little evidence to confirm its actual status. As a result, managers have limited visibility into employee work, find it difficult to understand why a project is falling behind, and identify potential delays only after deadlines have already been missed. Existing tools also provide limited centralized analytics that connect organizational goals with day-to-day task execution. To address these challenges, this paper proposes an Online Productivity and Goal Tracking Tool for Corporate Offices, a centralized web-based platform that links organizational goals, projects, milestones, tasks and employee work in a structured hierarchy. The system supports three roles. The Admin manages users, roles, departments and access control. The Manager creates projects, defines goals, divides projects into milestones and tasks, assigns employees, reviews submitted evidence and monitors analytics. The Employee works on assigned tasks, updates task status, submits supporting evidence, views manager feedback and resubmits rejected work. Managers approve or reject each submission with feedback, and only approved work contributes to project and milestone progress, so the system distinguishes clearly between reported work and manager-validated work. As the research component, an AI-based project risk prediction and recommendation module is integrated into the platform. The module analyzes project data such as task completion rate, overdue tasks, milestone progress, deadlines, remaining days, evidence approval and rejection status, number of resubmissions, blocked tasks and task dependencies. After data preprocessing and feature extraction, a machine learning model classifies each project as low, medium or high risk, explains the factors contributing to the prediction, and generates actionable recommendations that help managers act on potential delays at an earlier stage. The system is developed using HTML, CSS and JavaScript on the frontend and Node.js and Express.js on the backend, with SQLite for development and Amazon RDS, S3, SNS and IAM for cloud deployment. Security is provided through JWT authentication, role-based access control, input validation, file type validation and audit logging. The risk prediction model is evaluated against a rule-based baseline using accuracy, precision, recall, F1-score and confusion matrix, and compared with models such as Logistic Regression, Decision Tree and Random Forest. The proposed approach aims to improve transparency, evidence-based progress tracking, early risk identification and decision support in corporate project management.

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CloudPulse: A Serverless Cloud-Based Employee Productivity

and Goal Management System
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Abstract

Organizations often manage employee tasks, goals, and progress through spreadsheets, emails, and disconnected applications, which makes real-time tracking difficult for managers and leaves employees unclear about their goals and deadlines. In this paper, we propose CloudPulse, a cloud-based employee productivity and goal management system that provides a centralized platform for secure authentication, goal creation, task assignment, progress monitoring, and deadline tracking. The system is built on a serverless architecture using Amazon Web Services. Amazon Cognito provides secure user authentication and role-based access for employees and managers, while Amazon S3 and CloudFront deliver the web application with low latency. Amazon API Gateway and AWS Lambda handle secure API communication and backend processing, Amazon DynamoDB stores application data, and Amazon SNS sends automated notifications for task assignments and approaching deadlines. A productivity dashboard gives managers real-time visibility into employee progress. The proposed system is scalable, cost-efficient, and requires no dedicated server management, improving transparency and productivity in organizations.

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Cloud-Based Digital Growth Monitoring and Attendance Tracker for Children in Community Nutrition Centres

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Abstract

In community nutrition centres attendance and growth measurements for a large number of children are regularly recorded, but since the use of paper registers it is difficult to obtain past records, to keep track of changes over time and to follow up on children who require attention. The present paper describes a cloud-based digital system for the purpose of growth monitoring and the tracking of attendance in community nutrition centres. Authorized staff members access a mobile-friendly application, set up child profiles using a unique child identification number, and enter daily attendance together with height and weight measurements dated for that day. The records are kept in a cloud database, with offline storage and automatic synchronization taking place when internet connection is restored. The system calculates basic growth indicators on the basis of age, the measurements and previous records and classifies children as Healthy, At Risk or Critical, while a dashboard displays the growth status, attendance summaries and daily, weekly or
monthly reports. The advantages of the system when compared with manual record keeping are evaluated in terms of data accuracy, processing time, accessibility, storage, security and ease of use. By substituting dispersed paper records with structured cloud data the system enables organised monitoring and early follow-up, thus contributing to SDG 3 (Good Health and Well-Being).

Keywords: Cloud Computing, Growth Monitoring, Attendance Tracking, Offline Synchronization, Child Nutrition

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Development of Digital Mental Health and Psychological Support System for Students in Higher Education

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Abstract

Mental health problems among higher education students, including anxiety, depression, academic stress and social isolation, are rising, yet many students never seek help because of stigma, limited counsellor availability and delays in access. Most campus support services are reactive and reach students only after their condition has worsened. This paper presents the design and development of a digital mental health and psychological support system for students in higher education. The proposed system combines validated self-assessment screening tools (PHQ-9 and GAD-7), daily mood tracking, an AI-assisted conversational module for first-line guidance, anonymous counsellor appointment booking, and a curated library of coping resources. A risk-flagging mechanism escalates high-risk cases to professional counsellors, and privacy is maintained through consent-based, anonymised data handling. The system was implemented using [technology stack] and evaluated with [N] students over [duration]. Performance was measured using usability (System Usability Scale), user engagement and [response time /screening agreement with counsellor assessment]. Results show [key finding with a number], indicating that the system can support early detection and timely intervention. The proposed approach offers a scalable, accessible and confidential model for student well-being.

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Resonex: Predictive Maintenance System for MSME Textile Machinery Using Vibration and Thermal Data

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Abstract

Textile machinery operates continuously in manufacturing industries, and unexpected mechanical faults can lead to unplanned downtime, production interruption, maintenance delays and increased operating costs. Predictive maintenance plays an important role in identifying machine conditions before serious failures occur. The prediction of machine health is considered challenging since vibration and thermal sensor observations can vary between machines and operating conditions. This has created a need for intelligent systems that can analyse machine conditions and provide timely maintenance decisions. In this paper, we proposed Resonex, a software-based predictive maintenance framework for MSME textile machinery using historical vibration and thermal sensor data. The proposed system performs data validation and preprocessing, followed by statistical, Fast Fourier Transform (FFT)-based and temporal feature extraction. An Adaptive Health Engine combines machine-specific Health Index calculation, anomaly detection, degradation analysis and health-state classification. Further, an uncertainty-aware Remaining Useful Life (RUL) prediction and maintenance risk analysis are used to identify the severity and priority of potential failures. A context-aware Retrieval-Augmented Generation (RAG) module provides relevant maintenance knowledge, while a Large Language Model (LLM) generates evidence-based explanations and recommendations. Results demonstrate that the proposed approach can effectively analyse machine health and support predictive maintenance decision-making for textile machinery.

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Offline RAG Assistant with Self-Contained Attestation of Network Isolation for Air-Gapped Networks

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Abstract

Most AI assistants depend on cloud servers and constant internet access, which makes them unusable in air-gapped networks and in environments where data cannot leave the premises. In this paper, we present a conversational assistant that runs fully offline on commodity hardware. A quantized open-source language model is served locally through Ollama, and a Retrieval-Augmented Generation (RAG) pipeline grounds every response in a private document set. Documents are cleaned, split into overlapping chunks, embedded with a local embedding model, and stored in a local vector database. Each answer carries a source citation, and the assistant declines to answer when no relevant content is retrieved, which reduces hallucination. Offline operation was checked through network traffic analysis, which showed no outbound connections during model loading, retrieval, and generation. As an advancement, we propose a self-contained attestation mechanism. Each answer is recorded in a hash-chained entry that links the query hash, the retrieved-chunk hashes, a snapshot of the network interface state, and the previous entry's hash. The citations of an answer can therefore be verified later, without re-execution, to have been generated under confirmed network isolation. The mechanism needs no cloud key, blockchain anchor, or trusted hardware. The result is a private, verifiable, and reliable assistant for secure and restricted environments.

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Class-Weighted Transfer Learning with Explainability Validation for Dermoscopic Malignancy Detection

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Abstract

Skin cancer is among the most common cancers worldwide, and early detection of malignant lesions greatly improves patient outcomes. Dermoscopic diagnosis, however, depends on specialist expertise, and benign and malignant lesions can appear visually very similar. Deep learning models trained on public dermoscopic datasets are further affected by severe class imbalance, which biases predictions toward majority classes and can hide poor minority-class performance behind high overall accuracy. In addition, medical AI systems must be interpretable before they can be trusted, and visual inspection of saliency maps alone is not sufficient evidence of reliability. In this paper, we propose an explainable deep learning framework for 8-class skin-lesion classification on the ISIC 2019 dataset using transfer learning with a lightweight EfficientNet-B0 backbone pretrained on ImageNet. A Convolutional Block Attention Module (CBAM) is integrated to focus the network on lesion regions, and Focal Loss with class weighting is used to handle class imbalance. Grad- CAM heatmaps are generated for each prediction and are quantitatively validated against matched lesion segmentation masks using Intersection over Union (IoU) and Dice coefficient. Model performance will be evaluated using accuracy, precision, recall, specificity, F1-score and ROC-AUC, with ablation studies comparing configurations with and without CBAM and Focal Loss. A preliminary ResNet50 transfer-learning baseline for binary benign-versus-malignant classification has already been developed, and the proposed approach is expected to improve minority-class detection while providing reliable, lesion-aligned explanations.

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GlassBox: Personalized, Privacy-Preserving, and Explainable Multimodal Risk Assessment for Online Exam Proctoring

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Abstract

The rapid expansion of online examinations has increased reliance on automated proctoring, yet fixed thresholds, isolated events, and opaque decisions can confuse legitimate individual differences with potentially unusual behavior while providing limited evidence for instructor review. GlassBox addresses this limitation by shifting proctoring from uniform surveillance toward individualized, evidence-based risk assessment. We present a browser-based framework integrating personalized behavioral calibration, multimodal temporal reasoning, interpretable machine learning, privacy-preserving processing, and empirical robustness evaluation. During an initial calibration period, GlassBox establishes an individual baseline for gaze and head-pose behavior, allowing subsequent observations to be interpreted as deviations from the student's own behavioral norm. Real-time visual signals from MediaPipe Face Landmarker are combined with browser evidence, including tab visibility, focus changes, and clipboard activity, and organized into temporal behavioral episodes capturing persistence, recurrence, and cross-signal concurrence rather than isolated events. The resulting features are processed by an interpretable logistic-regression classifier that produces probabilistic risk estimates and exposes feature-level contributions underlying each assessment. For visual robustness and fairness analysis, an open-source RetinaFace detector is evaluated independently of the real-time MediaPipe pipeline using demographically diverse facial data to quantify detection performance and subgroup disparities. The completed framework is evaluated through student- and session-independent validation, fixed-threshold and rule-based baselines, personalized-versus-global calibration comparisons, modality and component ablations, predictive-performance and probability-calibration metrics, and demographic robustness analysis. GlassBox follows a privacy-by-design, human-in-the-loop approach, performing sensitive facial processing locally wherever feasible and treating automated predictions as decision support rather than proof of misconduct. By transforming opaque, one-size-fits-all monitoring into individualized, traceable, and auditable evidence-based assessment, GlassBox provides a practical foundation for more transparent, accountable, and scientifically testable online examination proctoring.

Keywords: Online Exam Proctoring, Personalized Calibration, Explainable Al, Multimodal Risk Assessment, Privacy-Preserving Al, Algorithmic Fairness, Behavioral Risk Assessment, Human-in-the-Loop Al

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DROPGUARD - AI-Powered Student Dropout Prediction System

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Abstract

Most school information systems only report historical statistics, which makes them reactive and leaves little time to support students who are about to drop out. In this paper, we present DropGuard, an AI-powered web system that predicts student dropout risk from academic, attendance, behavioural, family, health, and technology-access data. Student records are imported from CSV or Excel files with header mapping and dry-run validation, and a feature engineering layer converts them into 24 model inputs. A tuned Logistic Regression pipeline computes a dropout probability, a confidence score, and a Low, Medium, or Highrisk level for each student. Model coefficients identify the top five factors behind every prediction, and each factor triggers a targeted recommendation such as academic tutoring, attendance monitoring, financial support, or counselling. The model was trained on a synthetic dataset of 647 student records and achieved an accuracy of 85.32% and a ROC- AUC of 95.93%. As an advancement, we combine coefficient-based explainability with school-scoped data isolation. JWT authentication and role-based access ensure that administrators, headmasters, and teachers see only permitted records, and every prediction, update, and import is audit-logged. The result is a transparent, secure, and proactive early-warning tool for schools.

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Integrated CFD-Based Early-Warning Framework for Cavitation and Flow-Induced Vibration in an Oil & Gas Control Valve

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Abstract

V-type regulating ball valves used in oil and gas processing systems are exposed to complex flow disturbances that can promote cavitation, pressure pulsations, and flow-induced vibration (FIV), potentially affecting valve performance and service reliability. Early identification of these phenomena is therefore important for improving valve reliability and operational safety. This study presents the development of an integrated CFD-based framework for the assessment and early warning of cavitation and FIV in a V-type regulating ball valve. An engineering-reconstructed valve flow domain was developed using controlled geometric dimensions, followed by tetrahedral mesh generation and numerical quality assessment. The final computational mesh consisted of 755,361 cells and 133,061 nodes, with a minimum orthogonal quality of 0.2096 and maximum skewness of 0.7904. A steady single-phase CFD simulation was first established to obtain a stable baseline flow field. The developed baseline solution predicted an inlet mass flow rate of approximately 3.93 kg/s, with inlet and outlet static pressures of approximately 547.8 kPa and 49.9 kPa, respectively, and a maximum flow velocity of approximately 50.5 m/s. The resulting flow field was subsequently transferred to a multiphase mixture framework containing liquid and vapor phases. Controlled numerical investigations demonstrated successful transfer and restoration of the pressure, velocity, and turbulence fields in the multiphase environment when the cavitation mass-transfer mechanism was disabled, thereby identifying the numerical sensitivity associated with cavitation-model activation. Based on this computational foundation, the framework is being extended to identify low-pressure and vapor-generation regions, pressure pulsations, and flow-induced vibration characteristics. These parameters will subsequently be combined into a Unified Valve Condition Indicator (UVCI) for relative early-warning assessment of valve operating conditions. The study establishes an engineering CFD methodology for integrating cavitation and FIV indicators into an early-warning framework for regulating valve applications in the oil and gas industry

Keywords: V-type regulating ball valve; computational fluid dynamics; cavitation; flow-induced vibration; pressure pulsation; multiphase flow; Schnerr–Sauer; early warning; UVCI.

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Intelligent Student Dropout Prediction and Early Intervention System for Government Schools

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Abstract

Student dropout in government schools is influenced by academic, behavioural, family, health, financial and accessibility factors, yet conventional student information systems are largely reactive and do not predict risk early enough for intervention. In this paper, we present DropGuard, an AI-powered early-warning system that predicts student dropout risk and provides explainable, data-driven intervention recommendations. Student data covering academic, attendance, behaviour, family, health, financial and technology-access parameters are imported through CSV/Excel upload, validated, and transformed into 23 engineered features. A tuned Logistic Regression model, developed on 647 student records, classifies each student as Low, Medium or High risk and reports the dropout probability. The model coefficients are used to identify the major risk factors behind each prediction, which are mapped to targeted academic, attendance, financial and counselling interventions. The system is implemented with FastAPI, React with Vite and PostgreSQL, and is secured through JWT authentication, role-based access control, school-level data isolation and audit logs. The tuned model achieved an accuracy of 85.32% and a ROC-AUC score of 95.93%. The result is a secure, interpretable and cost-efficient platform that supports teachers, headmasters and administrators in taking timely preventive action.

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Hospital Asset Tracking System and Mobile Medical Equipment Geolocation Dashboard

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Abstract

Hospital environments depend on the timely availability and safe movement of mobile medical equipment, yet conventional asset registers and manual handoff procedures can make it difficult to know where equipment is located, who has custody of it, and whether it is available when required. This paper presents a full-stack Hospital Asset Tracking System and Mobile Medical Equipment Geolocation Dashboard, developed as MedTrack – AVS. The system combines QR-based asset identification, browser-based mobile GPS tracking, real-time WebSocket/STOMP communication, server-side geofencing, automatic alerts, transactional asset transfers, audit logging, and analytics. A React and Vite frontend with Bootstrap provides a responsive operational console, while a Java 21 Spring Boot backend with Spring Security, JWT, Google OAuth2, JPA/Hibernate, and MariaDB provides secure and persistent services. Accepted GPS telemetry is stored with UTC microsecond precision and published after successful database commit so live map updates and alerts reflect authoritative server data. Geofence entry and exit events are detected using the Haversine distance model, while transfer workflows maintain a single active asset assignment and preserve tracking history during handoff. The dashboard further introduces three project-specific indicators: Asset Movement Continuity (AMC), Handoff Friction Index (HFI), and Geo-Utilization Balance (GUB), enabling operational assessment of tracking continuity, handoff efficiency, and department-level supply-demand balance. The implementation was validated through physical mobile GPS and QR tests, real-time browser verification, regression testing, and production builds, demonstrating a practical and integrated approach to hospital equipment visibility and control.

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Flash Flood Prediction System for Hilly Regions using Multi-Source Data

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Abstract

Flash floods in hilly regions occur suddenly with very short warning times, causing severe loss of lives, property, and infrastructure. Existing forecasting systems often lack hyper-local prediction and the real-time environmental monitoring needed for effective disaster management. To address this challenge, this paper proposes a Flash Flood Prediction System for Hilly Regions using Multi-Source Data that combines IoT-based sensing with machine learning intelligence. IoT-enabled sensors deployed in vulnerable locations collect rainfall intensity, soil moisture, water level, temperature, and humidity, and transmit them over wireless networks to a centralized cloud platform. These readings are fused with external sources such as satellite imagery, terrain information, historical flood records, and meteorological datasets. After preprocessing, a machine learning model based on XGBoost predicts flash flood risk levels at village and ward levels, while Geographic Information System (GIS) technology visualizes flood-prone zones, monitors affected regions, and analyzes user exposure by location. The platform provides multilingual alerts, risk-zone mapping, safer evacuation route recommendations, and decision-support tools for disaster management authorities. The integrated software-hardware architecture ensures continuous monitoring, accurate prediction, and timely dissemination of warnings, thereby enhancing disaster preparedness, reducing response time, minimizing loss of life and property, and building resilient communities in flood-prone hilly regions.

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Intelligent Tourism Management Framework Combining Adaptive Grouping, Predictive Intelligence, and Multilingual Voice Assistance

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ABSTRACT

Tourism management increasingly requires coordinated planning, timely decision-making, and accessible communication among tourists, tour operators, and local service providers. Conventional tour management often relies on static itineraries and manual coordination, making it difficult to respond efficiently to changes in group preferences, weather, traffic, and local conditions. This paper proposes an Intelligent Tourism Management Framework that integrates adaptive tourist grouping, predictive intelligence, and multilingual voice assistance into a unified platform. The adaptive grouping component organizes tourists according to factors such as interests, travel preferences, schedules, and accessibility needs, helping operators coordinate suitable group activities. The predictive intelligence component uses available contextual data, including weather, traffic, and itinerary progress, to identify potential disruptions and support timely itinerary adjustments. A multilingual voice interface is designed to help tourists access travel information, receive updates, and communicate with the system in their preferred language. The framework can also provide tour operators with a consolidated view of group status and itinerary changes, reducing dependence on repeated manual communication. The proposed system is intended to improve flexibility, coordination, and inclusiveness in organized tourism. Its effectiveness can be assessed through measures such as grouping relevance, itinerary adjustment time, prediction performance, response accuracy, and user satisfaction. The framework offers a practical direction for applying intelligent and language-aware technologies to more responsive tourism services.

Keywords: Intelligent tourism, adaptive grouping, predictive intelligence, itinerary optimization, multilingual voice assistance

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ICU SHARE: A MULTI-HOSPITAL ICU BED RESOURCE COORDINATION AND BOOKING PLATFORM

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Abstract

Intensive Care Unit (ICU) bed availability is a critical resource management challenge because the availability of suitable beds can change continuously across hospitals. Patients and their attendants
may face difficulty identifying nearby hospitals with available ICU beds and required facilities, while hospitals require an organized mechanism to manage bed availability and respond to booking requests. In this paper, we present ICU Share, a full-stack web-based platform that connects users with hospitals through dedicated User and Hospital Portals. The User Portal allows users to search hospitals based on location, distance, hospital type, ICU availability, and required facilities, and submit structured ICU booking requests. The Hospital Portal enables hospitals to manage ICU beds using AVAILABLE, OCCUPIED, RESERVED, and MAINTENANCE states and respond to incoming booking requests. When a request is accepted, an available ICU bed is reserved and the user receives a booking confirmation and notification. When a request is rejected, the system provides suitable alternative hospitals based on availability, facility requirements, and distance, while leaving the final booking decision to the user. The system is implemented using React, Vite, FastAPI, PostgreSQL, SQLAlchemy, REST APIs, and JWT-based authentication. The current implementation uses controlled demonstration data and is designed as an academic full-stack prototype. The proposed system demonstrates how a centralized digital platform can improve ICU resource visibility, booking coordination, and communication between users and hospitals.

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INTELLIGENT AI AGENT PLATFORM FOR AUTOMATED FULL-STACK APPLICATION DEVELOPMENT

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Abstract

Developing full-stack applications involves several tasks such as requirement analysis, user interface design, frontend development, backend development, database creation, API integration, testing, and debugging. Performing these tasks manually can be time-consuming and requires knowledge of different programming technologies. This project introduces an Intelligent AI Agent Platform for Automated Full-Stack Application Development, a tool designed to simplify and automate the application development process. The system uses AI agents to understand user requirements and generate the required application components. It can create frontend pages, backend services, databases, APIs, and other necessary components based on the given instructions. The platform also checks the generated code for errors and provides suggestions for improvement. Different AI agents can work on specific development tasks and coordinate with each other to build a complete application. A dashboard displays the project progress, generated components, code status, and other important information. The main purpose of this platform is to reduce the time and effort required to develop full-stack applications. It helps users create applications using simple natural-language instructions without manually writing every part of the code. The system is designed to support developers and students by automating repetitive development tasks. It does not completely replace developers; instead, it acts as an intelligent assistant that helps in building, testing, and improving applications efficiently.

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Automated GST invoice fraud detection system using anomaly detection for MSME compliance

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Abstract

GST invoices contain essential information about taxable amounts, GST rates, and total values of the invoice.Manual verification of a significant number of GST invoices is a tiresome and error-prone task.This project, Automated GST invoice fraud detection system using anomaly detection for MSME compliance, aims to detect anomalies in GST invoices by leveraging machine learning techniques.These anomalies represent invoices that are suspicious in nature. Prior to the application of machine learning algorithms, the data contained within invoices is preprocessed to remove missing values, remove duplicates, and correct inconsistencies in the data type.In addition, discrepancies in the calculation of the Goods and Service Tax (GST) are highlighted.The Isolation Forest algorithm was used to isolate invoice records that isolating processes detected as anomalies.The system also exposes a fraud risk score which aids in the prioritization of the invoices for manual verification. Furthermore, the system provides supporting information for its anomaly detection and a dashboard that displays the data of invoices, anomalies detected, and the interpretation of the anomalies' risk scores.The proposed system can aid businesses categorized as Micro, Small and Medium Enterprises (MSME) in verifying their GST invoices.The anomalies detected by the system are merely indicative of potential areas of concern and should not be used to directly conclude the fraudulence of an invoice.

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AI-ASSISTED STRUCTURAL TOPOLOGY OPTIMIZATION OF LIGHTWEIGHT ELECTRIC VEHICLE CHASSIS SYSTEMS

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Abstract

Due to the increasing popularity of electric cars, the need for light but strong chassis is becoming more prominent. The purpose of this project is to design and optimize a four-seater electric vehicle chassis by applying Artificial Intelligence (AI) assisted topology optimization.The first stage will involve building a prototype of a car chassis in the SolidWorks program. Then, in order to assess the design's performance, a finite element analysis (FEA) will be conducted in the Fusion 360 software. Through the process of topology optimization, the problem areas will be identified and adjusted until a desirable balance between the removed material and structural integrity is reached. The obtained data, regarding the performance of different chassis configurations will be compiled into a machine learning model, which will help identify desirable properties for a set of lightweight chassis to be used in electric cars. Finally, the optimized chassis can be submitted to additional simulations in Fusion 360, where its performance will be compared to the initial design. By utilizing the power of AI, this project hopes to demonstrate the potential of AI assisted topology optimization in creating a desirable balance between removing unnecessary weight while maintaining a sufficient level of structural integrity.

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AI-Based PredictiveToolWear Monitoring for High-Speed CNC Machining Using Vibration and Spindle Current Signals

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Abstract

Tool wear is one of the most critical factors affecting product quality, dimensional accuracy, and production cost in high-speed CNC machining. Unexpected tool failure leads to poor surface finish, machine downtime, and material wastage. Conventional approaches such as scheduled tool replacement and manual inspection are either uneconomical or unreliable, since they do not reflect the actual condition of the cutting tool. Online monitoring of tool condition is considered difficult because the wear process is nonlinear and the sensor signals are affected by noise, varying cutting parameters, and machine dynamics. This phenomenon has drawn a lot of attention to research on intelligent and predictive tool condition monitoring. In this paper, we propose an AI-based model for predictive tool wear monitoring using vibration and spindle current signals acquired during high-speed CNC machining. Time-domain, frequency-domain, and time frequency features are extracted from both signals and used to train a Convolutional Neural Network (CNN), a Long Short-Term Memory (LSTM) network, and a hybrid CNN LSTM model to estimate the progressive tool wear and predict the remaining useful life. The models are compared using RMSE, MAE, R2 and F1-score. Results prove that the proposed approach predicts tool wear efficiently and supports timely tool replacement in smart manufacturing environments.

Keywords: Tool wear monitoring, High-speed CNC machining, Vibration signals, Spindle current, Deep learning, Remaining useful life

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Dynamic Tour Operator Group Coordination Engine with Real-Time Location and Itinerary Adjustment Dashboards

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ABSTRACT

Managing multiple tourist groups across changing locations and schedules is challenging when coordination depends on fixed itineraries and manual communication. Unexpected traffic, weather changes, delays, and differences in group requirements can disrupt planned activities and make it difficult for tour operators to maintain a clear view of each group's progress. This paper proposes a Dynamic Tour Operator Group Coordination Engine that combines real-time location tracking with itinerary adjustment dashboards to support more responsive tour management. The framework is designed to present group locations and itinerary status on a centralized dashboard, helping operators monitor progress and identify schedule deviations. When disruptions occur, the system can use available contextual information, such as estimated travel times, weather updates, and planned activity durations, to support the generation of revised schedules and route suggestions. The dashboard provides a shared view of updated plans so that tour operators can coordinate changes and communicate relevant updates to group members. The proposed engine aims to reduce coordination overhead, improve visibility across groups, and support timely decisions without relying solely on repeated phone calls or manual itinerary revisions. Its performance can be evaluated using measures such as location update latency, itinerary adjustment time, schedule adherence, and operator and tourist feedback. The framework offers a practical approach to building more flexible and coordinated group tourism services.

Keywords: Tour operator coordination, real-time location tracking, dynamic itinerary adjustment, tourism management, dashboards

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Privacy-Aware Faculty Email Management Using Local Large Language Models and Sender-Based Reminders.

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Abstract

The growing amount of emails received by faculty members makes it hard to find important information, keep track of due dates, and follow up on activities. Manually going through emails is time consuming and can lead to missing crucial instructions and tasks. To solve these problems, this paper proposes a privacy aware AI based faculty email management system that combines automated email summarization, extraction of important information, and generation of sender based reminders into a web application.The proposed system utilizes a DeepSeek-Coder large language model that runs locally in Ollama to summarize email content succinctly. Gmail API is used to connect the application with emails, while FastAPI backend manages email processing, summarization, and reminder activities. React.js frontend provides an interface for viewing email summaries and managing sender based reminders. The reminder feature makes use of APScheduler to setup follow up activities and notify the user. By doing AI summarization locally, the need for third party cloud based AI is reduced thus improving the privacy of email content. The proposed system can be evaluated based on summary quality, processing time, accuracy of extracting key information and reminder reliability. The solution would help reduce the effort involved in manually managing emails while making it easy to find key information for the faculty.

Keywords: Large Language Models, Email Summarization, DeepSeek-Coder, Natural Language Processing, Privacy-Aware AI, Information Extraction, Sender-Based Reminders.

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Intelligent Campus Energy Demand Forecasting Using Machine Learning

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Abstract

Problem: Campus electricity demand is volatile. It follows class timetables, examination and vacation periods, occupancy, laboratory activity, weather, and air-conditioning load. Unanticipated peaks raise energy costs, strain local distribution infrastructure, and hinder efficiency planning, yet many forecasting studies overstate accuracy by validating with random splits that leak future information into training.
Objective: This work presents a calendar-aware machine-learning framework for hour-ahead campus electricity demand forecasting, together with a leak-free protocol for comparing models fairly.
Methods: Historical energy-meter readings are combined with calendar, academic-calendar, lag, and rolling-statistic features, plus weather and occupancy variables where available. After data-quality checks and missing-value treatment, Linear Regression, Random Forest, and gradient-boosting regression are benchmarked against a seasonal-naïve baseline. Evaluation uses chronological train–validation–test splits with walk-forward validation. Accuracy is reported with MAE, RMSE, and MAPE, supplemented by sMAPE for zero or near-zero loads, with peak-hour detection assessed separately. Permutation-based feature importance interprets the main demand drivers.
Key contributions:
1. A leak-free, reproducible benchmark protocol for campus load forecasting.
2. Quantification of the value of academic-calendar features over generic time features.
3. Peak-load-focused evaluation that links forecast accuracy to scheduling decisions.
4. Interpretable identification of the factors driving campus consumption.
Impact: The approach gives campus facility teams a practical decision-support tool to plan energy use, schedule energy-intensive activities, and evaluate conservation measures, supporting smarter and more sustainable campus energy management.

Keywords: campus energy management; electricity demand forecasting; machine learning; time-series validation; feature engineering; peak-load prediction; sustainable campus

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Automated Tamil language speech-to-text engine for government document transcription

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Abstract

The Automated Tamil Language Speech-to-Text Engine for Government Document Transcription is a project designed to convert Tamil speech into written text to support the preparation of government documents. Manual transcription of audio recordings can be time-consuming and requires considerable effort, especially when dealing with lengthy meetings, official discussions, and administrative recordings. This project aims to simplify the transcription process using Automatic Speech Recognition (ASR) technology. The system uses the IndicWhisper Tamil speech recognition model to convert spoken Tamil audio into text. It accepts audio files in formats such as WAV, MP3, OGG, M4A, and FLAC. The uploaded audio is preprocessed by converting it into a mono-channel, 16 kHz audio signal and applying amplitude normalization. The processed audio is then passed to the speech recognition model to generate the Tamil transcription. A text post-processing module improves readability by removing unnecessary spaces and correcting spacing around punctuation marks. The generated transcript can be reviewed and edited through a user interface developed using React, while a FastAPI backend connects the interface with the speech recognition pipeline. The system also supports generating downloadable PDF and DOCX documents from the transcribed text. The model was evaluated on an initial five-sample subset of the Tamil Kathbath validation dataset, achieving an overall Word Error Rate (WER) of 9.8% and a Character Error Rate (CER) of 1.6% on that subset. These preliminary results demonstrate the potential of the system for Tamil speech transcription, although evaluation on a larger and more diverse dataset is required for broader performance assessment. Overall, the project provides a foundation for reducing manual transcription effort and improving the preparation of Tamil-language government documents through automated speech recognition and document generation.

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A Machine Learning Based Student Performance Monitoring and Reward Management System for Institutions

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Abstract:-

Educational institutions are increasingly encouraging students to take part in competitive programming on external platforms such as LeetCode, Codeforces, and CodeChef. However, this activity is typically not monitored in a consistent way. Faculty and administrators often rely on spreadsheets and manual reports, which makes it difficult to identify students who are consistently active, detect those who need more support, or create targeted training programs. This paper presents a web-based system for managing rewards and tracking performance within colleges. The system gathers student activity data from multiple competitive programming platforms and consolidates it into a single institutional database. It transforms completed levels and milestones into reward points and uses a Random Forest model to classify students based on how often they participate. Students are grouped using K-Means clustering, with the number of clusters determined through silhouette analysis, to assist in planning training programs for each group. The system also applies item-based collaborative filtering with cosine similarity to recommend additional problems or topics based on what similar students have solved. Administrators and mentors can select specific student groups using plain-English queries, which are translated into SQL by a large language model, checked using AST parsing, and run on a read-only database connection. If the language model is not available, a rule-based approach is used as an alternative. Role-based dashboards offer customized views for administrators, mentors, and students, and student records are maintained for long-term tracking. The system's modular design enables the addition of new platforms and the retraining of models as institutional data grows.

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AI-Assisted BIS/IS Compliance Analysis Using Multimodal Processing and Clause-Level Hybrid Retrieval

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Abstract:
Compliance verification against Bureau of Indian Standards (BIS) and Indian Standards (IS) requirements is essential for ensuring the quality, safety, and reliability of engineering products and systems. However, manually reviewing technical documents, engineering drawings, and test data sheets against lengthy standard documents is time-consuming and challenging. This paper proposes an AI-Assisted BIS/IS Compliance Analysis System using Multimodal Processing and Clause-Level Hybrid Retrieval to support efficient and traceable compliance verification. The proposed approach processes user- provided technical documents to extract relevant textual, visual, and numerical information. A hybrid retrieval mechanism combining keyword-based and semantic search is used to identify applicable clauses from the relevant standards. The extracted technical evidence is then compared with the retrieved requirements to identify potential compliance gaps, mismatches, and areas requiring further review. The system organizes the analysis into clause-level results with supporting evidence and standard references, enabling users to understand how compliance conclusions are derived. By integrating multimodal information extraction, hybrid retrieval, and evidence-based analysis, the proposed system aims to reduce manual review effort, improve traceability, and support consistent compliance assessment. The framework can be extended to additional engineering standards, document formats, and domain-specific compliance applications.
Keywords: BIS/IS Standards, Multimodal Document Processing, Hybrid Information Retrieval, Clause-Level Analysis, Compliance Verification, Artificial Intelligence.

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AI - Based Handwritten Character and Numerical Recognition for Digitization and Retrieval of School Answer Sheets

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Abstract :

The management of handwritten school answer sheets presents challenges in digitization, storage, searching, and retrieval, particularly during the revaluation process. Manually locating specific answer sheets can be time-consuming when examination records are maintained in physical form. This paper proposes an artificial intelligence-based system for handwritten character and numerical recognition to support the digitization and retrieval of school answer sheets. The proposed system integrates image preprocessing, handwritten text recognition, database management, and web-based access. Scanned or uploaded answer sheets are processed using image processing techniques to improve image quality before recognition. Optical Character Recognition (OCR) and deep learning-based methods are considered for identifying handwritten characters and numerical values and converting them into digital information. The extracted information, along with relevant examination details and references to the original answer-sheet images, is stored in MongoDB, a NoSQL document database. A Flask-based web interface is proposed to allow authorized users to upload answer sheets, search for stored records, and retrieve the corresponding digital documents.

The proposed system aims to reduce the manual effort involved in locating examination records and improve access to answer sheets during verification and revaluation. It is intended to assist teachers rather than automatically evaluate answers or assign marks. The system's effectiveness will be assessed using recognition accuracy, processing time, and retrieval performance.

Keywords :
Handwritten Character Recognition, Numerical Recognition, Optical Character Recognition, Deep Learning, Image Processing, Answer Sheet Digitization, MongoDB, Revaluation Support.

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VR Space Station Exploration: A Mission-Based Virtual Reality Training and Survival Experience

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Abstract:

Virtual Reality (VR) provides an immersive environment for interactive learning, exploration, and simulation. The proposed project, VR Space Station Exploration, aims to simulate life aboard an orbital space station using Unreal Engine 5. The system allows users to experience a virtual space environment and complete various engineering, scientific, and emergency-response missions. VR headset tracking and motion controllers are used to provide realistic interactions, including object grabbing, movement, and exploration. The project incorporates a mission-based progression system in which users complete objectives, earn experience points (XP), collect medals, and unlock new tools, astronaut suits, and station areas.In addition to the main missions, five game modes—Time Attack, Puzzle Mode, ExplorationMode, Find the Object, and Survival Mode—provide varied gameplay experiences. The system also includes a tutorial, interactive user interface, and save system to maintain progress across sessions. By combining immersive VR interaction, mission-based challenges, and a rewarding progression system, the proposed project delivers an engaging and replayable space station simulation that promotes exploration, problem-solving, and interactive learning.

Keywords: Virtual Reality, Unreal Engine 5, Space Station Simulation, VR Interaction, Mission-BasedGameplay, Immersive Learning.

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AI - Based Handwritten Character and Numerical Recognition for Digitization and Retrieval of School Answer Sheets

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The proposed system aims to reduce the manual effort involved in locating examination records and improve access to answer sheets during verification and revaluation. It is intended to assist teachers rather than automatically evaluate answers or assign marks. The system's effectiveness will be assessed using recognition accuracy, processing time, and retrieval performance.

Keywords :
Handwritten Character Recognition, Numerical Recognition, Optical Character Recognition, Deep Learning, Image Processing, Answer Sheet Digitization, MongoDB, Revaluation Support.

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SIGNVISION: AI-powered Hand Sign Translation System

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ABSTRACT:-

Communication has become easier with technologies such as chatbots and virtual assistants, but people who are deaf or mute still face difficulties while communicating with others. Existing sign-language recognition systems mainly focus on recognizing individual signs and often have limited support for continuous communication. To overcome this challenge, the project aims to develop an AI-powered Sign Language Assistant to facilitate real-time communication between sign language users and the broader community using computer vision, natural language processing, and a conversational AI interface.The system employs computer vision for hand detection and recognition of sign language gestures to detect them in real-time. It is trained with Indian Sign Language (ISL) dataset that consists of the standard sign samples for recognition. Recognized signs are translated to text and fed to an AI chatbot that is able to produce appropriate responses based on the context of the conversation. These responses are delivered via spoken word and captioning, thus making the interaction more natural and accessible.The accuracy, precision, recall, F1-score and recognition latency are used to assess the performance of the sign-recognition module. The accuracy, relevance, response time, and semantic similarity of the chatbot responses are assessed. This system is stable under normal circumstances with limited vocabulary, but might have problems when the signer's hand is occluded, the signing is very fast, and the vocabulary is large. In the future, AI-based sign- language learning, sign-language personalized feedback, sign-language 3D hand-pose estimation, and sign-language animation with AI avatars are expected to be added, with the goal of enhancing sign-language recognition and user experience.

Keywords: Sign Language Recognition; Dual-Hand Detection; Continuous Recognition; AI Chatbot; AI Avatar; Voice Assistant

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An Explainable Multilingual Framework for Grievance Categorization and Automated Departmental Dispatch in Government Portals

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Abstract:

Government grievance portals receive complaints covering a wide range of public services, making manual classification and departmental assignment time-consuming and difficult to manage consistently. These challenges become more pronounced when citizens submit grievances in different languages or use informal expressions. This paper presents an intelligent multilingual grievance management framework that combines machine learning-based categorization, priority assessment, explainable predictions, and automated departmental routing within a unified workflow. The system applies language and script detection to incoming text, followed by text preprocessing and feature extraction using Term Frequency–Inverse Document Frequency (TF-IDF). A Logistic Regression classifier predicts one of nine grievance categories, while a separate priority assessment component considers category severity and urgency-related terms. Based on the predicted category, the framework recommends the responsible department and provides feature-based explanations to support transparency. The application also incorporates role-based access control for citizens, officers, and administrators, enabling controlled grievance tracking, department-specific handling, and authorized correction of classification results. The prototype has been evaluated using 675 synthetically generated grievance samples across nine categories, alongside automated software tests. Since the dataset is synthetic, the evaluation provides preliminary evidence of implementation feasibility rather than proof of generalization to real government complaints. The framework demonstrates an integrated approach to explainable grievance triage and dispatch, with further validation on authentic, multilingual grievance data required before operational deployment.

Keywords: Multilingual NLP, Grievance Management, Text Classification, TF-IDF, Logistic Regression,Explainable AI, Priority Assessment, Automated Departmental Routing.

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Authenticity Validator for Academia: A Secure Academic Document Verification System

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Abstract:

Academic credentials are routinely submitted to institutions and employers for verification, yet manual checking is slow and depends on correspondence with issuing bodies, while forged or altered documents threaten the integrity of academic records. Authenticity Validator for Academia is a web-based platform that supports secure academic document verification through document validation,source authenticity checking, and the generation of structured verification results. The frontend is built with React.js, and Spring Boot services expose REST APIs that carry the verification workflow, with PostgreSQL providing persistent storage of user, document, and verification records. Security is addressed at the application level: Spring Security protects service endpoints, JWT authentication manages stateless sessions, BCrypt hashing safeguards stored passwords, and role-based access control limits sensitive operations to authorized users. The platform is containerized with Docker, maintained on GitHub, and deployed on Render. The aim is to give verifiers a consistent and secure workflow for checking a document against its issuing source, rather than relying on informal manual confirmation.The system is expected to improve verification reliability, reduce manual effort, and support better academic document management. No accuracy or detection-rate claims are made at this stage; any performance results will be reported only after testing. The platform is intended to assist verification officers, not to replace the authority of the issuing institution in confirming a credential.

Keywords: academic document verification; source authenticity; Spring Boot; JWT authentication; role-based
access control

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AI-Powered Mamp;A Due Diligence Assistant: A Retrieval-Augmented Generation Approach

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Abstract:

Mergers and acquisitions (M&A) require due diligence across large volumes of financial, legal, regulatory, tax and compliance documents, and manual review is slow, costly and often inconsistent in locating relevant evidence. In this paper, we propose an AI- powered M&A Due Diligence Assistant that uses Retrieval-Augmented Generation (RAG) to support document analysis. The system extracts text from PDF documents, splits it into manageable chunks, generates semantic embeddings and indexes them in a vector store for similarity-based retrieval. For each user query, the most relevant passages are retrieved and supplied as context to a language model, which generates a grounded answer that can be traced back to its source content. The assistant supports summarization, question answering and preliminary identification of potential due-diligence concerns. The prototype is implemented using Python, Streamlit, PyMuPDF, Sentence Transformers, FAISS and a locally hosted language model through Ollama, enabling an offline-oriented workflow that keeps sensitive deal documents on local infrastructure. Evaluation criteria include retrieval relevance, answer faithfulness, response time and expert review of generated findings. The proposed approach aims to make document-intensive due diligence more searchable, explainable and efficient, and is intended to assist rather than replace professional review.

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Multi-Hospital Shared ICU Beds Resource Allocation Platform with Real-Time Patient Severity Indexing.

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Abstract:

Timely access to intensive care unit (ICU) beds is critical for emergency and accident patients, yet hospitals still coordinate bed availability and transfers largely through manual calls, causing delays when the nearest hospital lacks a suitable ICU resource. This paper proposes a centralized web-based platform that connects ambulances and multiple hospitals for real-time ICU bed monitoring, patient severity assessment and transfer coordination. Beds across connected hospitals are tracked as available, occupied, reserved or unavailable and updated live through WebSocket communication. Ambulance crews register emergency patients, with a temporary ID generated for unidentified accident victims so that care is not delayed by missing personal details. Vital signs, namely SpO_2, heart rate, blood pressure, respiratory rate, temperature and level of consciousness, are retrieved from compatible medical devices over Bluetooth/BLE or USB, or entered manually, and are used to generate a clinically reviewed severity index and emergency priority. Using this priority, the required facilities and bed availability, the platform identifies potentially suitable ICU beds, searches other connected hospitals when the current hospital cannot accommodate the patient, sends pre-arrival alerts to the receiving hospital and supports transfer requests. Insurance, government scheme and payment information is kept separate from clinical severity
and priority. The system is built using React.js, Node.js, Express.js and MongoDB. The proposed platform aims to improve ICU bed visibility, reduce manual coordination and support faster critical-care decisions, while serving as a decision-support tool that complements, rather than replaces, clinical judgement.

Keywords: ICU Bed Allocation, Patient Severity Index, Emergency Medical Services, Multi-Hospital Coordination, Real-Time Monitoring, Pre-Arrival Alerts, Clinical Decision Support, Healthcare Resource Management

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AI-Driven Critical Patient Monitoring, Alert Management, and Clinical Dashboard

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Abstract:-

Intensive care unit (ICU) healthcare professionals must continuously monitor multiple critically ill patients and their vital parameters. Managing patient information, vital signs, ICU bed status and alerts through separate or manually maintained systems increases workload and makes it difficult to identify critical changes quickly. This paper presents an AI-driven critical patient monitoring system that provides a centralized digital platform for monitoring ICU patients efficiently, accessing patient information quickly and improving the visibility of critical conditions. The system is developed as a full-stack web application using React.js and Tailwind CSS for the frontend, Node.js and Express.js for backend REST APIs, and MongoDB with Mongoose for data storage. The clinical dashboard supports
patient management, patient details, vital information, ICU bed monitoring and alert management. An AI layer is proposed to analyze patient vital data, identify abnormal patterns, generate risk indicators, prioritize alerts and provide concise clinical summaries, following the pipeline of patient data, backend, AI analysis, risk assessment, alert or summary, and doctor dashboard. The AI module acts as a clinical decision-support tool that assists, rather than replaces, clinical decisions. The current implementation focuses on the full-stack application, while AI-based risk analysis, intelligent alert prioritization and
clinical summarization can be integrated as advanced modules. Compared with a conventional dashboard that only displays raw patient data, the proposed approach provides scope for more intelligent monitoring.

Keywords: AI-Driven Patient Monitoring, ICU, Critical Care, Clinical Dashboard, Alert Management, Risk Assessment, Clinical Decision Support, Vital Signs Monitoring, Clinical Summarization, MERN Stack

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RAG-Powered Automated Due Diligence System for Mergers and Acquisitions in Indian Markets

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Abstract:
Mergers and acquisitions require extensive review of unstructured company documents — annual reports, financial statements, legal filings — to identify risks and verify facts before a transaction proceeds. This process is traditionally manual, time-consuming, and prone to overlooking information buried across hundreds of pages. Simple keyword search, the most common existing tool, fails to surface semantically related content phrased differently from the search term. This project presents a prototype AI assistant that addresses this gap using a Retrieval-Augmented Generation (RAG) pipeline. Uploaded PDF documents are parsed page-by-page using pypdf, split into overlapping text chunks via LangChain's text splitter, and converted into 384-dimensional semantic embeddings using the pre-trained Sentence-Transformers model all-MiniLM-L6-v2. These embeddings are indexed using FAISS for efficient similarity search. When a user poses a natural-language question, the system retrieves the most relevant chunks and passes them to a large language model (accessed via Groq's API), which is explicitly constrained to answer only from the retrieved context — every answer is returned with its exact source document and page number, and the system is designed to state when information is insufficient rather than fabricate an answer.
Beyond question-answering, the system performs rule-based classification of questions into Financial, Legal/Regulatory, Business/Operational, and Risk categories; calculates basic financial ratios (Net Profit Margin, Debt-to-Asset, Debt-to-Equity) from user-supplied figures; and flags potential risk indicators using keyword-based evidence matching, explicitly avoiding any fabricated numerical risk score. All findings are compiled into a downloadable due diligence summary report. Implemented with an intentionally compact, well-understood technology stack and validated through 29 automated unit tests, this project demonstrates practical application of embeddings,vector search, and retrieval-augmented generation to a realistic, document-intensive business problem, while remaining fully explainable and reproducible within an undergraduate academic context.

Keywords: Retrieval-Augmented Generation, Semantic Search, FAISS, Sentence Embeddings, Large Language Models, Due Diligence, Mergers and Acquisitions

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Machine Learning–Based Anomaly Detection for Automated GST Invoice Fraud Detection to Support MSME Compliance

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Abstract:

Input Tax Credit (ITC) fraud is a major challenge in the Goods and Services Tax (GST) ecosystem, with Indian tax authorities detecting 30,162 ITC fraud cases totaling ₹74,781.56 crore in FY 2025–26. Micro, Small and Medium Enterprises (MSMEs) process thousands of GST invoices every month, which makes manual verification time-consuming and error-prone. Incorrect invoice values, duplicate invoices, unusual transaction patterns and vendor compliance risks often remain undetected, leading to financial losses and compliance penalties. In this paper, we propose a machine learning-based GST invoice anomaly detection system for MSMEs. Invoices are uploaded through a user-friendly application, relevant numerical features are extracted automatically, and the data is preprocessed before being analysed by an Isolation Forest model for unsupervised anomaly detection. Each invoice is assigned a risk score so that high-risk invoices can be prioritised for further review. The system is designed as a screening and decision-support tool, with human review retained in the decision process, and not as an automatic fraud confirmation system. Future enhancements include GSTIN verification, duplicate invoice detection, reconciliation with GSTR-1 andGSTR-3B, and historical vendor profiling.

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SMART LAB CALIBRATION SYSTEM
A Priority-Based Approach for Intelligent Laboratory Equipment Monitoring

ABSTRACT:

Laboratory equipment needs regular calibration to ensure accurate and reliable results. However,managing calibration schedules and equipment records manually can be difficult, especially when multiple instruments are used in a laboratory. Missing calibration deadlines or using equipment with expired calibration may affect the reliability of laboratory results. To address these issues, this project proposes a Smart Lab Calibration System, a web-based application designed to organize equipment information and simplify calibration monitoring.

The system will maintain equipment details, previous calibration records, upcoming calibration dates, and current calibration status in one place. Its main feature is a priority-based alert mechanism that considers how soon calibration is due, whether a deadline has passed, and the importance of the equipment. A dashboard with colour-coded indicators will help users identify equipment that needs attention first. The system will also provide calibration history and basic reports, making records easier to access and maintain.

The proposed system aims to reduce manual record-keeping, help prevent missed calibration schedules, and support better planning of laboratory activities. It is intended as an affordable software-based solution that can be developed without additional hardware, making it suitable for college laboratories, research centres, and small testing facilities. By combining calibration tracking with priority-based monitoring, the system offers a practical way to make laboratory equipment management more organized and efficient.

Keywords: Laboratory Calibration, Equipment Management, Calibration Scheduling, Priority-Based Alerts, Web-Based Monitoring.

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NLP-Based Research Paper Recommendation and Evidence-Grounded Research Intelligence System

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Abstract:

The rapid growth of scientific publications makes it increasingly difficult for researchers to identify relevant papers, compare existing methodologies, understand research developments, and discover potential research gaps. Traditional keyword-based search systems often fail to capture the semantic relationships between research papers and provide limited support for systematic literature analysis. To address these challenges, NLP-Based Research Paper Recommendation and Evidence-Grounded Research Intelligence System is designed to provide an integrated platform for academic paper discovery, analysis, comparison, and continuous research monitoring. The proposed system utilizes Natural Language Processing (NLP) techniques and the scientific document embedding model to represent research papers in a 768-dimensional semantic vector space. FAISS enables efficient similarity-based retrieval, while PostgreSQL stores paper metadata and application data. The system supports multiple input formats, including research queries, titles, abstracts, keywords, and PDF manuscripts. Academic metadata is collected from multiple scholarly sources, with duplicate detection based on identifiers and title similarity to improve the consistency of search results. The platform provides explainable paper recommendations, structured paper comparison, and research summaries to assist users in understanding the objectives, methodologies, datasets, evaluation metrics, results, and limitations of selected publications.A key component of the system is the Dynamic Research Intelligence and Update Tracking(DRIUT) mechanism, which monitors newly ingested publications against a researcher saved papers and identifies potential relationships under six categories: Extension, Improvement, Challenge, Reproduction,Alternative, and Potential Replacement. The planned enhancement of DRIUT incorporates supporting textual evidence, relationship confidence, and research evolution tracking to make these classifications more transparent and reliable. The system also aims to provide evidence-grounded research-gap exploration,personalized research collections, and secure user access through Firebase Authentication, subject to integration and validation. By combining semantic retrieval, scholarly metadata integration, structured comparison, and continuous research monitoring, ResearchLens aims to move beyond conventional paper recommendation systems toward evidence-grounded research intelligence. The platform is intended to help researchers reduce literature review effort, understand connections between scientific publications, track emerging developments,and formulate potential research questions.

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TRUST-AWARE TEMPORAL AI: An Evidence-Fusion Framework for Crowdsourced Rural Infrastructure Monitoring

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Abstract:

Monitoring rural infrastructure progress across multiple villages can be difficult when inspections are periodic and progress information comes from different sources. Photographs and videos submitted by citizens could help provide more frequent updates on public infrastructure. However, repeated uploads, missing location details, outdated photographs, can make it difficult to determine a reported development status is supported by reliable evidence. This paper proposes Trust-Aware Temporal AI, a framework for assessing crowdsourced infrastructure evidence over time. Its central component, a Project-Level Evidence Confidence Engine, examines whether submitted observations are relevant to a particular project, consistent with its recorded location and timeline, and compatible with the expected development stage. It also identifies duplicate submissions and conflicting observations instead of treating every upload as independent confirmation. Based on these checks, the system groups evidence as corroborative, contradictory, or insufficient to support an assessment and identifies cases that require further human verification. The resulting information is presented through a geospatial analytics portal to help monitor village-level development and prioritise inspections. The system is designed to assist monitoring and inspection decisions; it does not treat citizen submissions or AI-generated assessments as official proof of project completion.

Keywords: Rural Infrastructure Monitoring; Crowdsourced Evidence; Temporal Computer Vision; Geospatial Analytics; Evidence Fusion; Confidence Assessment; Human Verification.

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Reward Points Management System

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Abstract:

Managing reward points for student achievements and institutional activities involves evidence verification, allocation decisions, approval tracking, and conversion into academic benefits. Manual records and spreadsheets can make these processes difficult to coordinate and monitor. This paper presents a Reward Points Management System that centralizes student activity-based reward administration through a role-based web application. The system provides dedicated interfaces for students, faculty reviewers, and administrators.

Students submit evidence of competition achievements and completed institutional activities, track submission status, and view their reward balances and internal-mark outcomes. Faculty reviewers assess evidence and allocate points using either predefined activity values or submission-specific decisions. Allocated points undergo verification and approval before being credited to student balances. Administrators manage users,activity categories, reward policies, and institutional dashboards. A central feature is configurable, tiered conversion of reward points into internal marks. Administrators define successive mark ranges and the points required per mark within each tier, using the average reward points of a relevant batch or semester as a policy reference. This approach supports different conversion costs across mark ranges instead of applying a single fixed rate. The institution schedules a redemption date, on which the system automatically initiates conversion and deducts consumed points from student balances. Redemption is limited to internal marks, and students cannot modify conversion policies or control the redemption process. The application uses React.js with Vite
for the frontend, Node.js for backend operations, and MySQL for persistent storage. Dashboards, communication facilities, and analytics support review coordination and monitoring of allocations and redemption. The implemented system provides a unified workflow designed to improve transparency, traceability, and consistency in institutional reward administration. Formal testing and institutional evaluation remain necessary to establish its reliability and operational benefits.

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AI-Assisted Clinical Decision Support and Location-Aware Hospital Referral System for Patient Management in Tamil Nadu

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Abstract:
Public and private healthcare networks face critical challenges in efficiently allocating patients to appropriate medical specialties and matching them with hospitals possessing the requisite clinical infrastructure, bed capacity, and specialized equipment, particularly across regions in Tamil Nadu. Conventional manual referrals often result in bottlenecks and delays in specialized care, highlighting the need for automated, data-driven decision support. This
paper proposes a secure, research-oriented AI-Assisted Hospital Referral and Patient Management System built on a robust backend of Node.js, Express.js, and MongoDB, with data validation handled via Joi and security enforced through JWT and HttpOnly cookies. The system integrates a planned Python and scikit-learn machine learning service for predicting medical specialties from patient symptoms, vital signs, and laboratory findings, alongside a geographic matching service utilizing Haversine distance calculation. Eligible hospitals are filtered and ranked according to verified clinical capabilities, capacity, tier, and accreditation data. Emergency cases are systematically routed through specialized emergency procedures, ensuring safety and compliance. The proposed system aims to provide healthcare staff with a scalable, secure, and transparent framework to support clinical decision-making,
optimize patient allocation, and improve healthcare logistics without replacing medical professionals.

Keywords: Hospital Referral System, Clinical Decision Support, Medical Specialty Prediction, Geographic Matching, Haversine Distance, Patient Management, Healthcare Logistics, Node.js, MongoDB

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POLICYPILOT — ADAPTIVE RAG AND MULTI-AGENT AI SYSTEM FOR BANKING LOAN COMPLIANCE

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Abstract:

Bank loan compliance is a critical process in the banking sector, where employees must review customer documents, financial information, and applicable banking and RBI policies before making lending decisions. Manual verification and policy lookup can be time-consuming, inconsistent, and difficult to audit, particularly when relevant regulatory information is distributed across lengthy policy documents. Inadequate retrieval of relevant policy evidence may also lead to incomplete or unsupported compliance assessments. In this paper, we propose PolicyPilot, an adaptive Retrieval-Augmented Generation (RAG) and multi-agent AI system designed to assist bank employees in loan compliance analysis. The system supports loan application and document management, retrieves relevant policy information using semantic embeddings and a ChromaDB vector database, and employs an Adaptive RAG engine that evaluates retrieved evidence and rewrites queries when the evidence is insufficient. Gemini is used for structured compliance reasoning through validated Pydantic output models, while specialised Document, Policy, Compliance, and Explanation agents coordinate the analysis. The system produces evidence-grounded compliance findings with status, confidence, rationale, and audit information to improve transparency and traceability. PolicyPilot is designed as a decision-support system, with human review retained throughout the workflow and final loan decisions remaining the responsibility of authorised bank employees. Future enhancements include advanced OCR and document intelligence, expansion of the RBI policy knowledge base, production-grade embedding providers, continuous policy version management, and a human feedback loop.

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Neuromorphic computing in smart industry

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ABSTRACT

The rapid rise of Industry 4.0 has led to a transformation of traditional manufacturing into an intelligent and interconnected industrial environment involving machines, sensors, robots, and cyber-physical systems, which produce and process massive amounts of data. Conventional computing architectures for industrial artificial intelligence encounter challenges related to energy consumption, data movement, processing delays, and decision-making latency, especially with the use of AI at the factory edge. Neuromorphic computing, inspired by the information-processing mechanisms of the human brain, is an innovative technology that enables addressing these limitations through event-based computing and parallel processing. This article focuses on the potential of neuromorphic computing and artificial intelligence in the smart industry, with a specific focus on Spiking Neural Networks and neuromorphic edge intelligence. The study highlights the importance of neuromorphic computing in smart industry applications, including predictive maintenance, quality inspection, anomaly detection, robotics, and the Industrial Internet of Things through the temporal processing of industrial data, neuromorphic sensing, edge AI, and hybrid approaches, including neuromorphic processors and conventional computing systems, for complex industrial needs. The research also considers the opportunities and challenges associated with memristive devices, neuromorphic processors, and event-based sensors for energy-efficient industrial intelligence. Finally, the article addresses the challenges of temporal processing using Spiking Neural Networks (SNNs), hardware-software co-design solutions, and SNN training, while highlighting the opportunities for smart industrial systems and robotics. Keywords: Neuromorphic Computing, Artificial Intelligence, Smart Industry, Spiking Neural Networks, Neuromorphic Edge Intelligence, Industry 4.0, Edge AI, Industrial Internet of Things, Predictive Maintenance, Event-Based Computing

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Aegis: Calibrated Discovery of Unrecorded Entity Relationships in Document Corpora

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Abstract:

Retrieval-augmented systems answer questions about what a corpus states, but they do not look for the relationships it leaves unstated. This paper presents Ageis, a local-first intelligence engine written in C++20 that ingests document corpora, embeds them with a multilingual transformer, builds an entity–relation graph, and runs a cognition loop that proposes links between entities that lie close in vector space yet have no path between them in the graph. Each entity is represented by the centroid of the passages that are evidence about it rather than by the embedding of its name, which raised sector-recovery AUC on eight SEC issuers from 0.794 to 0.894. Candidate pairs are scored on semantic similarity and co-occurrence and judged against an empirical null fitted over the corpus's own pairs, with Benjamini–Hochberg false-discovery control at 5%, in place of a hand-tuned cosine threshold. A closed-form permutation null for the combined statistic, with a separate null per power class, brings a shuffled-label control from rejecting on up to 12 of 12 seeds to 0 of 12. On a labelled company register this raised the precision of certified links from 68.5% to 73.7% and the ranking AUC from 0.949 to 0.956. In a forward-looking backtest over the 500 largest SEC filers (4,943 annual reports, 1.1 million passages), the engine predicts which companies will later name each other with AUC 0.69–0.73 across three time splits, outperforming centroid cosine at every split and co-mention counting at two of three, though counting still yields the better top-30 shortlist. Surviving proposals are argued by a large language model and triaged by a human analyst.

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Predictive Dynamic Pricing and Revenue Management System for Eco-Tourism Hospitality Incorporating Local Event Scraping

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Abstract:
The hospitality industry faces challenges in determining suitable room prices due to fluctuations in customer demand, seasonal variations, weather conditions, and local events. Traditional pricing methods may not effectively respond to changes in demand, which can affect hotel revenue and occupancy rates. This paper proposes a predictive dynamic pricing and revenue management system for eco-tourism hospitality by incorporating local event data. The proposed system integrates data collection, local event scraping, data preprocessing, machine learning-based demand prediction, and dynamic pricing recommendations. Information about local events, hotel booking patterns, seasonal trends, and relevant weather conditions is collected and processed to identify factors influencing hotel demand. Machine learning techniques are used to predict demand variations and support suitable pricing decisions. The system aims to help eco-tourism hospitality providers understand periods of increased or decreased demand and make informed pricing decisions. The proposed system is intended to improve revenue management by providing data-driven pricing recommendations based on predicted demand and local event information. Its effectiveness can be evaluated using prediction metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared, along with the relevance of the generated pricing recommendations.

Keywords: Dynamic Pricing, Revenue Management, Eco-Tourism Hospitality, Local Event Scraping, Machine Learning, Demand Prediction, Hotel Booking, Data Analysis.

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EVALUATION OF YOLOv8-SMALL FOR HANDGUN AND KNIFE DETECTION UNDER CLEAR AND FOGGY CONDITIONS

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Abstract:
Artificial Intelligence and computer vision play an important role in video surveillance by enabling the automatic detection of objects and potential threats. However, detecting small weapons such as handguns and knives remains challenging when they occupy only a small portion of a video frame or are affected by poor visibility, motion blur, and reduced image clarity. Foggy conditions may further affect the accuracy and reliability of weapon detection systems. The existing YOLOSR method combines the YOLOv8-small object detection model with a super- resolution branch to improve the detection of small weapons. This branch assists feature learning during training and is removed during inference to maintain computational efficiency. However, the performance of lightweight weapon detection under foggy conditions requires further evaluation. Therefore, this study evaluates YOLOv8-small for handgun and knife detection under clear and foggy surveillance conditions using an existing labelled dataset. The model will be trained and tested on surveillance images or recorded videos, and its performance will be assessed using precision, recall, F1- score, Average Precision, false detections, missed detections, and processing time. The study aims to analyse the effect of fog on weapon detection performance and identify the strengths and limitations of the model in reduced-visibility environments.

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Prompt Injection Attack Detection and Defense Framework for Production LLM Systems

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Abstract:
Large Language Models (LLMs) are now widely deployed in production applications such as chatbots, enterprise assistants, retrieval-augmented generation (RAG) pipelines and autonomous agents. This rapid adoption has exposed a serious security weakness known as prompt injection, where malicious instructions hidden in user input or in external content such as web pages, documents and emails override the intended behaviour of the model. Successful attacks can lead to leakage of system prompts, exposure of confidential data, unauthorized tool or API calls and the generation of harmful content. Existing safeguards such as static keyword filters and manual prompt hardening are easily bypassed by obfuscated, multilingual and multi-turn attacks. In this paper, we propose a layered Prompt Injection Detection and Defense Framework designed for production LLM systems. The framework combines input sanitization and normalization, a hybrid detector that uses rule-based pattern matching together with a fine-tuned transformer classifier, and semantic similarity analysis against known attack signatures. It also includes privilege separation between trusted instructions and untrusted content, and an output validation layer that blocks policy violations and data exfiltration before a response reaches the user. The framework is evaluated on direct and indirect injection benchmarks, and its performance is measured using accuracy, precision, recall, F1-score, false-positive rate and added latency. The results show that the proposed approach detects prompt injection attacks effectively while keeping overhead low enough for real-time deployment.

Keywords: Prompt Injection, Large Language Models, LLM Security, Jailbreak Detection, Retrieval-Augmented Generation, Adversarial Defense

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AI POWERED MULTILINGUAL VOICE BOT FOR RURAL GOVERNMENT SERVICE DELIVERY

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Abstract:
Access to government services remains a major challenge for rural communities due to language barriers, limited digital literacy, and the complexity of existing online portals. Citizens often face difficulties in identifying suitable government schemes, understanding eligibility criteria, and obtaining information about the required documents and application procedures. To address these challenges, this paper proposes an AI-Powered Multilingual Voice Bot for Rural Government Service Delivery that enables citizens to access government-related information through natural voice interactions in their preferred languages. The proposed system integrates Automatic Speech Recognition (ASR), language detection, intent and entity identification, Retrieval-Augmented Generation (RAG), machine translation, and Text-to-Speech (TTS) synthesis into a unified conversational framework. Users can ask questions using their regional languages without needing to know the exact names of government schemes. The system analyses the user's query, identifies the intended service, and retrieves relevant information from a structured knowledge base containing government schemes, eligibility conditions, required documents, and application procedures. The retrieved information is then used to generate simple, context-aware responses that can be delivered through both text and speech. A conversational session manager supports follow-up questions, while a relevance-based safety mechanism helps prevent unsupported responses when reliable information is unavailable. The proposed architecture is designed to support Tamil, English, Hindi, Telugu, and Kannada, with scope for future expansion to additional regional languages. By combining multilingual speech processing with retrieval-based information generation, the system aims to improve the accessibility, usability, and reliability of government service information for rural citizens. The evaluation framework considers speech recognition error rate, intent detection accuracy, retrieval precision and recall, response correctness, end-to-end response time, and user satisfaction. Quantitative evaluation and real-world user testing are planned as future work. This research presents a human-centered approach to inclusive digital governance by reducing language-related barriers and making government information more accessible through voice-based interaction.

Keywords: Multilingual Voice Bot, Artificial Intelligence, Speech Recognition, Retrieval-Augmented Generation, Intent Detection, Regional Languages, Rural Digital Governance, Government Service Accessibility.

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An AI-Assisted, Knowledge-Graph-Driven System for Prerequisite-Aware Curriculum Sequencing and Adaptive Learning

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Abstract:
Traditional educational systems often rely on predefined curriculum structures and fixed learning sequences, which may not adequately represent dependencies between concepts or address individual learners' knowledge gaps. Existing learning platforms also recommend educational resources without systematically verifying whether they cover the required curriculum concepts. To address these challenges, this paper proposes CurriculumMind AI, an AI-assisted, knowledge-graph-driven framework for automated curriculum knowledge graph construction, prerequisite-aware semester-wise sequencing, intelligent resource discovery, and adaptive learning path generation. The proposed system uses Large Language Models (LLMs) to extract curriculum information from syllabus documents and free-text learning objectives, transforming unstructured content into structured representations of courses, units, topics, concepts, and prerequisite relationships. These representations are organized into a knowledge graph using Neo4j, enabling the identification of conceptual dependencies and meaningful learning sequences. To improve reliability, the framework incorporates schema validation, cycle detection, redundant relationship identification, disconnected concept checks, and human-in-the-loop verification. The validated knowledge graph supports semester-wise and week-wise learning plans while considering prerequisite constraints and workload requirements. Furthermore, CurriculumMind AI integrates semantic embedding-based resource discovery to identify educational materials relevant to specific curriculum concepts. It evaluates concept-level resource coverage, detects missing learning materials, and performs targeted searches within a maximum of three iterations. Quiz-based mastery estimation helps identify learners' knowledge gaps and recommend remedial learning paths based on prerequisite dependencies. By connecting curriculum structure, resource coverage, and learner performance, the system provides a unified approach to personalized learning support. The implemented prototype has undergone component testing, integration testing, and end-to-end functional verification. A preliminary development-stage experiment using a six-unit Java programming syllabus generated 37 concepts and 25 prerequisite relationships, demonstrating the feasibility of automated curriculum representation while highlighting the importance of validation and human review. Quantitative accuracy measurements, comparisons with baseline approaches, and evaluations of learner outcomes remain future research directions.

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CARDIAC DISEASE PREDICTION USING AST + CONFORMER IN DEEP LEARNING

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ABSTRACT:

Cardiovascular diseases are one of the most common health issues today and they often produce abnormal heart sounds. Conditions like heart murmurs, Aortic Stenosis (AS) and Mitral Regurgitation (MR) can be identified through these sounds. Early detection of such abnormalities using Phonocardiogram (PCG) signals helps in non-invasive monitoring of heart health. In this work, PCG signals are collected from four different valve locations - Aortic Valve (AV), Mitral Valve (MV), Pulmonary Valve (PV) and Tricuspid Valve (TV). The raw recordings are pre-processed and converted into Mel-spectrograms to extract meaningful patterns from the heart sounds. The proposed system first identifies whether a murmur is present or not. For this, Audio Spectrogram Transformer (AST) and Conformer models are combined to capture both global and local features of the PCG signal. Apart from murmur detection, two separate modules are trained specifically to detect Aortic Stenosis (AS) and Mitral Regurgitation (MR) using labelled PCG data. All three modules are integrated into a single web-based platform. The user can upload the PCG recording and the system will provide the predicted result instantly. This approach offers an automated, non-invasive method for heart sound analysis.

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Design and Development of an Autonomous Mecanum-Wheeled Search and Rescue Robot for Disaster Response Applications

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Abstract

Disaster response operations often expose rescue personnel to hazardous environments, unstable structures, and areas that are difficult to access. This work presents the design and development of an autonomous Mecanum-wheeled search and rescue robot intended to support remote inspection and victim search in disaster scenarios. The robot uses four Mecanum wheels to provide omnidirectional movement, enabling forward, backward, lateral, and rotational manoeuvres in confined spaces. An ESP32 microcontroller coordinates the hardware, while an ESP32-CAM provides visual monitoring. Ultrasonic sensing supports obstacle detection and avoidance, and a YOLOv8-based vision module is intended for victim detection. Environmental monitoring is supported by DHT11 temperature-humidity sensing and an MQ-2 gas sensor, while an MPU6050 inertial measurement unit provides motion and orientation data. LoRa communication is used for remote data transmission. The proposed system integrates mobility, visual monitoring, environmental sensing, and remote supervision in a compact robotic platform. The prototype is intended to reduce direct human exposure during preliminary inspection and to assist rescue teams in assessing potentially dangerous locations. Further performance evaluation can quantify navigation, obstacle-avoidance, communication, and victim-detection effectiveness.
Keywords: Search and rescue robot; Mecanum wheels; Omnidirectional mobility; ESP32; YOLOv8; Obstacle avoidance; Disaster response; Environmental monitoring

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A Robust Deep Learning Ensemble System for Social Media Rumour Classification

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Abstract

Organizations need to monitor competitors, but relevant information is often scattered across public news, company websites, product announcements, hiring pages, and other permitted sources. Reviewing these signals separately can make it difficult to identify meaningful developments, understand their possible relevance to a particular organization, and keep earlier intelligence up to date. This paper presents the design of a Multi-Agent Corporate Intelligence and Competitive Monitoring Platform that coordinates specialized agents to collect and normalize public information, combine related signals, and produce evidence-linked intelligence summaries. The proposed workflow includes source tracking, entity resolution, event formation from related signals, mapping of detected events to defined internal business assets, and identification of previously generated conclusions that may require reassessment. Rather than regenerating every report after each new signal, the platform is designed to identify and re-evaluate only conclusions connected to the affected assets, while retaining source references and distinguishing observed facts from analytical interpretations. The system is intended to support human decision-making rather than make strategic decisions autonomously. The proposed design will be evaluated using test scenarios that assess event relevance, evidence traceability, update scope, processing effort, and the correctness of affected-conclusion identification. The platform uses publicly accessible and permitted data sources and is designed with source attribution and verification as core requirements.

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Confidence-Aware Occupancy Estimation Approach And Predictive Maintenance Framework For Classroom

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Abstract:

Accurate occupancy information is essential for energy-efficient classroom automation, yet low-cost door-level counters accumulate errors from simultaneous entries, hesitation at doorways, and repeated triggering, and these errors silently propagate into appliance control and attendance records. This paper presents KnoT, a low-cost smart classroom automation and control platform built around a sensor-consistency-based occupancy error detection framework. Unlike conventional approaches that either trust door-level counts or depend on costly dedicated occupancy sensing, KnoT cross-validates sequential entry and exit events against temperature and humidity trends, which serve as independent, low-cost physical evidence of occupancy change. A confidence-aware decision layer assigns a reliability score to each occupancy estimate and flags uncertain cases for review rather than correcting them automatically, which limits false corrections. The platform further links occupancy context with electrical current monitoring, so that appliance operation in an empty room is identified as energy waste and abnormal current behaviour is identified as a potential fault, supporting predictive maintenance. The system is implemented as an embedded controller that automates classroom fans and lights, together with a digital-twin-style web dashboard presenting real-time room status, a classroom comfort index, energy information, fault alerts, attendance records, and learning- support features. The framework was evaluated against conventional counting under controlled and real- classroom conditions, using manually verified occupancy as ground truth. An ablation across sensing configurations quantifies the contribution of each environmental signal. The findings indicate that low-cost environmental sensing can improve occupancy reliability and support energy-aware classroom automation without expensive infrastructure.

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INTELLIGENT AIR QUALITY MONITORING AND HEALTH ADVISORY PLATFORM

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Abstract:

Air pollution is a major environmental problem that affects human health, particularly children, pregnant women, senior citizens, and people with respiratory conditions. Existing air-quality monitoring systems mainly display pollution levels, but users may not always understand the health risks or precautions required. To address this issue, the proposed project develops an IoT-based Intelligent Air Quality Monitoring and Health Advisory Platform for real-time environmental monitoring and health awareness. The system uses an ESP32 connected to suitable air-quality, temperature, and humidity sensors to collect environmental data. The collected readings are processed and transmitted through Wi-Fi to a backend system for storage and visualization. The platform aims to classify air quality using the Air Quality Index (AQI) and display the results through a user-friendly web dashboard. Based on the detected pollution levels, it provides health advisories and alerts to help users take appropriate precautions. Historical data can also be used to identify pollution trends and support future AI-based prediction. The proposed system aims to improve air- quality awareness, encourage preventive actions, and support healthier environments in homes, educational institutions, and public spaces. Sensor calibration and testing will be important to ensure reliable readings and effective operation.

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AI And Edge-Computing Based Wearable System For Real-Time Fall Detection And Health Monitoring Of Elderly Individuals

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Abstract:

The present invention relates to an AI and edge-computing based wearable system for real-time fall detection and health monitoring of elderly individuals. The system comprises a waist-worn primary unit incorporating an inertial measurement sensor, a satellite positioning module,an on-device neural network classifier that classifies a rolling time-window of motion data into normal-activity, pre-fall, and fall-impact classes without dependency on continuous cloud connectivity for the core detection function, and a local audible alarm buzzer configured to sound immediately upon on-device confirmation of a fall event. A window-duration calibration methodology resamples benchmark training data so that the real-time duration of an inference window matches the deployment hardware's native sampling rate, enabling reliable transfer-learning based deployment onto low-cost embedded hardware. A multi-stage false-alarm suppression mechanism, combining a classifier confidence threshold with a requirement of agreement across consecutive inference windows, reduces false alerts arising from transient non-fall motion. A wrist-worn secondary unit continuously senses heart rate and blood oxygen saturation, displays the same to the wearer, and switches to a fall-confirmation screen upon receipt of a confirmed fall event. A sensor-fusion alert-dispatch logic corroborates a confirmed fall event with concurrently sensed physiological anomalies and satellite-positioning location data prior to transmitting a confidence-graded alert package to a remote server presenting the same on a caregiver dashboard. The combined system provides a low-cost, low-latency, low-false-alarm, and scalable solution for unattended fall detection and health monitoring of elderly individuals in domestic and assisted-living settings.

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Smart Campus Footfall Analysis and Predictive Modeling Using Data Analytics: A Case Study of BIT Campus

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Abstract:

Accurate campus footfall information is essential for effective utilization of institutional facilities, resource planning, crowd management, and the development of smart campus services. This paper presents a data-driven approach for analyzing and predicting footfall patterns at Bannari Amman Institute of Technology (BIT) Campus. The proposed framework collects and organizes campus footfall data and applies data analytics techniques to identify variations across time periods, days, locations, and activity patterns. Exploratory analysis is used to discover peak and low-footfall periods, recurring trends, and location-specific occupancy patterns. The processed data is further used to develop a predictive modeling framework for estimating future footfall and supporting proactive campus planning. The analytical results can assist administrators in optimizing common facilities, transportation, security, energy usage, and other campus resources according to expected demand. The approach also provides a structured dashboard-oriented view of historical and predicted footfall patterns, enabling data-supported decision making. The proposed methodology demonstrates how campus-generated footfall data can be transformed into actionable insights through data analytics and predictive modeling, contributing toward a smarter, more efficient, and responsive campus environment.

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Smart Donation Connect: An AI Based Platform for Reducing Waste and Connecting Donors with People in Need

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Abstract:

Millions of people in India are struggling to meet their daily needs even though there is plenty of surplus food, clothes, electronic devices and money available in the country. Every day, hotels, restaurants, event halls, businesses and households generate surplus food and other reusable items that are wasted. Meanwhile, orphanages, old-age homes, NGOs, poor families, and people living in remote villages are in dire need of resources. Although many wealthy individuals, business owners, and socially responsible citizens are willing to donate to charitable causes, they cannot find reliable beneficiaries since the needs of the underprivileged are hidden. Similarly,many middle-class families have clothes and household items that are rarely used but can dramatically improve the lives of others.

This project proposes Smart Donation Connect – an AI-enabled web application that connects verified donors with verified beneficiaries nationwide. The platform allows individuals, hotels,restaurants, businesses, and organizations to donate surplus food, clothing, electronic items, and monetary assistance, while orphanages, old-age homes, NGOs, educational institutions, and economically disadvantaged families can register their requirements through a secure verification process.

The proposed system features an AI-based food demand prediction model that analyzes historical sales data, seasonal trends, festivals, weather conditions, and customer behavior to estimate daily customer arrivals and surplus food quantities. This allows restaurants and hotels to optimize food preparation and reduce food wastage while distributing safe surplus food to nearby beneficiaries before it is discarded. The platform also features an AI-powered recommendation system that smartly matches donors with suitable recipients based on their location, urgency, donation type, and beneficiary requirements. Real-time notifications, transparent donation tracking, and verified user profiles foster trust among users. By connecting donors with people whose needs are often unseen, the proposed solution reduces waste, increases charitable engagement, and promotes sustainable development across the country.

Keywords: Artificial Intelligence; Web Development; Food Waste Prediction; Smart Donation Platform; Donation Matching; Resource Sharing; NGOs; Social Welfare; Real-Time Notifications; Sustainable Development.

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Adaptive Multi-Tenant Learning Management Ecosystem.

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Abstract:

Traditional Learning Management Systems deliver the same content to all learners, offering limited personalization, weak examination security, and little support for institutions that need separate and scalable platforms. This work presents the Adaptive Multi-Tenant Learning Management Ecosystem, an AI-powered platform that provides personalized, secure, and scalable digital learning for students and educational institutions. The system uses AI-driven cognitive profile assessment to evaluate individual abilities and learning patterns and to generate a profile for each learner. Based on these profiles, it provides personalized learning recommendations and adaptive content that match the needs of every learner. An automated AI-based proctoring module uses computer vision techniques to monitor online examinations and help detect potential malpractice. The multi-tenant architecture allows multiple educational institutions to use the same platform while maintaining separate data and administration for each institution. Analytics dashboards present useful insights into learner performance and learning progress for students, teachers, and administrators. The system is developed using React for the user interface, Node.js/Django for the backend, PostgreSQL for data storage, and Python with TensorFlow and OpenCV for the AI and computer vision components, deployed on cloud platforms such as AWS/Azure. Overall, the proposed ecosystem aims to improve learning quality, examination security, personalization, and scalability across educational institutions.

Keywords: Learning Management System, Multi-Tenant Architecture, Adaptive Learning, Cognitive Profiling, AI Proctoring, Computer Vision, Learning Analytics, Cloud Computing

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A Metaheuristic Guided Feature Optimization with Heterogeneous Ensemble Stacking framework for dimensionality reduction.

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Abstract:

Traditional feature selection algorithms and classifiers often face issues with premature convergence, which causes them to get stuck at local minima. This research introduces a novel framework, which processes input from UCI clinical CVD datasets through a three-stage approach. First, the data is pre-processed using KNN imputation to handle missing values, the Interquartile Range (IQR) technique to remove noise and outliers, and Z-score normalization. In the second stage, a population-based metaheuristic algorithm that incorporates swarm intelligence methods through the Binary Squirrel Search Algorithm, along with the Genetic Algorithm evolutionary operators, is employed to explore the complex binary feature space. This strategy optimizes both global and local search processes to identify the best subset of 7–10 crucial clinical factors, such as serum cholesterol, chest pain type, and maximum heart rate, from an initial set of 14-53 features. In the final stage, a heterogeneous stacking meta-classifier, comprising the Random Forest Bagging Method (RFBM) and an optimized SVM, is trained with these selected features. This framework demonstrates exceptional performance, achieving over 98.8% accuracy, more than 98.5% sensitivity, less than 1.2% false positive rate, and a zero false negative rate.

Keywords : KNN Imputation, Zscore, Meta heuristic algorithm, Genetic algorithm.

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Smart AI-Based Anaerobic Biorefinery for Organic Waste Valorization and Renewable Energy Generation in BIT Smart Campus

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Sathyamangalam, Erode, Tamilnadu, India-638401

Abstract:

Organic waste generated in university campuses is often landfilled or dumped, causing greenhouse gas emissions, odour nuisance and loss of valuable resources. This project proposes a Smart AI-Based Anaerobic Biorefinery for the BIT Smart Campus to convert organic waste into renewable energy and value-added products, promoting sustainable waste management. The study will first characterize and quantify the biodegradable organic waste generated across the campus. A smart anaerobic digestion system will then be designed and developed for efficient biogas production from mixed organic waste. Artificial intelligence-based monitoring and control will be integrated into the system to optimize key process parameters, including pH, temperature, organic loading rate and gas yield. Process optimization and microbial management strategies will be applied to maximize methane production and maintain stable digester performance. The digestate will be converted into biofertilizer for campus landscaping and agricultural applications, closing the nutrient loop. Finally, a scalable waste-to-energy model will be established to support circular
economy practices and carbon reduction initiatives within the campus. The proposed biorefinery is expected to improve waste management efficiency, supply clean energy, reduce dependence on fossil fuels and lower greenhouse gas emissions, while serving as a replicable demonstration for smart and sustainable campuses.

Keywords: Anaerobic digestion; Artificial intelligence; Biogas; Biofertilizer; Circular economy; Smart campus

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Smart Machine Health Clip – An Edge AI-Based Predictive Maintenance System

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Abstract:

The Smart Machine Health Clip is a compact and affordable system designed to monitor the operating condition of industrial machines and detect possible faults at an early stage. The system combines machine-condition sensing, an ESP32 controller, Edge AI, and IoT communication in a portable monitoring unit. It measures three important parameters: vibration, temperature, and current. An ADXL345 sensor detects vibration, a DS18B20 sensor measures temperature, and an INA219 sensor monitors current. The sensor data is collected and processed by the ESP32 microcontroller. A reference condition is established during normal machine operation, and subsequent readings are analysed to identify unusual changes in machine behaviour. Based on the analysis, the machine condition is classified into three levels: Normal, Warning, and Critical. An OLED display shows the machine health status, while an RGB LED and buzzer provide visual and audible alerts during abnormal conditions. The ESP32 also uses Wi-Fi connectivity to transfer machine information and alerts to an IoT-based mobile or web dashboard for remote monitoring. The proposed system provides a simple, low-cost, and easy-to-install solution for predictive maintenance. It can be applied to motors, pumps, compressors, generators, HVAC systems, manufacturing equipment, and other rotating machines. By detecting abnormal behaviour early, the system can support timely maintenance and reduce unexpected breakdowns and downtime.

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Multi-Hospital Shared ICU Beds Resource Allocation Platform with Real-Time Patient Severity Indexing.

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Abstract:

Timely access to intensive care unit (ICU) beds is critical for emergency and accident patients, yet hospitals still coordinate bed availability and transfers largely through manual calls, causing delays when the nearest hospital lacks a suitable ICU resource. This paper proposes a centralized web-based platform that connects ambulances and multiple hospitals for real-time ICU bed monitoring, patient severity assessment and transfer coordination. Beds across connected hospitals are tracked as available, occupied, reserved or unavailable and updated live through WebSocket communication. Ambulance crews register emergency patients, with a temporary ID generated for unidentified accident victims so that care is not delayed by missing personal details. Vital signs, namely SpO_2, heart rate, blood pressure, respiratory rate, temperature and level of consciousness, are retrieved from compatible medical devices over Bluetooth/BLE or USB, or entered manually, and are used to generate a clinically reviewed severity index and emergency priority. Using this priority, the required facilities and bed availability, the platform identifies potentially suitable ICU beds, searches other connected hospitals when the current hospital cannot accommodate the patient, sends pre-arrival alerts to the receiving hospital and supports transfer requests. Insurance, government scheme and payment information is kept separate from clinical severity and priority. The system is built using React.js, Node.js, Express.js and MongoDB. The proposed platform aims to improve ICU bed visibility, reduce manual coordination and support faster critical-care decisions, while serving as a decision-support tool that complements, rather than replaces, clinical judgement.

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Bare-Metal STM32 Multi-Protocol Machine Condition Monitor with DMA Support

YUGASHREE K N SWARNYA R

Abstract:

Embedded systems play an important role in monitoring physical parameters and processing sensor data in real time. This project focuses on the development of a bare-metal STM32-based monitoring system that integrates sensor interfacing, multiple communication protocols, and Direct Memory Access (DMA) for efficient data handling. The proposed system aims to monitor parameters such as temperature and vibration using suitable sensors connected to an STM32 microcontroller. Bare-metal programming is considered to provide direct control over the microcontroller peripherals without depending on a high-level operating system.The system is planned to incorporate communication interfaces such as Universal Asynchronous Receiver-Transmitter (UART), Inter-Integrated Circuit (I2C), and Serial Peripheral Interface (SPI), depending on the requirements of the selected sensors and peripherals. DMA support is intended to facilitate data transfers with reduced CPU involvement, thereby improving the efficiency of communication-related operations. The collected sensor data is planned to be processed and evaluated against predefined threshold values to identify variations in the monitored parameters.

The primary objective of this project is to explore the integration of bare-metal programming, sensor interfacing, peripheral communication, and DMA-based data transfer within a single embedded system. The project is currently in the development stage, and further work will involve hardware integration, firmware development, functional verification, and performance evaluation. The implementation aims to provide practical understanding of STM32 peripheral control and efficient data handling in embedded applications.

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AI-BASED CUFFLESS BLOOD PRESSURE ESTIMATION USING PPG SIGNALS WITH RESNET-50 AND CROSS-ATTENTION

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Abstract:
Blood pressure is usually checked with an inflatable cuff. This is a common way of measuring blood pressure, but using the cuff many times can be uncomfortable. Because of this, cuffless methods for blood pressure estimation are being studied. PPG signals are useful for this purpose because they contain information about changes in blood flow. Artificial Intelligence can be used to study this information and estimate blood pressure. Our study of existing methods shows that some systems use more than one physiological signal. Some also collect signals from different parts of the body. These requirements can make the system more difficult to use. Based on our literature study, we identified the need for a method that uses only one PPG signal for estimating Systolic Blood Pressure (SBP) and Diastolic Blood Pressure (DBP). In our system, the raw PPG signal is first passed through a Butterworth filter to remove unwanted noise. PPG, vPPG and aPPG are then obtained from the filtered signal. These three signal forms are converted into images. The images are given to ResNet-50 to obtain the required features. Multi-Head Cross-Attention is then used to combine the features, and a fully connected layer is used to estimate the SBP and DBP values. The system provides a software-based and non-invasive approach to cuffless blood pressure estimation. It uses different forms of a single PPG signal instead of depending on signals from different body locations. The proposed approach can be further studied for comfortable and continuous blood pressure monitoring.

Keywords: Cuffless Blood Pressure Estimation; Photoplethysmography (PPG); ResNet-50; Deep Learning; Cross-Attention; SBP; DBP; Signal Processing.

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Smart Assistive Device for Converting Sign Language to Text and Speech

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Abstract:
Communication barriers between deaf or hard-of-hearing individuals and non-signers remain a significant accessibility challenge. This project presents ZSign Detection, a real-time sign language recognition system built on the YOLOv5 object detection architecture. The system detects hand gestures such as Hello, Yes, No, Thanks, Please, and IloveYou from webcam streams or static images, and translates them into bounding boxes with text labels and confidence scores. A custom dataset was collected with OpenCV under varied angles, lighting conditions, and distances, then annotated in YOLO format using labelImg. A pre-trained YOLOv5s model was fine-tuned on this dataset through transfer learning, with data augmentation (mosaic, random flips, and scaling) to improve generalization. Training was run in Google Colab and Jupyter Notebook and monitored with TensorBoard. To improve reliability during inference, the system applies Non-Maximum Suppression, class-agnostic filtering, a confidence threshold of 0.80, and aspect-ratio checks that reduce false positives, such as faces being mistaken for the IloveYou gesture. The resulting system runs on live video with low latency, using PyTorch, OpenCV, NumPy, and PyYAML. It has potential applications in assistive technology, healthcare communication, education, public service kiosks, video conferencing captioning, and touchless human-computer interaction.

Keywords: Sign Language Recognition, YOLOv5, Computer Vision, Deep Learning, Object Detection, Real-Time Inference, Transfer Learning, OpenCV, PyTorch

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SafeGenX: A Constraint-Guided Generative AI Framework for Automated Multilingual Digital Safety Awareness Content for Children

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Abstract:
Manual creation of children& digital-safety education material does not scale across languages and age groups, and generic large language model (LLM) content pipelines are optimized for fluency rather than for the two constraints that matter most for child- directed material: age-appropriate readability and harm-safety. This paper presents SafeGenX, a constraint-guided generation framework that models multilingual safety-content production as a two-phase pipeline — an age-band-conditioned drafting phase that shapes vocabulary and narrative structure to a target reading level, and a post-generation verification phase that applies a harm/toxicity classifier, a readability re-scorer, and back- translation fidelity checking before content is released for human sign-off. We benchmark SafeGenX against three baselines — direct machine translation of generic (non-child-specific) content, age-conditioned prompting without a safety filter, and a safety filter applied to unconditioned generation — on a controlled, simulated content-generation pipeline at batch sizes of 50, 150 and 300 items, spanning three age bands and six target languages. Across all three scales, SafeGenX reduces the harmful-content rate by 84.1% to 89.4% relative to the two unfiltered baselines, but by only 18.8% to 27.6% relative to the safety-filter-only baseline — indicating that the post-generation classifier, not the age-conditioned prompt, is the dominant lever for harm reduction. Age-band prompting shows the mirror-image pattern: it accounts for the large majority of the reduction in reading-level mismatch, while the safety filter alone contributes comparatively little to age- fit. We describe the mathematical formulation of the age-fit, safety and translation-fidelity scores, the experimental protocol, and outline how learned toxicity classifiers and reinforcement-learning-based prompt adaptation could extend the framework. Because the underlying generation runs were simulated for controlled, reproducible comparison rather than drawn from a live production LLM deployment, we present this as a specific, falsifiable framework and evaluation protocol suitable for a course research project, and not as a claim of deployed-system performance.

Keywords: Generative AI, Large Language Models, Child Online Safety, Multilingual NLP, Content Moderation, Readability-Controlled Generation, Digital Literacy, Localization

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Design and Development of a DL-Based System for Enhanced Brain Tumor Detection Using Multimodal MRI

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Abstract:

Brain tumor detection is a critical task in medical imaging, where early and accurate diagnosis significantly improves patient survival rates. This research work presents the design and development of a deep learning–based system for enhanced brain tumor detection using multimodal MRI images. By integrating multiple MRI modalities such as T1, T2, and FLAIR, the proposed system captures comprehensive tumor characteristics. DL are employed to automatically extract discriminative features and classify tumor regions with high accuracy. The system aims to assist radiologists by reducing manual effort, minimizing diagnostic errors, and enabling faster and more reliable tumor identification.

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CareerCompass AI: Intelligent Skill Demand Aggregation andCareer Path Recommendation

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Abstract:
Students often find it difficult to connect their existing skills with suitable career roles and identify the knowledge needed to prepare for those roles. This paper presents CareerCompass AI, a system designed to combine machine-learning-based career-role recommendation with an AI-assisted learning roadmap. The career prediction module preprocesses student-entered skill keywords and uses Term Frequency–Inverse Document Frequency (TF-IDF) features to classify profiles into 11 job-role categories. On a dataset of 440 records, Logistic Regression achieved a mean accuracy of 78.86% (±1.54%) using stratified five-fold cross-validation. The module returns a predicted role with a confidence score, provides up to three alternative roles when confidence is below 40%, and identifies matched skills, missing skills, and skill coverage. The learning-roadmap module supports AI-generated prerequisite-linked knowledge graphs, learning resources, practice questions, answer grading, and progress-based next-step recommendations. Together, these functions are intended to help students move from career exploration to structured skill development. The current predictor is limited to the roles represented in its dataset and does not use live labour-market data. Unified end-to-end integration and validation remain ongoing work.

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Fabrication and Thermal–Mechanical Characterization of MgO and Nano- SiO2 Reinforced PET Composite for 3D Printing Applications

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Abstract:
Polymer composites have gained significant attention in additive manufacturing due to their potential to improve the thermal and mechanical properties of conventional polymers. This study aims to fabricate and evaluate a magnesium oxide (MgO) and nano-silicon dioxide (Nano-SiO2) reinforced polyethylene terephthalate (PET) composite for 3D printing applications. PET is selected as the polymer matrix, while MgO and Nano-SiO2 are incorporated as ceramic fillers to investigate their influence on the thermal and mechanical performance of the composite. Different weight percentages of MgO and Nano-SiO2 will be mixed with PET using a suitable blending technique. The prepared composite will be processed into filament suitable for fused deposition modelling (FDM) 3D printing, subject to successful filament fabrication and diameter control. The printed specimens will be evaluated through thermal analysis, hardness testing, and mechanical tests to determine the effects of filler concentration. The properties of the composite specimens will be compared with those of pure PET to assess the contribution of the ceramic fillers. This study is expected to provide insights into the development of PET-based hybrid composites with improved thermal stability and suitable mechanical properties for 3D printing applications.

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Smart Internship Placement and Geofenced Attendance Management for Local Industries

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Abstract:
Internship opportunities are essential for developing practical skills and connecting students with relevant industries, yet conventional internship placement and attendance processes often depend on manual coordination, scattered information, and basic attendance records. This paper presents a smart internship placement and geofenced attendance management platform designed to connect students with suitable local industries while improving the reliability of internship attendance verification. The proposed system provides a centralized platform through which students can view and apply for relevant internship opportunities, while organizations can manage internship requirements and student participation. A geofencing-based attendance mechanism verifies whether a student is physically present within the permitted internship location before recording attendance, reducing the possibility of false or proxy attendance. The platform further maintains internship, placement, and attendance information in a structured digital environment, enabling easier monitoring by students, industries, and academic coordinators. The system is intended to support localized industry participation and simplify the internship management process through location-aware digital verification. The proposed approach can improve placement coordination, attendance transparency, and industry–student connectivity while providing a scalable foundation for efficient internship management.

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Development and Characterization of Recycled PET–Al2O3–Graphite Reinforced Composite

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Abstract:

This study focuses on developing a sustainable composite material using recycled polyethylene terephthalate (PET) reinforced with aluminium oxide (Al2O3) and graphite for potential automotive applications. Recycled PET is used as the matrix to reduce plastic waste, while aluminium oxide improves hardness, stiffness, and wear resistance. Graphite is added to enhance thermal conductivity and tribological performance.The composite will be fabricated using melt compounding and extrusion to produce material suitable for Fused Deposition Modelling (FDM) 3D printing. Mechanical and tribological properties will be evaluated through tensile, flexural, hardness, and wear tests. Scanning Electron Microscopy (SEM) will be used to study filler distribution, surface morphology, and interfacial bonding. The results will help identify a suitable filler composition with improved mechanical performance, thermal properties, and printability. The developed composite may be used in lightweight, non-structural automotive components, promoting sustainable material development and effective plastic waste utilization.

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AI-Based Dynamic Pricing and Revenue Management for Eco-Tourism Hospitality

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Abstract:

Eco-tourism lodges operate with few rooms and demand that changes with season, weekends, festivals and local events, yet many still use fixed or manually adjusted room rates. This paper presents an AI-based dynamic pricing and revenue management prototype that forecasts room demand and recommends a nightly price for each date. Because real eco- tourism booking data was unavailable, three years of daily bookings for a 20-room lodge were generated from assumed demand rules. A seasonal baseline, Random Forest, price-monotonic Gradient Boosting and a smooth demand-curve model were compared on a held-out test year. The demand-curve model performed best (MAE 1.18, RMSE 1.86, R2 0.793, against R2 0.205 for the baseline) and was paired with an optimizer that selects the revenue-maximizing rate between ₹2,000 and ₹9,000. In simulation, dynamic pricing earned about 7.9% more revenue than a fixed rate. The system is delivered as a Flask and MySQL web application with an owner dashboard and a public customer price page. As the data is synthetic, the results show feasibility rather than real-world performance.

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Smart Fintech Skill Tracking and Automated Micro-Credential Management System

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Abstract:

The rapid evolution of financial technology requires students and professionals to continuously develop and demonstrate relevant technical and industry-oriented skills, yet conventional skill tracking and certificate management approaches are often fragmented, manual, and difficult to maintain. This paper presents a Smart Fintech Skill Tracking and Automated Micro-Credential Management System designed to organize skill development and digital credential records within a unified platform. The proposed system maintains individual skill profiles by tracking completed courses, assessments, projects, certifications, and micro-credentials, while mapping these achievements to relevant fintech skill categories. An automated micro-credential management layer records credential status, validates achievement information, and provides a structured digital representation of the learner's competencies. The platform further supports skill-gap identification by comparing existing competencies with target fintech roles, enabling users to identify areas requiring additional learning. A dashboard provides an organized view of skill progress, credential history, completion status, and recommended development areas. The proposed approach reduces manual record maintenance, improves credential visibility, and supports continuous skill development through data-driven tracking. The system is intended to provide a scalable and efficient mechanism for managing fintech-related skills and micro-credentials while helping learners demonstrate their competency readiness for emerging financial technology roles.

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AI-Based Predictive Maintenance for Industrial Motors Using IoT Sensor Data and Random Forest Classification

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Abstract:
Industrial motors are widely used in manufacturing, processing, and automation systems, where unexpected failures can cause production interruptions, increased maintenance costs, and reduced equipment availability. Conventional reactive and time-based preventive maintenance approaches may not identify developing faults early enough to avoid unplanned downtime. This work presents an Artificial Intelligence (AI)-based predictive maintenance approach that combines motor-condition data, machine learning, and dashboard-based monitoring to support timely maintenance decisions. The proposed system considers key motor operating parameters, including vibration, temperature, current, voltage, and rotational speed. The collected or dataset-based measurements are preprocessed, and relevant statistical features can be extracted to represent changes in motor behaviour. A Random Forest classification model is trained using labelled motor-condition data to classify operating conditions and identify patterns associated with abnormal behaviour. The model output is intended to be integrated with a monitoring dashboard that displays input measurements, predicted condition categories, and alerts to help users interpret the motor's status. The dashboard provides a user-friendly interface for viewing model predictions and supporting maintenance planning. The proposed approach aims to improve the visibility of motor operating conditions, assist in early identification of potential abnormalities, and reduce dependence on purely reactive maintenance. Its performance should be assessed using suitable classification metrics, including accuracy, precision, recall, and F1-score, together with tests on representative motor data. Where a physical prototype is used, sensor readings and model predictions can be compared with observed operating conditions to evaluate practical feasibility. The system is intended as a cost-conscious framework for integrating AI and IoT concepts into industrial motor monitoring. Actual performance and fault-detection capability depend on the quality and representativeness of the training data and require experimental validation before deployment in safety-critical industrial environments.

Keywords: Artificial Intelligence, Predictive Maintenance, Industrial Motors, Internet of Things (IoT), Random Forest, Condition Monitoring, Machine Learning.

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AI-Based Career Intelligence: A Personalized Framework for Competency Gap Detection and Job-Role Alignment

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ABSTRACT:

The growing mismatch between academic learning outcomes and evolving industry skill requirements has created a significant challenge in assessing student employability. Conventional student evaluation methods primarily rely on academic performance, self-reported skills, and isolated assessments, often overlooking practical exposure and the dynamic skill requirements of specific job roles. To address these limitations, this paper proposes an IoT-Enabled AI Framework for Student Skill Gap Detection and Personalized Career Readiness. The framework integrates Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and the Internet of Things (IoT) to construct a comprehensive, role-oriented student skill profile. The proposed framework collects heterogeneous student data, including academic records, resumes, project descriptions, certifications, assessment scores, and IoT- based practical activity records. NLP techniques are employed to extract and categorize skills from unstructured textual information, while assessment results and project evidence contribute to estimating skill proficiency. An ESP32-based IoT module integrated with RFID technology can record participation in practical learning activities, providing supplementary evidence of hands-on exposure rather than treating participation alone as proof of competency. The resulting skill profile is compared against a structured industry skill database containing job-specific competencies and required proficiency levels. A weighted skill-matching mechanism identifies missing and underdeveloped competencies, prioritizes critical skill gaps, and generates a role-specific skill-match score. Based on these findings, a recommendation module proposes suitable courses, certifications, practical projects, and progressive learning activities. Unlike approaches that depend exclusively on academic scores or resume-based screening, the proposed framework combines textual skill extraction, assessment- based proficiency estimation, and IoT-supported practical activity tracking within a unified architecture. The system is designed to present interpretable skill-gap results through a dashboard, enabling students to understand their competency gaps and supporting educators in planning targeted skill-development activities. The framework is intended to promote data-informed career preparation and strengthen alignment between academic development and industry expectations. Its effectiveness can be evaluated through skill-extraction accuracy, proficiency- estimation performance, recommendation relevance, and usability testing. The proposed approach offers a foundation for scalable, personalized, and evidence- supported student skill development.

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Intelligent Helpdesk Ticketing System with Automated Root-Cause Category Mapping

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Abstract:
College helpdesk services often receive complaints related to network connectivity, biometric systems, student portals, attendance, examinations, and academic projects. When these requests are handled manually, tickets may be sent to the wrong department, require repeated clarification, or take longer to resolve. This paper proposes an Intelligent Helpdesk Ticketing System with Automated Root-Cause Category Mapping to organize complaint submission, classification, and department-level handling through a centralized digital platform. Students can submit a ticket describing their issue, while a text-classification component is intended to identify the likely category and recommend the responsible department based on the complaint details. The initial department mapping covers NMC for network and biometric issues, PS for PS Portal, PCDP App, and attendance issues, COE for examination-related issues, and Office of Academics for mini- project and project-related matters. Each department can access and update only the tickets assigned to it, while an administrator can monitor tickets across departments and review their status. The proposed system aims to reduce incorrect routing, improve ticket visibility, and support more consistent complaint tracking. Its effectiveness will depend on the quality of the complaint data, the accuracy of category mapping, and evaluation through practical testing. The platform provides a foundation for improving campus support workflows through structured ticket management and automated classification.

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VeriSelf: Neutralizing AI Threats to Personal Privacy

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Abstract:
Personal photographs shared online can be copied, misused for impersonation, or altered without the owner permission, leaving individuals with limited ways to investigate suspicious content and organize supporting evidence. VeriSelf is a full-stack web application developed to support personal digital identity protection through image analysis, identity verification, public-web monitoring, and evidence management. The system uses React and Vite to provide the user interface, FastAPI to handle backend operations, and SQLAlchemy with SQLite to manage application records. For identity verification, it employs MTCNN for face detection and an InceptionResnetV1 model pretrained on VGGFace2 to generate facial embeddings for comparison. Perceptual hashing helps identify visually similar images, while image and video analysis modules examine characteristics such as noise inconsistency, edge density, texture variation, and colour-channel differences to identify potential manipulation indicators. The monitoring module scans a user-specified public webpage for candidate images, and the evidence module organizes investigation findings for further review. Additional functions support photograph protection, risk assessment, alerts, and the preparation of draft legal-response notices. By bringing these capabilities together, VeriSelf provides a structured workflow for examining suspected digital identity misuse and documenting relevant findings. Its results require human review because facial similarity does not guarantee identity, perceptual matching cannot establish ownership, and heuristic forensic indicators alone cannot confirm image or video manipulation.

Key words: Digital Identity Protection, Face Detection, Facial Embeddings, Perceptual Hashing, Public-Web Monitoring, Image Forensics, Video Analysis, Risk Assessment, Digital Evidence Management

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Student resolution platform

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Abstract:
The Student Query Resolution Platform is a Generative AI-based system designed to provide students with quick, accurate, and reliable answers to common institutional queries related to admissions, academics, examinations, fees, and hostel information. In the existing process, students often have to contact administrative staff or faculty through email or phone for repetitive questions such as fee details and examination schedules. This can increase response time and administrative workload. Conventional LLM-based chatbots may also generate incorrect institutional information when they do not have access to the institution's official documents. The proposed platform addresses these issues by using Retrieval-Augmented Generation (RAG) to ground responses in verified institutional information. The system first collects institutional source documents such as student handbooks, FAQs, syllabi, examination schedules, fee structures, and past query logs. These documents are divided into smaller chunks of approximately 300–500 tokens and converted into vector embeddings using sentence-transformers. The embeddings are stored in a vector database such as ChromaDB or FAISS, allowing the system to perform semantic similarity searches when a student submits a query. When a student enters a question through the chat interface, the backend receives and processes the request. A query classifier determines whether the question is within the platform's knowledge scope. For an in-scope query, the system performs vector-based retrieval and selects the most relevant document chunks. These retrieved passages are then provided as context to a Large Language Model (LLM), which generates a natural- language answer based only on the retrieved institutional information. The generated response is returned to the student together with source citations, helping the student identify the information used to produce the answer. For queries that are out of scope, complex, or associated with low retrieval confidence, the platform does not provide a speculative answer. Instead, the query is converted into a ticket and escalated to a human/faculty dashboard, where the appropriate staff member can resolve it. The resolved information can subsequently be incorporated into the knowledge base. A feedback mechanism using thumbs-up/thumbs- down responses is also included to help analyze answer quality and improve future retrieval. The proposed platform uses FastAPI for backend services, React or Streamlit for the user interface, sentence-transformers for embeddings, ChromaDB/FAISS for vector storage, and an LLM such as Claude/GPT, with Llama or Mistral through Ollama as local alternatives. PostgreSQL or MongoDB can be used to store users, query logs, and feedback. JWT-based authentication can restrict access to enrolled students. Overall, the Student Query Resolution Platform provides an integrated approach for automated student support using RAG and Generative AI while maintaining a connection to official institutional information.

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An Adaptive Retinal State Evolution Framework for Intelligent Disease Progression Analysis Using Digital Twin

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Abstract:
Diabetic retinopathy is one of the major complications of diabetes that can affect a person's vision if it is not detected and monitored at an early stage. With the development of artificial intelligence and deep learning, retinal fundus images can be analysed to identify different stages of diabetic retinopathy. However, many existing systems mainly focus on analysing individual images and do not effectively represent the changes in a patient's retinal condition over time.This project proposes ARTEMIS-DT, an AI-based framework for longitudinal diabetic retinopathy monitoring using a patient-specific retinal Digital Twin. The system combines fundus image preprocessing, retinal feature extraction, EfficientNet-B0 based deep learning, and clinical information to estimate the current retinal condition. The proposed Adaptive Retinal Twin Update (ARTU) mechanism is designed to update the patient's retinal state when new observations become available. The system also incorporates explainable AI techniques to provide a better understanding of the prediction. The overall aim of the project is to develop a framework that can support continuous monitoring of diabetic retinopathy and provide useful information about changes in retinal condition over multiple visits.

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CORPORATE SUSTAINABLE PROCUREMENT AUDITOR WITH AUTOMATED VENDOR ESG COMPLIANCE VERIFICATION

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Abstract:

Organizations increasingly need to evaluate the environmental, social, and governance (ESG) performance of vendors before making sustainable procurement decisions. However, manually reviewing vendor documents can be time-consuming and may lead to inconsistent assessments or overlooked discrepancies. This paper presents a Corporate Sustainable Procurement Auditor with Automated Vendor ESG Compliance Verification, a system designed to assist in reviewing vendor ESG documents and supporting procurement decisions. The proposed system accepts vendor documents and applies optical character recognition (OCR) to extract relevant information. The extracted data is checked against defined ESG compliance rules, while document details and available evidence are examined for potential inconsistencies. The system combines these checks to produce a compliance score and a verification summary that helps users identify documents requiring further review. A dashboard presents the uploaded documents, verification results, and vendor compliance status in a structured format. By reducing repetitive manual checking and providing a consistent review process, the proposed system aims to improve transparency, efficiency, and traceability in sustainable procurement. The system is intended to support human decision-making; flagged documents and scores should be reviewed against the original evidence before final procurement decisions are made.

Keywords: Sustainable procurement, vendor assessment, ESG compliance, optical character recognition, automated verification, compliance scoring

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Design and Development of PET/CaCO3/Nano-SiO2 Composite for AutomotiveTrim Components

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Abstract:
The automotive industry increasingly demands lightweight, durable, and cost- effective materials with improved mechanical and thermal properties. This research focuses on the development of a hybrid polymer composite using polyethylene terephthalate (PET) as the matrix, calcium carbonate (CaCO3) as a mineral filler, and nano-silicon dioxide (Nano-SiO2) as a reinforcing additive for potential automotive interior trim applications. Different filler compositions will be investigated to understand their influence on the mechanical strength, hardness, and thermal properties of the composite. The specimens will be fabricated using a suitable blending and compounding technique and evaluated through appropriate experimental tests. The properties of the developed composites will be compared with those of unfilled PET to assess the effects of the fillers. The study aims to identify a suitable composition that balances mechanical performance, thermal stability, durability, and material cost. The proposed composite could offer potential for selected automotive interior trim components, subject to experimental validation.

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HACKFORGE AI: AN AI-POWERED HACKATHON PROBLEM GENERATION AND EVALUATION PLATFORM USING MACHINE LEARNING

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Abstract:

Hackathons encourage innovation and collaborative problem-solving; however, creating suitable problem statements and evaluating participant submissions consistently can be challenging. This paper proposes Hackforge AI, an integrated platform designed to support both hackathon problem generation and participant solution evaluation using generative artificial intelligence, machine learning, and natural language processing. The Generator module produces customized problem statements based on organizer-defined constraints, including application domain, technology, difficulty level, duration, and available resources. It incorporates adaptive difficulty selection, resource-aware generation, semantic novelty checking, and problem-specific evaluation rubric generation. The Evaluator module analyzes participant submissions using requirement coverage analysis, semantic similarity, innovation indicators, feasibility assessment, and rubric-based scoring. Sentence embeddings and cosine similarity are employed to estimate semantic alignment between problem statements and proposed solutions, while rule-based processing and machine learning techniques support structured assessment. The platform also provides explainable evaluation results and automated feedback to help participants identify strengths and areas for improvement. By integrating problem generation and evaluation within a unified workflow, Hackforge AI aims to improve consistency, transparency, and efficiency in hackathon management. The proposed system is intended to serve as a decision-support tool for organizers and evaluators rather than replace human judgment. Its effectiveness will be assessed through functional testing and evaluation using representative problem statements and project submissions.

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Smart Wearable ECG Monitoring and Arrhythmia Detection System Using IoT and Machine Learning

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Abstract:
Cardiovascular diseases are among the leading causes of mortality worldwide, making early detection and continuous monitoring of cardiac abnormalities essential. This project presents a Smart Wearable ECG Monitoring and Arrhythmia Detection System using the Internet of Things (IoT) and Machine Learning (ML) to support real-time cardiac monitoring. The proposed system uses three disposable ECG electrodes to acquire the heart's electrical activity. An AD8232 ECG sensor module conditions the acquired signal, and an ESP32 microcontroller processes the signal for heart-rate estimation and wireless data transmission. The system integrates with the Blynk IoT platform to display ECG waveforms and heart rate in real time, enabling remote monitoring. For arrhythmia detection, ECG signal characteristics, including R-peaks, RR intervals, and QRS complex features, are considered for machine-learning- based classification. The machine-learning model is intended to identify potential irregularities in cardiac rhythm and generate alerts through visual indicators, an audible buzzer, and IoT notifications. The proposed system aims to provide a cost-effective, portable, and accessible approach to continuous ECG monitoring, particularly for applications requiring remote observation. Further testing and validation are required to assess the system's accuracy and reliability under practical conditions.

Keywords: Electrocardiogram (ECG), Arrhythmia Detection, Internet of Things (IoT), Machine Learning, ESP32, AD8232, Blynk, Remote Patient Monitoring.

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AI-Powered Source Code Analysis and Vulnerability Detection System

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Abstract:
Software vulnerabilities such as SQL injection, cross-site scripting, hardcoded credentials and insecure deserialization continue to reach production code, particularly in projects built by students and small teams without a dedicated security review. Rule-based static analysis tools detect known patterns precisely but cannot reason about context, raise many false positives and offer little explanation, while large language models (LLMs) reason about intent but can produce inconsistent or unsupported findings when used alone. This paper proposes an AI-powered source code analysis and vulnerability detection system that combines both approaches for Python and JavaScript code. Submitted code is first scanned by static analyzers, including Bandit for Python and security rules for JavaScript, to produce candidate findings mapped to CWE and OWASP categories. An LLM pipeline built with LangChain and the Groq API then reviews the code together with these findings, explains each issue in plain language, assigns a severity and suggests a remediation. Results are ranked and shown as inline annotations in a browser-based Monaco editor, with scan history stored in PostgreSQL through a FastAPI backend. Anchoring the LLM to concrete static evidence is intended to reduce unsupported findings while retaining contextual reasoning. The system is under development; its evaluation will compare static-only, LLM-only and hybrid detection on labelled vulnerable code samples using precision, recall and F1-score.

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CGA: An AI-Assisted Mobile Application for Personalized Career Guidance and Skill Development

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Abstract:
Students often find it difficult to identify suitable career paths, understand the skills required for their goals, and organise their learning activities into a structured plan. This work presents CGA, an AI-assisted Android mobile application that brings career assessment, personalized career recommendations, learning roadmaps, practice activities, and progress monitoring together in a single platform. The application is developed for Android using Kotlin and an MVVM-based architecture, with Supabase providing backend services and authentication. The career assessment supports personalized career recommendations, while structured learning roadmaps help students identify the relevant skills and learning steps for a chosen career. Practice activities and progress monitoring enable learners to follow their development over time. CGA also incorporates an AI learning companion intended to support learning from educational materials, including PDFs, presentations, and video transcripts, through source-grounded tutoring and adaptive assessments. These capabilities are described as part of the intended system design, and no experimental or user-study results are reported in this abstract. By combining career guidance, structured planning, and practice within one application, CGA has the potential to support personalized career planning and continuous skill development for students.

Keywords: Career Guidance, Artificial Intelligence, Personalized Career Recommendations, Android Application, Learning Roadmap, Adaptive Assessment, AI Learning Companion.

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Design of an Automated Liquid Blending Station Utilizing a Synchronized Continuous-Motion Gantry Dispenser

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Abstract:

Conventional industrial liquid blending systems often experience production bottlenecks due to intermittent conveyor operations, where containers must stop during the dispensing process. Additionally, the use of individual flow sensors for multiple liquid reservoirs increases instrumentation costs and system complexity. This paper presents the design of a Programmable Logic Controller (PLC)-based automated liquid blending station that addresses these limitations through an integrated fluid control and motion synchronization architecture. To reduce hardware requirements, a shared manifold configuration with time-multiplexed flow measurement is employed, allowing multiple liquid reservoirs to utilize a single pulse-output flow sensor through electrically controlled solenoid valves. Furthermore, a synchronized continuous-motion gantry dispensing mechanism is proposed to enable liquid filling without interrupting conveyor movement. The gantry dynamically tracks the moving container by synchronizing its linear motion with the conveyor velocity, enabling dispensing during continuous transportation. The proposed system integrates automated liquid selection, flow- based volumetric measurement, conveyor control, and synchronized gantry motion within a unified PLC-controlled architecture. This approach aims to reduce conveyor idle time, optimize instrumentation requirements, and improve production throughput while maintaining dispensing accuracy. The proposed design offers a scalable solution for automated liquid blending applications in high-speed industrial production environments.

Keyword: Industrial Automation, Programmable Logic Controller (PLC), Liquid Blending, Shared Flow Measurement, Continuous-Motion Dispensing, Motion Synchronization, Mechatronics.

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Deep Learning Techniques for Early Melanoma Detection

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Abstract:

Skin Cancer is one of the fastest spreading disease worldwide, A Deep Graph Learning Framework for Precise Automated Skin Lesion Classification to Aid Dermatologists in Detecting Skin Cancer. The paper describes the development of a Deep Graph Learning framework for automated and accurate classification of skin lesions to help dermatologists in early skin cancer detection. The method uses an innovative approach to pre-process dermoscopic images and detects subtle differences between cancerous and non-cancerous lesions. The CNN–GNN model uses the relationship between objects in the image, specifically the spatial connection of lesion areas and the information from these area, shape, color and texture. The research provides a detailed analysis of the impact of various factors on the accuracy of classification, such as attention mechanisms, context managers and adaptive learning rates. The model achieves this by focusing on the key objects and reducing false positives, thereby increasing the accuracy of suspicion. The tool also allows generating easily interpretable visual information in a variety of formats. Finally, the evaluation of the model using different classification metrics – accuracy, sensitivity, and specificity, as well as efficiency, shows that the new framework is a cost-effective solution for dermatology practice in detecting skin cancer at an early stage.

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AI-Powered Student Academic Performance Tracking and Mentoring Mobile Application (EduMentor AI)

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Abstract:
Students often find it difficult to monitor their academic performance across semesters, recognise their strengths and weaknesses, and identify the subjects that need additional attention. Conventional tracking methods rarely provide personalised feedback or a structured way to follow continuous improvement. This work presents EduMentor AI, a mobile application that combines academic performance tracking with AI-based mentoring on one platform. After registration and login, students can maintain semester-wise records for eight semesters (S1–S8), covering subjects, internal marks, and end-semester marks. Academic performance is calculated by weighting internal marks at 40% and end-semester marks at 60%, and is classified as Good, Average, or Needs Improvement. Attendance can be recorded, with warnings when it falls below the required level, and a Practice Zone offers subject-wise questions based on the student's own academic records. An AI Mentor is intended to give personalised guidance and recommendations using academic records, examination marks, attendance, and practice performance. The application is built with Google AI Studio, while Supabase provides backend services, the database, and authentication. These AI-based capabilities represent the intended design, and this abstract reports no experimental or user-study findings. EduMentor AI aims to help students track progress, recognise areas needing improvement, practise subject questions, and receive academic guidance.

Keywords: EduMentor AI, Academic Performance Tracking, Artificial Intelligence, Student Mentoring, Attendance Monitoring, Personalized Recommendations, Mobile Application.

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AI-Powered Continuous Code Review and VulnerabilityDetection System

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Abstract:

Modern web applications built with Python (Flask), React/JavaScript, Node.js, and Java are exposed to critical OWASP Top 10 vulnerabilities such as SQL Injection (CWE-89), Cross-Site Scripting (CWE-79), Command Injection (CWE-78), and Hardcoded API Secrets (CWE-798). Fixing such flaws after deployment can cost up to 30 times more than catching them during development, while traditional rule-based SAST tools produce high false-positive rates and leave remediation to developers. This project presents Severa Defender AI, a full-stack, multi-language continuous code review platform that applies a shift-left security approach by combining real-time static analysis with AI-assisted vulnerability remediation inside the developer workbench. The system uses a hybrid dual-engine design. A lightweight Abstract Syntax Tree (AST) pattern- matching engine with more than 50 OWASP Top 10 and MITRE CWE rules scans Python, JavaScript, TypeScript, Java, Go, Dockerfile, and SQL code in under 50 ms, while LLM contextual reasoning (Qwen-2.5-Coder-32B / Gemini), accessed through a serverless AI proxy with a deterministic local fallback, generates context-aware patches. The workbench, built with React 19, Vite, and Tailwind CSS, provides line-level vulnerability markers, parallel multi-file remediation using Promise.all, and a side-by-side diff viewer for one-click patch application. The platform also includes a CI/CD quality gate simulator that evaluates pull requests against Critical/High flaw thresholds and a Security Health Score of at least 80%, Software Composition Analysis (SCA) that detects known CVEs in package.json dependencies, and an executive dashboard that tracks OWASP flaw distribution and security debt hours. The system achieved an average scan time of under 45 ms per file, 98.4% detection accuracy on standard OWASP benchmark test suites, and a 75% reduction in remediation time, while reducing critical vulnerabilities to zero and improving code maintainability from grade D/F to A+. It has potential applications in DevSecOps pipelines, secure software development education, enterprise code auditing, and pull request security gating, with future scope in GitHub Actions and GitLab CI/CD integration, Tree-Sitter WASM-based taint analysis, and container image and Infrastructure as Code scanning. Keywords: Continuous Code Review, Static Application Security Testing (SAST), Vulnerability Detection, Large Language Models, Abstract Syntax Tree, Software Composition Analysis, CI/CD Quality Gate, OWASP Top 10, DevSecOps

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Generative AI-Powered Citizen Rights Information System

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Abstract

Most citizens remain unaware of their legal and constitutional rights, entitlements, and grievance redressal mechanisms, because legal texts are lengthy, technical, and hard for the general public to interpret. In this paper, we present a Generative AI-Powered Citizen Rights Information System, a conversational platform that answers questions on fundamental rights, consumer protection, labour and employment, women and child safety, the Right to Information, and government welfare schemes in simple, easy-to-understand language. The system is built on a Retrieval-Augmented Generation (RAG) pipeline: authoritative sources such as the Constitution of India, central and state acts, government notifications, and official scheme documents are cleaned, divided into chunks, converted into vector embeddings, and stored in a vector database. For each user question, the most relevant passages are retrieved and passed to a large language model, which generates a grounded answer together with references to the source provisions. As an advancement, we combine source-grounded generation with safety filters and prompt design, which reduce hallucination and ensure the system offers general legal information rather than binding legal advice. A web-based chat interface with regional language support provides step-by-step guidance on filing complaints and approaching the appropriate authorities. The result is an accessible, transparent, and reliable tool for citizen awareness, legal aid support, and e-governance services.

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Generative AI Based Alumni Networking Platform with Personalized Connection Recommendations

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Abstract:
Students and graduates often find it difficult to identify alumni who can offer career guidance, mentorship, professional connections, internship opportunities, job-related information, and industry insights. Traditional alumni networking platforms largely serve as static directories and offer limited personalization or intelligent connection recommendations, which restricts meaningful interaction between alumni and students. This project proposes a Generative AI Based Alumni Networking Platform with Personalized Connection Recommendations that aims to connect students, alumni, and graduates based on their profiles, skills, interests, academic backgrounds, professional experience, and career goals. The proposed platform is intended to support alumni and student profile management, personalized alumni connection recommendations, and intelligent matching based on skills, interests, and career aspirations. Generative AI-assisted networking and interaction support is envisioned to help users initiate and sustain professional conversations, while alumni-student mentorship and career guidance features aim to facilitate structured and purposeful engagement. The platform is expected to help users discover relevant professional connections, obtain career guidance, and improve mentorship opportunities. By encouraging professional knowledge sharing and sustained interaction, it is also expected to strengthen alumni engagement and institutional connectivity, creating a supportive network that benefits both current students and graduates.

Keywords: Generative AI, Alumni Networking, Personalized Recommendations, Intelligent Matching, Career Guidance, Mentorship, Alumni Engagement

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Smart Alumni Networking Platform with AI Recommendations

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Abstract:
Students often lack structured access to the alumni whose experience could guide their academic and career decisions, and existing alumni directories rarely go beyond static listings. This project presents a full-stack Generative AI Alumni Networking Platform for Bannari Amman Institute of Technology that connects students with alumni through personalized recommendations, mentorship requests, secure messaging, and AI-assisted career guidance. The platform provides three role-based dashboards for students, alumni, and administrators. A recommendation engine ranks alumni by combining a student's department, skills, interests, projects, and desired industry with an alumnus's expertise, professional experience, and mentoring interests, and considers mutual connections where appropriate. CGPA is never used as the sole ranking factor, and each recommendation is accompanied by a plain-language explanation of the matching factors. Generative AI features explain profile matches, produce career guidance and learning roadmaps, suggest skills for a chosen career path, summarize alumni industry updates, and help students prepare mentorship questions. All AI responses are grounded in profile and platform data, never invent alumni qualifications or job openings, and clearly separate suggestions from verified information. The system is built with React.js, Vite, and Tailwind CSS on the frontend, Node.js and Express.js on the backend, and MongoDB for storage. Security is handled through JWT authentication, bcrypt password hashing, role-based authorization, input validation and sanitization, and API keys kept in environment variables. Alumni can publish industry updates, jobs, and career advice, and earn admin-controlled contribution points, while administrators verify alumni, moderate content, manage events, and view platform analytics. The platform aims to strengthen student–alumni connections, career development, and knowledge sharing.

Keywords: Generative AI, Alumni Networking, Recommendation System, Career Guidance, Mentorship, MERN Stack, Role-Based Access Control, JWT Authentication

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AI-Powered Disaster Intelligence

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Abstract:
The AI-Powered Disaster Intelligence and Emergency Response Platform is an intelligent digital system designed to improve the monitoring, analysis, and response to disaster situations. The system integrates modern technologies such as Artificial Intelligence (AI), Deep Learning, Satellite Imagery, Geographic Information System (GIS), and data analytics to provide better visibility into disaster-affected regions. The proposed system collects and processes important information from satellite imagery and geospatial data related to disaster-affected areas. This data can be analyzed using AI and deep learning techniques to identify affected regions, support situation assessment, and generate useful information for emergency response. GIS can further assist in visualizing affected locations and presenting disaster-related information geographically. AI-based analysis can help reduce the time and manual effort required for disaster assessment and support the identification of critical areas. The integration of satellite imagery with GIS provides a centralized view of disaster-related information, enabling users to understand the affected region more effectively and support timely decision-making. The main objective of the system is to improve disaster intelligence, enhance emergency response support, reduce manual monitoring, and provide a centralized platform for managing disaster-related information. The proposed system can be further enhanced with advanced AI models, real-time data sources, improved satellite-image analysis, cloud infrastructure, and predictive analytics. Overall, the AI-Powered Disaster Intelligence and Emergency Response Platform provides a technology-driven approach for smarter, faster, and more effective disaster management.

KEYWORDS: Artificial Intelligence, Deep Learning, Disaster Management, Satellite Imagery, Geographic Information System, Emergency Response, Disaster Intelligence, Data Analytics, Disaster Monitoring, Risk Assessment.

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Live Customer Churn Risk Assessment Engine with Real-Time Recommendation Overrides for E-Commerce

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Abstract:
E-commerce platforms may lose customers when early signs of disengagement are not detected and relevant retention actions are delayed. This project proposes a Live Customer Churn Risk Assessment Engine with Real-Time Recommendation Overrides for E-Commerce to identify changing churn risk and support timely, personalized interventions. The system is designed to use customer behaviour signals, such as browsing activity, cart abandonment, session drop-offs, and purchase history, to derive risk indicators and update customer risk assessments as events occur. The proposed architecture uses Python for churn assessment, FastAPI for backend services, and Kafka and Redis for real-time event processing and caching. Based on the assessed risk, a recommendation override component is intended to adjust recommendations or present suitable retention actions, helping the platform respond to potentially disengaged customers. The engine aims to connect risk scoring with actionable personalization rather than relying only on periodic churn analysis. The project is currently under development; therefore, the described capabilities represent the system's intended design, and no experimental or user-study results are claimed. The approach has the potential to support proactive customer retention, more responsive recommendation strategies, and ongoing engagement in e-commerce environments.

Keywords: Customer Churn Prediction, Artificial Intelligence, E-Commerce, Real-Time Recommendation System, Personalized Retention, Risk Assessment Engine, Machine Learning.

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FRAUDLENS: Automated GST invoice fraud detection using anomaly detection for MSME compliance

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Abstract:

Investigative compliance and fraud detection often require working with disparate datasets scattered across multiple invoices, enterprise entities, and tax filing periods. While advanced machine learning and AI-generated summaries significantly reduce manual cross-examination, standard retrieval models frequently blur the line between a fact recorded in a source document and a relationship inferred from statistical irregularities. To resolve this, this project introduces an automated GST invoice fraud detection platform integrated with an investigation-specific Evidence Provenance Ledger. The system combines natural language querying and retrieval-augmented generation (RAG) with machine learning anomalies like Isolation Forest and Local Outlier Factor (LOF) to uncover illicit Input Tax Credit (ITC) claims, circular trading, and shell company transactions. The proposed ledger dynamically connects each flagged fraud alert or transactional lead back to its exact source records and the specific analytical operations that generated it. By design, the ledger classifies platform outputs into three strict tiers: source-recorded facts, system-derived associations, or unverified anomalies. It attaches transaction-level citations, flags missing or conflicting tax filings, and actively withholds definitive compliance summaries when the available invoice evidence is insufficient. The primary objective is to preserve a transparent, auditable trail from an investigator's initial query down to the underlying database records and downstream machine learning outputs. This architecture will be evaluated against standard RAG and unsupervised anomaly detection workflows using a controlled set of synthetic and de- identified GST transactional records. Evaluation metrics will strictly measure claim-level evidence support, citation correctness, correct classification of facts, unsupported-lead rates, and abstention quality. Ultimately, this framework provides tax authorities and MSMEs with a proactive, scalable, and highly verifiable tool designed to assist professional compliance reviews without treating unverified algorithmic inferences as absolute facts.

Keywords: Automated GST fraud, retrieval-augmented generation, evidence provenance anomaly detection, responsible AI, MSME compliance, tax auditing.

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Spatio-Temporal Rain-Aware Predictive Routing for Millimetre-Wave 5G Backhaul Mesh Networks in Tropical Climates: A Kalman Filter-Based Approach

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Abstract:
Millimetre-wave (mmWave) wireless backhaul is a key enabler of dense 5G small-cell deployments, but its links are highly vulnerable to rain attenuation, especially in tropical regions such as southern India where intense monsoon rainfall can cause sudden outages. Conventional routing is reactive: traffic is moved only after a link has already degraded. Moreover, most studies examine single links using temperate-climate rain statistics and ignore that moving storm cells fade different links of a network at different times. This paper proposes a spatio-temporal, rain-aware predictive routing scheme for mmWave backhaul mesh networks. A moving storm cell is simulated over a 20-node E-band (73 GHz) mesh, and per- link rain attenuation is computed with the ITU-R P.838 specific attenuation and P.530 path reduction models. A Kalman filter tracks the attenuation of each link and extrapolates it forward over the routing reconfiguration delay, so that traffic is moved away from links likely to fade before they fail. The scheme is compared with static shortest-path routing and reactive rerouting over 30 randomised storms, using packet loss, outage probability, end-to-end delay, throughput and route changes. Static routing loses 8.65% of traffic, whereas reactive rerouting reduces this to 0.49%. Predictive rerouting reduces loss and outage probability by a further 14.6% and 10.9% respectively at a 5 s reconfiguration delay, and by about 19% in loss at 15 s, showing that the benefit grows with control delay. This gain comes at the cost of roughly 80% more route changes. The approach is lightweight, physics-based and requires no training data.

Keywords: mmWave backhaul, rain attenuation, predictive routing, Kalman filter, mesh networks, 5G, tropical climate

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TechPilot: An AI-Guided Personalized Learning and Career Readiness Platform for Technology Students

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ABSTRACT:
In today competitive job market, many students and aspiring developers struggle to find a structured path to learn technology skills and secure suitable employment. This project presents TechPilot, an intelligent career guidance application that bridges the gap between self-learning and career readiness in the technology domain. The application offers curated learning tracks across multiple in-demand domains, including Web Development covering HTML, JavaScript, React, and Projects; Machine Learning covering Python, Statistics, core ML concepts, and Projects; UI/UX Design covering design principles, Figma, UX research, and Portfolio building; Cybersecurity covering Networking, Linux, Security, and Projects; and Data Science covering Python, SQL, Statistics, and Projects. Based on the user's performance, interests, and periodic skill-assessment results, the system recommends the most suitable career path and highlights matching job openings or internship opportunities, along with resume-building guidance, interview preparation resources, and progress-tracking dashboards. By integrating structured learning modules with data-driven career recommendations, TechPilot removes the need for learners to depend on scattered resources and generic advice. The system was implemented using a client-server architecture with a responsive front-end and a backend database storing user progress, skill levels, and job-matching data. Testing with sample users showed that the guided learning approach improved skill retention and helped users identify clearer career goals compared to unstructured self-learning. This project demonstrates how combining career counselling with skill development across multiple technology domains can create an effective, accessible pathway for students to learn in- demand skills and achieve employment in the technology industry. Key words: Career Guidance, Machine Learning, UI/UX Design, Job Recommendation, Skill Assessment

Institute of Computer Science and Engineering, SIMATS Engineering, Chennai.
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Edge AI-Based Energy Monitoring and Intelligent Load Optimization for Small Industries

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Abstract:
The process of smart energy management is gaining importance since small scale industries are looking for ways to cut costs and maximize energy efficiency. Traditional energy monitoring systems mainly rely on cloud-based computing, which causes delays in communication and a decrease in the level of reliability at times when there are problems with connection. The study provides a solution in the form of the Edge AI-Based Smart Energy Monitoring and Intelligent Load Optimization that is using Raspberry Pi or ESP32 for edge data processing in real-time. Energy consumption data is taken from voltage, current and power sensors so that the system can analyze the information and identify abnormal energy consumption patterns and optimize load scheduling at the same time. Only the summarized information and alerts about problems are sent to the cloud so that the system would have a minimum need in bandwidth and dependency on the internet connection in order to achieve faster response. As a result, the proposed system leads to increased energy efficiency, all the while minimizing the latency and monitoring.

Keywords: Edge AI, Internet of Things (IoT), Smart Energy Monitoring, Intelligent Load Optimization, Raspberry Pi, ESP32, Small Industries, Energy Management.

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SmartPark - IOT and ML Based Real-Time Parking Slot Availability System

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ABSTRACT:
Parking areas can be hard to use when drivers know which spots are open or when reserved spots will be free. This paper introduces SMART PARK, a system that brings together Internet of Things (technology with machine learning to predict space availability. An infrared (IR) sensor is attached to each parking spot. An ESP32 device sends updates via Wi-Fi to Firebase indicating whether the spot is taken. A mobile app shows the status of each spot and allows people to make reservations based on a start date, start time and end time. The app checks existing reservations before confirming one. The times when parking sessions start and end are also saved to create a history that can be looked at later. A Random Forest model uses information such as the day of the week, the time, the weather and past space usage to predict how many spots will be free at a given time. This system brings together checking, planning ahead, booking and predicting all in one process. Testing with a planned prototype is still needed, and possible improvements could include automatic number-plate recognition and more ways to handle reservations.

Key Words: IOT; Smart Parking; ESP32; IR Proximity Sensor; Cloud/Firebase; Real-time Occupancy; Time- Reservation; Parking Session Management; Machine Learning.

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Smart focus- Admin Controlled Computer Monitoring and Management System

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Abstract:
Managing large numbers of connected workstations in modern educational institutions is increasingly difficult because administrators must monitor system health, application usage, user sessions, and policy compliance across distributed devices. This paper presents a distributed workstation monitoring and management system designed for college laboratories and institutional environments. The proposed architecture combines endpoint monitoring agents, secure telemetry channels, centralized reporting, and administrative dashboards to improve visibility, reduce operational effort, and support proactive response to system anomalies. The system is designed to provide real-time health indicators, application awareness, and secure remote administrative controls while preserving data integrity and accountability. Experimental evaluation and system design analysis show that the proposed framework is scalable, practical, and suitable for deployment in academic lab environments with moderate to large numbers of computers. Index Terms-workstation monitoring, distributed systems, system administration, proactive alerting, secure telemetry, academic labs.

Key words: Real-Time Monitoring,proactive Alerting,Remote Management, Academic Laboratories

ISBN No.: 978-93-344-7370-4

LearnXAI: An AI-Powered Personalized Learning and Career Intelligence Platform

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Abstract:
Traditional learning platforms often provide standardized learning content that may not effectively address the individual learning needs, skill gaps, and career goals of students. To overcome these limitations, this paper proposes LearnXAI, an AI-powered personalized learning and career intelligence platform designed to improve learning efficiency through adaptive learning recommendations and skill development. The proposed system evaluates students' knowledge through initial assessments and topic-wise quizzes to identify their strengths and weaknesses. Based on assessment results and performance history, the platform calculates mastery scores, performs dynamic gap detection, and generates personalized learning paths. Students can access recommended learning resources, receive AI-powered tutoring assistance, practice quizzes, and revise weak concepts while progressing toward advanced topics in areas where they demonstrate proficiency. Furthermore, LearnXAI recommends suitable projects, identifies relevant skill-development opportunities, and provides career roadmaps aligned with students' learning progress and career interests. The system is designed using Flutter and Dart for the mobile application, Spring Boot with Java for backend services, PostgreSQL for data management, and the Gemini API for AI-powered assistance. By integrating continuous performance monitoring with adaptive recommendations, LearnXAI aims to provide a flexible, student-centered learning experience that supports academic improvement and career readiness. The proposed platform offers a unified approach to personalized education by connecting learning assessment, skill-gap identification, practical development, and career planning within a single system.

Keywords: Artificial Intelligence, Personalized Learning, Adaptive Learning, Dynamic Gap Detection, Skill Development, Career Intelligence, Gemini API, Educational Technology.

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AI-Driven Multilingual Smart Grievance System with Location-Based Local Issue Detection and Adaptive Priority for Local Governance

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Abstract:

Public grievance management is essential for improving the delivery of government services, particularly in rural and digitally underserved areas. However, citizens may face difficulties accessing online grievance platforms, while repeated complaints from the same locality may be processed independently without identifying a common local issue. This paper proposes an AI-driven multilingual smart grievance system that integrates a physical grievance kiosk with a mobile and web-based interface. The proposed system supports Tamil and English grievance input and uses natural language processing and machine learning to classify complaints into appropriate categories. A location-aware related grievance detection mechanism identifies similar complaints using category, geographical information, and text similarity. Based on the number of related complaints detected within a locality, a local issue detection module determines the level of a recurring problem. An adaptive priority mechanism then adjusts the initial grievance priority according to predefined rules. The system also supports department routing, centralized dashboard monitoring, grievance status tracking, and escalation of unresolved complaints. A government grievance dataset is used to develop and evaluate the grievance classification component. The proposed approach aims to improve grievance accessibility, identify recurring local problems, and assist authorities in prioritizing complaints more effectively. Further evaluation of the integrated system is required to establish its overall effectiveness in real-world governance environments.

Keywords: Artificial Intelligence; Machine Learning; Natural Language Processing; Multilingual Grievance Management; Smart Kiosk; Local Issue Detection; Location-Aware Similarity; Adaptive Priority; Local Governance.

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Live Customer Churn Risk Assessment Engine with Real-Time nRecommendation Overrides for E-Commerce

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Abstract :

E-commerce platforms may lose customers when early signs of disengagement are not detected and relevant retention actions are delayed. This project proposes a Live Customer Churn Risk Assessment Engine with Real-Time Recommendation Overrides for E-Commerce to identify changing churn risk and support timely, personalized interventions. The system is designed to use customer behaviour signals, such as browsing activity, cart abandonment, session drop-offs, and purchase history, to derive risk indicators and update customer risk assessments as events occur. The proposed architecture uses Python for churn assessment, FastAPI for backend services, and Kafka and Redis for real-time event processing and caching. Based on the assessed risk, a recommendation override component is intended to adjust recommendations or present suitable retention actions, helping the platform respond to potentially disengaged customers. The engine aims to connect risk scoring with actionable personalization rather than relying only on periodic churn analysis. The project is currently under development; therefore, the described capabilities represent the system's intended design, and no experimental or user-study results are claimed. The approach has the potential to support proactive customer retention, more responsive recommendation strategies, and ongoing engagement in e-commerce environments.

Keywords: Customer Churn Prediction, Artificial Intelligence, E-Commerce, Real-Time Recommendation System, Personalized Retention, Risk Assessment Engine, Machine Learning.

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ML-Based Counterfeit Currency Detection Device

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Abstract :

The circulation of counterfeit currency poses a significant threat to global economies, financial institutions, and retail businesses. Identifying forged banknotes using traditional manual methods is often time-consuming and prone to human error, which creates a pressing need for automated, reliable, and highly accurate detection mechanisms to prevent financial fraud. This project proposes an ML-Based Counterfeit Currency Detection Device that aims to authenticate banknotes through advanced computer vision and machine learning algorithms. The proposed system is intended to utilize high- resolution image acquisition and image-processing techniques to extract critical security features from currency, such as watermarks, micro-printing, and ultraviolet properties. A trained machine learning model is then envisioned to analyze these extracted features to accurately classify the currency as either genuine or counterfeit. The device is expected to help retail businesses, cashiers, and everyday users quickly and reliably verify currency, thereby reducing the financial risks associated with counterfeit money. By providing an automated, real- time authentication solution, the system aims to improve transaction security, though its physical implementation will depend on the quality of the image sensors and the comprehensiveness of the underlying training dataset.

Keywords: Machine Learning, Counterfeit Currency Detection, Image Processing, Currency Authentication, Feature Extraction, Classification Model

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SMART AND SAFE BUCKET WATER HEATER USING TEMPERATURE CONTROL TECHNOLOGY

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Abstract :
Conventional bucket water heaters require continuous manual monitoring to maintain a safe water temperature and prevent overheating. This project presents a Smart and Safe Bucket Water Heater Using Temperature Control Technology for automatic temperature monitoring and control. The system uses a temperature sensor to measure the water temperature continuously and a microcontroller to process the measured value. Based on a predefined temperature limit, the controller automatically switches the heating element ON or OFF through a relay, helping to maintain the required temperature and reduce unnecessary heating. The system provides a simple indication of the temperature and heating status, allowing the user to monitor the operating condition easily. When the desired temperature is reached, the heating element is switched OFF, and it is switched ON again when the temperature falls below the set level. This automatic control reduces the need for continuous supervision and helps improve energy efficiency. The proposed system demonstrates a simple and low-cost approach to safe water heating using embedded electronics. It combines temperature sensing, microcontroller-based processing, automatic relay control, and safety-oriented temperature regulation. The system is suitable for domestic use, educational demonstrations, laboratory applications, and prototype development. Keywords: Smart Water Heater, Temperature Control, Temperature Sensor, Microcontroller, Relay, Heating Element, Safety, Energy Efficiency.

Keywords: IoT, ESP32, ESP8266, Energy Theft Detection, Energy Monitoring, ACS712, ADS1115, Current Sensor, OLED, Relay Module, Abnormal Load Detection.

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An Intelligent Legal Document Drafting Assistant Using Generative AI and Retrieval-Augmented Generation

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Abstract:
Preparing legal documents such as rental agreements, affidavits, and legal notices is challenging for individuals and small organisations, requiring accurate information, proper structure, and attention to applicable legal requirements. Traditional templates cannot address every user's needs or jurisdictional variations, and general-purpose AI tools may omit important information. This project proposes an Intelligent Legal Document Drafting Assistant using Generative Artificial Intelligence (GenAI), Large Language Models (LLMs), Natural Language Processing (NLP), and Retrieval-Augmented Generation (RAG). The system is a web-based application where users select the document type and Indian state or jurisdiction. A guided conversational interface collects details such as purpose, party information, dates, and terms, asking follow-up questions when details are unclear and avoiding invented personal details.RAG connects the language model with a curated knowledge base of legal references and approved templates. Relevant passages retrieved by jurisdiction and document type guide the LLM to generate a structured draft with headings, clauses, and signature fields. The system explains clauses in simpler language, though this is informational, not legal advice. Users can review and export documents in PDF or DOCX format. The assistant supports preliminary drafting rather than replacing a qualified lawyer, and final legal verification remains necessary.

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SMART TOURISM AND AR BASED HERITAGE

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Abstract:
The Nilgiri Smart Tourism and AR-Based Heritage Application is a web-based platform that focuses on promoting sustainable tourism, preserving indigenous heritage and culture, and supporting local business ventures in the Nilgiris district of Tamil Nadu. There are some popular destinations like Ooty and Coonoor that become overcrowded with tourists in peak seasons. However, there are many other locations in the area that go unnoticed by tourists. The main idea of the application is to help tourists find both popular locations and hidden gems in Nilgiris – such as Ooty, Coonoor, Kotagiri, Gudalur, Pandalur, and Kundah. The platform offers the tourists information about tourist destinations, ideal visiting times, safety measures, possible risks, nearby hotels, restaurants, chill-out places, and even taxi services. There is also a local business promotion module included that will allow businesses to promote their products and services for tourists in a special section. There is an admin dashboard where authorized administrators can control information about tourism in the region and other related content and make sure that appropriate information is shown on the interface of the application. Another part of the platform is the AR-based heritage module aimed at the promotion of the history, traditions, languages, and practices of indigenous communities of the Nilgiris, including Todas, Kotas, Badagas, Irulas, Kurumbas, and Paniyas. The module promotes indigenous traditional products and handicrafts like Toda embroidery and Kota crafts. Moreover, it incorporates a Badaga typing and language learning module that supports learning of the Badaga language by tourists and younger generations. Additionally, the platform provides information about environmental awareness that will help tourists to follow responsible tourism principles by not littering and protecting natural resources. Thus, the application combines tourism information, location-based services, local business promotion, digital heritage, language learning, and environmental awareness in one platform.

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AI Agent-Based Full-Stack Application Generation Platform

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Abstract:
Developing full-stack applications requires considerable effort in requirement analysis, system architecture design, frontend development, backend programming, database design, component integration, and software testing. Conventional AI-assisted coding tools primarily support individual code-generation tasks, often requiring developers to coordinate different components manually and resolve inconsistencies during integration. This project proposes an AI Agent-Based Full-Stack Application Generation Platform that aims to automate and coordinate the software development lifecycle using specialized Artificial Intelligence (AI) agents and Large Language Models (LLMs). The proposed web- based platform accepts natural-language application requirements and processes them through a structured workflow involving requirement analysis, task decomposition, architecture planning, and technology selection. Specialized agents are responsible for generating frontend components, backend APIs, database schemas, and supporting project files. A central agent orchestrator manages task dependencies, while a shared project context maintains consistency across the generated components. Furthermore, the platform incorporates code validation, functional testing, error analysis, and targeted refinement to identify and address implementation issues through an iterative feedback mechanism. The system also provides a project dashboard, generated-file management, downloadable project output, and performance evaluation metrics to assess requirement coverage, validation results, test success rate, and generation time. The proposed platform aims to reduce manual coordination, improve modularity, and streamline the transformation of natural-language requirements into structured full-stack application prototypes. The effectiveness of the approach will be assessed through prototype implementation and experimental evaluation. The platform is intended to support software development and prototyping rather than eliminate the need for developer review and verification.

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Data Driven Bibliometric Trajectory Analysis and Technology Trend Prediction Using Cloud Big Data Pipelines

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Abstract:
Researchers, analysts, and technology strategists often track emerging technology areas through manual literature searches, spreadsheets, and disconnected reports, which makes it difficult to observe trajectories across scientific literature and patents and to anticipate which domains are emerging or declining. In this paper, we propose a cloud-based bibliometric trajectory analysis and technology trend prediction system built on a serverless big data pipeline that tracks a curated panel of twenty technology domains across scientific papers indexed by OpenAlex and patents obtained from PatentsView. Scheduled ingestion functions collect raw publication and patent records, which are preprocessed into analytics-ready columnar datasets, from which bibliometric indicators such as publication and citation growth are computed. The system classifies each domain into emerging, growing, maturing, or declining trajectories, and forecasts one to three years ahead using classical time-series models that are backtested against a naive baseline before their results are published. An emergence score combines growth signals into a single comparable ranking, and an interactive React dashboard presents trends, forecasts, and rankings in real time. Hosted on Amazon Web Services using S3, Lambda, API Gateway, and DynamoDB, the proposed system is scalable, cost-efficient, and requires no dedicated server management, providing an objective, data-driven view of technology trends for researchers and decision makers.

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CloudPulse: A Serverless Cloud-Based Employee Productivity and Goal Management System

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Abstract :
Organizations and educational institutions often lack real-time visibility into campus energy consumption, leading to high peak demand charges and HVAC waste in empty classrooms. In this paper, we propose Smart Campus Energy Analytics Platform, a web-based system built on the MERN stack that provides real-time peak detection using a Z-Score algorithm, HVAC load curtailment simulation with 30% rebound modeling, and savings calculation in kWh, rupees, and CO2. The platform offers role-based dashboards for Energy Manager, Building Supervisor, and Energy Analyst, enabling classroom AC control based on student strength. The system is scalable, cost-efficient, and requires no hardware investment, improving energy transparency and reducing electricity costs in educational campuses.

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AI-POWERED MULTILINGUAL TEXTBOOK SUMMARIZATION AND ADAPTIVE LEARNING SYSTEM

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Abstract:

Students today often struggle with long textbooks and PDFs, with identifying important topics, with understanding complex concepts, and with language barriers. They also get no personalised guidance on what to study next. This project presents an AI-powered multilingual textbook summarization and adaptive learning system that addresses these problems in one platform. The student uploads a textbook in PDF format, and the system extracts the text using Python and PDF processing techniques. Natural language processing and large language model techniques then generate concise chapter-wise summaries with key terms. The summaries can be translated from English to Tamil to support multilingual learning. A Retrieval-Augmented Generation (RAG) based chatbot lets the student ask questions and receive answers drawn from the uploaded textbook, using embeddings to retrieve the relevant content. The system also generates quizzes automatically, including multiple-choice, true/false and short-answer questions, and evaluates the student's score. A machine learning module analyses the quiz performance topic by topic, identifies weak topics, and recommends what the student should revise next. The results are shown on a student dashboard, and an optional study planner builds a day-wise schedule for an upcoming exam. The system is developed in Python with Streamlit for the interface and SQLite for storing users and results. By combining summarization, translation, question answering, assessment and adaptive recommendation, the project helps students move from uploading a textbook to understanding, practising, analysing and improving.

Keywords: Textbook summarization, Multilingual learning, Retrieval-Augmented Generation, Quiz generation, Adaptive learning, Natural language processing

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AI-Powered Campus Companion for Automated Academic Audit and Personalized Skill-Gap Analysis

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Abstract:

Educational institutions generate academic information that can help students understand their performance and plan their career development. However, reviewing academic records manually and identifying the skills required for a chosen career can be time-consuming and may not provide students with clear, personalized guidance. This project proposes an AI-Powered Campus Companion that automates academic auditing and performs personalized skill-gap analysis through a unified web-based system. The application is designed to organize student academic details, evaluate academic performance, and present a structured summary of strengths and areas that require improvement. By comparing a student's current skills with relevant target skills, the system can identify skill gaps and provide recommendations to support focused learning and career preparation. The proposed solution uses a backend service with REST APIs and a database to manage student information and analysis results, while AI/ML- based functionality supports intelligent analysis and personalized recommendations. The system aims to reduce manual effort, improve the accessibility of academic insights, and help students make informed decisions about their learning and placement readiness. The project brings academic auditing and skill development guidance together in a single campus companion platform.

Keywords: Academic Audit, Skill-Gap Analysis, Artificial Intelligence, Machine Learning, REST API, Personalized Learning

Institute of Computer Science and Engineering, SIMATS Engineering, Chennai.
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Generative AI Threat Intelligence Aggregator and Automated Incident Response Playbook Synthesizer

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Abstract

The increasing number of cyberattacks has created a need for intelligent and automated cybersecurity solutions. Traditional security systems often require manual analysis to identify threats and determine appropriate response actions, which can increase incident response time. The proposed Threat Guard AI system aims to improve cyber threat detection and incident handling by integrating Artificial Intelligence (AI), Machine Learning (ML), and Generative AI techniques. The system collects and processes threat-related information from sources such as security logs, suspicious IP addresses, URLs, domains, and Indicators of Compromise (IOCs). Different machine learning algorithms, including Random Forest and Support Vector Machine (SVM), will be evaluated for threat classification, while anomaly detection techniques will be explored to identify suspicious activities. The performance of candidate algorithms will be compared using suitable evaluation metrics to select the most appropriate approach. Based on the detected threats and their severity, the system aims to generate security alerts and use Generative AI to synthesize relevant incident response playbooks containing recommended investigation, containment, and recovery actions. The proposed system is expected to reduce manual effort, improve threat analysis efficiency, and support security analysts in making faster and more informed incident response decisions. Human verification will be considered for critical response actions to improve the reliability and safety of the system.

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DIGITAL VILLAGE ADMINISTRATION SERVICES PORTAL WITH CITIZEN IDENTITY VERIFICATION

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Abstract

Village-level (Gram Panchayat) administration in India still depends largely on in-person visits, handwritten registers and manual comparison of paper identity documents. Citizens often travel to the office more than once because of missing paperwork, officers spend time transcribing and cross-checking identity details where spelling variations, reordered names and faded copies cause errors and delays, and applicants have little visibility of the status of their requests, while paper records offer limited accountability. To address these challenges, this paper presents VillageConnect, a web-based Digital Village Administration Services Portal with Citizen Identity Verification. The system supports two roles. The Citizen registers, uploads an identity document for verification, applies for village services through a three-step wizard once verified, tracks application status, receives in-app notifications and downloads issued certificates. The Administrator reviews identity verification submissions in a split-screen workspace, approves or rejects them with a recorded reason, processes service applications, assigns grievances and inspects an append-only audit log. Five services are supported: Residence, Income and Family certificates, Water Connection requests and General Village Grievances. Applications follow state-controlled lifecycles, approved certificate requests generate PDF certificates, and a certificate is revoked if its approval is later reversed.

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Development of an Intelligent Agricultural Yield Prediction and Crop Health Monitoring Framework

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Abstract:

Agricultural productivity depends on timely monitoring of soil and environmental conditions and early detection of crop diseases. Manual observation is slow, subjective, and costly for small-scale farmers. This work presents an intelligent agricultural yield prediction and crop health monitoring framework integrating the Internet of Things (IoT), Artificial Intelligence (AI), and Machine Learning (ML). Soil and environmental parameters are acquired in real time using sensors interfaced with an ESP32 microcontroller. Machine learning algorithms analyse historical agricultural and environmental data to estimate crop yield. A Convolutional Neural Network (CNN) classifies crop leaf images to identify possible diseases. Results are displayed on a dashboard that lets farmers monitor crop conditions, understand disease risks, and make informed decisions. The system achieves [XX]% yield prediction accuracy (R^2 = [X.XX]) and [XX]% disease classification accuracy, evaluated using precision, recall, and F1-score. The framework provides an affordable, practical solution supporting precision agriculture.

Keywords: Precision agriculture; IoT; ESP32; Machine learning; Convolutional neural network; Crop yield prediction; Crop disease detection

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VIKSHANA: An Evidence-Provenance Framework for Traceable AI-Assisted Crime Investigation

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Abstract:

Investigators often work with records stored separately across cases, entities, locations, and investigation notes. Search tools and AI-generated summaries can reduce manual comparison, but they may blur the difference between a fact recorded in a source and a relationship inferred from shared details. VIKSHANA is an AI-assisted crime-intelligence platform that combines natural-language querying and retrieval-augmented generation (RAG) with similar-case retrieval, relational-record analysis, DBSCAN-based spatial clustering, trend analysis, and investigation summaries. This work proposes an investigation-specific Evidence Provenance Ledger that connects each reported lead to its source records and the analytical operations that produced it. The ledger will classify outputs as source- recorded facts, system-derived associations, or unverified leads; attach record-level citations; flag missing or conflicting support; and withhold definitive summaries when the available evidence is insufficient. The aim is to preserve the path from an investigator's question to retrieved records and downstream analytical results, rather than treating all generated statements as equally established. The proposed layer will be evaluated against the existing RAG workflow using a controlled set of synthetic or appropriately authorized, de-identified records. Measures will include claim-level evidence support, citation correctness, correct classification of facts and associations, unsupported-lead rate, and abstention quality. Any performance improvement will be reported only after testing. VIKSHANA is intended to assist professional review, not determine guilt or make operational decisions.

Keywords: crime intelligence; retrieval-augmented generation; evidence provenance; analytical leads; responsible AI

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A Machine Learning Based Student Performance Monitoring and Reward Management System for Institutions

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Abstract:

Educational institutions are increasingly encouraging students to take part in competitive programming on external platforms such as LeetCode, Codeforces, and CodeChef. However, this activity is typically not monitored in a consistent way. Faculty and administrators often rely on spreadsheets and manual reports, which makes it difficult to identify students who are consistently active, detect those who need more support, or create targeted training programs. This paper presents a web-based system for managing rewards and tracking performance within colleges. The system gathers student activity data from multiple competitive programming platforms and consolidates it into a single institutional database. It transforms completed levels and milestones into reward points and uses a Random Forest model to classify students based on how often they participate. Students are grouped using K- Means clustering, with the number of clusters determined through silhouette analysis, to assist in planning training programs for each group. The system also applies item-based collaborative filtering with cosine similarity to recommend additional problems or topics based on what similar students have solved. Administrators and mentors can select specific student groups using plain-English queries, which are translated into SQL by a large language model, checked using AST parsing, and run on a read-only database connection. If the language model is not available, a rule-based approach is used as an alternative. Role-based dashboards offer customized views for administrators, mentors, and students, and student records are maintained for long-term tracking. The system's modular design enables the addition of new platforms and the retraining of models as institutional data grows.

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AI Assisted Hybrid Indoor and Outdoor Campus Navigator with Privacy Preserving Geofenced Attendance System

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ABSTRACT

The rapid growth of large, multi-building campuses has made navigation difficult for students, staff, and visitors, while existing attendance systems raise privacy concerns from continuous location tracking. Prior navigation systems rely mainly on outdoor GPS, offer little indoor wayfinding, and use inefficient algorithms like Dijkstra's that can't adapt to real-time changes. Prior attendance systems depend on continuous GPS surveillance and centralized biometric storage, creating battery, privacy, and security risks. The proposed project provides a unified website-based platform addressing both problems together to investigate navigation and attendance while reducing unnecessary location tracking. An efficient hybrid indoor-outdoor navigation system has been designed to develop a privacy-preserving geofence-based attendance mechanism. An easily accessible and offline-capable solution has been built with measurable benchmarks. Using Global Positioning System (GPS) for outdoor pathfinding and QR-code or Bluetooth Low Energy (BLE) checkpoints for indoor localization. The algorithms have been used for computing the routes that adapt to blocked paths or crowd density. The built-in features include wheelchair-friendly routes and locally cached offline maps. Attendance uses event-triggered geofenced check-ins instead of continuous tracking, with an optional Convolutional Neural Network (CNN)-based facial-verification step storing encrypted embeddings. Built with a Python backend (Flask/Django) and a responsive website front end. Lower battery/privacy risk, better indoor wayfinding, offline usability, accessibility, and encrypted data storage — evaluated via route computation time, positioning accuracy, and check-in reliability. Indoor accuracy depends on checkpoint density, facial verification needs good lighting, and the system is untested on a live, large-scale deployment.

Keywords: Global Positioning System, Centralized biometric storage, Bluetooth Low Energy, Convolutional Neural Network, Facial verification, Geofenced attendance.

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Medical Waste Evaluation

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Abstract:

Medical waste such as gloves, syringes, dressings and plastics must be segregated correctly to protect healthcare workers and the environment, yet manual sorting is slow, unhygienic and prone to human error. Conventional bin sensors cannot distinguish organic from inorganic waste, and cloud-only systems suffer from delay and data loss during network failures. This paper presents a low-cost Edge-AI smart bin that identifies waste automatically and monitors its own fill level. A VL53L0X time-of-flight sensor detects an item at the bin inlet, triggers a camera, and measures the remaining bin capacity. The captured image is pre-processed with OpenCV and classified by a YOLOv5 model running directly on a Raspberry Pi 4 as biodegradable or non-biodegradable waste. Each event is recorded with its timestamp, class, confidence and distance, and is stored on a MicroSD card and uploaded to a cloud dashboard, so no records are lost without Wi-Fi. The model was trained and validated in Google Colab and deployed on the Raspberry Pi. Live-video waste identification and proximity-triggered capture have been tested, while frame drops and misclassification are being reduced through additional data and threshold tuning. The system reduces manual contact with waste, lowers sorting errors and enables remote tracking of waste type and bin level using off-the-shelf components. Final accuracy evaluation and extension to more medical waste categories, such as sharps and infectious waste, remain future work.

Keywords: Medical waste, YOLOv5, Edge AI, Raspberry Pi, VL53L0X, smart bin

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Know MyCampus: An Interactive Campus Discovery Platform

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Abstract :

Campus information is often distributed across websites, notices, and different departments, making it difficult for students, visitors, and parents to locate accurate details about departments, facilities, and important places. This paper presents KnowMyCampus, a web-based campus discovery platform designed to bring essential campus information together in a single, accessible interface. The platform organizes department and facility details and provides search and campus location links to help users discover relevant information and identify places on campus. The frontend is developed using HTML5, CSS3, and JavaScript, with the website hosted through GitHub Pages. A Generative AI campus assistant is proposed as a further component to answer campus-related questions in natural language by retrieving information from a verified campus knowledge base. Retrieval-based grounding could help improve the relevance and factual reliability of responses, although the assistant must be integrated and tested before being described as an implemented feature. The platform can be evaluated using search success rate, answer relevance, factual accuracy, response time, and user feedback. By combining campus information discovery and location guidance in one interface, KnowMyCampus aims to reduce the effort required to find campus information and provide a more convenient experience for students and visitors. The usefulness of the platform depends on the completeness, accuracy, and timely maintenance of the campus information it contains.

Keywords: Campus information system, campus discovery, web-based platform, department search, campus navigation, conversational AI, information retrieval. ICAI-C

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NeuroFusionAI: A Multimodal Foundation Model for Schizophrenia Detection

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Abstract

Schizophrenia is a complex psychiatric disorder in which language and communication can show clinically relevant patterns, and automated analysis of text may support early screening. This paper presents NeuroFusionAI, a multimodal transformer-based framework for classifying text as schizophrenia or normal control. Two pretrained DistilBERT encoders embed a patient-authored text stream and a clinical-note text stream, and a MulT-style bidirectional cross-modal attention block exchanges information between them before a dense layer with softmax produces the class probabilities. The Reddit dataset supplies the supervised labels (13,270 samples after cleaning), while MTSamples is used as an unlabeled clinical-text corpus that is pseudo-paired with each Reddit sample, because the two datasets share no patient identifiers. Under stratified 5-fold cross-validation the model achieves a mean accuracy of 94.12%, F1-score of 94.76% and ROC-AUC of 0.9858, and 10-fold cross-validation gives comparable results. SHAP is applied through a random-forest surrogate as a simple explainability analysis. Because the clinical text is pseudo-paired and the labels reflect community membership rather than clinical diagnosis, the results demonstrate the feasibility of the architecture and not clinical diagnostic accuracy.

Keywords: Schizophrenia Detection, Multimodal Learning, DistilBERT, Transformer, Cross-Modal Attention, MulT, Explainable AI, SHAP, Natural Language Processing, Transfer Learning.

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Two-Stage Brain Tumor Detection and Classification from MRI Using a One-Class SVM Gatekeeper and a Convolutional Neural Network

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ABSTRACT:

Manual interpretation of brain Magnetic Resonance Imaging (MRI) scans is time-consuming and may lead to variations in diagnostic assessment. This paper presents a two-stage deep learning system for brain tumor detection and classification using MRI images. The first stage uses ImageNetpretrained VGG16 features and a One-Class Support Vector Machine (OCSVM) to identify and reject images that do not resemble brain MRI scans. The second stage employs a Convolutional Neural Network (CNN) to classify accepted images into four categories: glioma, meningioma, pituitary tumor, and no tumor. The model is trained using data augmentation on 5,712 MRI images and evaluated on a test set of 1,311 images. The proposed CNN achieves an overall test accuracy of 94.97% and a macro-averaged F1-score of 0.947. The complete pipeline is deployed through a Streamlit web application, allowing users to upload MRI images and view the predicted class, confidence score, and classification results. The system aims to support research in automated brain tumor classification and improve the efficiency of MRI image analysis. However, further validation using independent datasets and non-brain images is required to establish its generalizability and reliability. The proposed system is intended for educational and research purposes and is not a substitute for professional medical diagnosis. Brain Tumor Detection, Magnetic Resonance Imaging, Convolutional Neural Network, One-Class SVM, VGG16, Deep Learning, Streamlit.

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VidhaAI: Student Opportunity and Career Guidance Platform

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Abstract :
Students who look for jobs, internships, scholarships, free learning programs, competitions and conferences usually have to visit many different websites and check each one's eligibility requirements by hand, which takes time and makes it easy to miss suitable opportunities. This paper presents VidhaAI, a web-based platform that collects these opportunities in one place and shows each student the ones that fit their profile. The front end is built with React and Vite, the back end with Java 21 and Spring Boot, and the data is stored in a MySQL database. A student enters education, graduation year, state, district, skills, preferred location, and the kind of opportunity wanted, and then browses searchable opportunity cards that can be filtered by category and carry a personalised match percentage. To keep the information current, job listings are imported automatically every day from official company career sources; listings that are no longer available are marked inactive, and records whose deadline has passed are marked expired. An administrative page allows opportunity records to be added and edited. The paper reviews related work on job recommender systems, describes the architecture and workflow of the platform, and presents its current interface. The match percentage indicates how closely a profile resembles an opportunity's requirements and does not guarantee eligibility or selection.

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HOSTELIA: A ROLE-BASED SMART HOSTEL AND CAMPUS LIFE MANAGEMENT PLATFORM

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ABSTRACT:

Hostel administration involves managing student records, room allocation, fee payments, mess attendance, complaints, and leave permissions. Handling these activities through separate records and manual communication can make it difficult to maintain accurate information and track approval status. This paper presents Hostelia, a Role-Based Smart Hostel and Campus Life Management Platform, a web-based application designed to coordinate hostel operations through a centralized system. The platform provides five role-based accounts: Management, Head of Department (HOD), Warden, Student, and Parent, each with specific responsibilities and access permissions. The major modules include student profile management, room allocation, fee tracking, mess attendance, complaint management, notifications, and reports. A key feature is the multi-stage leave approval workflow, in which students submit leave requests, parents provide permission, the HOD reviews the requests, and the Warden makes the final decision. The platform aims to simplify hostel administration, improve coordination among stakeholders, and provide better visibility into student and hostel-related information. The proposed system architecture and current interface are described, including the management and student dashboards. Further functional testing, security verification, and performance evaluation are required to validate the implementation and assess its effectiveness. Hostelia aims to provide a more organized, transparent, and accessible approach to smart hostel and campus life management. Smart Hostel Management, Role-Based Access Control, Student Information System, Leave Approval Workflow, Flask, Database Management.

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Pet service management system

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Abstract:

Pet owners often struggle to manage their pets' health records, vaccination schedules, grooming appointments, and veterinary visits across multiple disconnected sources. Missing vaccination dates and lost paper records lead to poor health outcomes and increased veterinary costs. This paper presents a Pet Service Management System, a web-based platform designed to centralise pet care management and automate appointment reminders. The frontend uses HTML5, CSS3, and JavaScript to provide a responsive, app-like user interface. The backend is implemented using Python 3 with the Flask framework, while SQLite stores application data through SQLAlchemy ORM. The system features comprehensive pet profile management, appointment scheduling for veterinary visits, grooming sessions, and vaccination schedules. Its core innovation is an automated background scheduler that sends personalised WhatsApp reminders to pet owners exactly one day before every appointment without requiring any manual administrative intervention. Additionally, the interface includes an analytics dashboard, seven customizable colour themes, dark mode support, and complete medical record tracking. This paper describes the motivation, related work, methodology, and system architecture of the proposed platform. Measured evaluation results will be added after structured testing.

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Web-Based Smart Visitor Management System with QR Code Authentication

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Abstract:

Managing visitors in institutions and workplaces through manual registers can lead to long waiting times, data-entry errors, difficulty verifying visitor identities, and challenges in retrieving past visit records. This project proposes a Web-Based Smart Visitor Management System with QR Code Authentication to make visitor registration and entry verification more organized, efficient, and secure. The system provides a centralized web interface for recording visitor details, the purpose of the visit, and the person or department to be visited. After registration, a unique QR code can be generated for the visitor and scanned at the entry point to verify the visit details before check-in. The system can maintain digital records of visitor check-in and check-out times, allowing authorized staff to search and review visit history when required. An administrative dashboard can help staff monitor visitor activity and manage records, reducing dependence on paper-based logs. QR code authentication is intended to support quick verification; access decisions should still follow the institution's rules and any required identity checks. By combining web-based record management with QR-based verification, the proposed system aims to improve the speed, traceability, and convenience of visitor handling. Its effectiveness depends on accurate registration, appropriate access permissions, and secure handling of visitor data.

Keywords: Visitor management, QR code authentication, web-based system, visitor registration, digital check-in, access verification, visitor records.

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CAMPAIGN PERFORMANCE PREDICTION AND BUDGET OPTIMIZATION SYSTEM

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ABSTRACT

Digital advertising plays a significant role in modern marketing, enabling businesses to reach target audiences through multiple advertising channels. However, campaign performance varies depending on campaign characteristics, channel selection, and budget allocation, making it challenging to utilize advertising resources effectively. This paper proposes a system that integrates machine-learning-based campaign performance prediction with constrained advertising budget optimization to support data-driven marketing decisions. Using historical campaign data, the system applies supervised learning techniques to analyze campaign characteristics and predict expected performance across different campaigns and advertising channels. The system employs a Gradient Boosting Regressor to predict campaign revenue based on relevant campaign characteristics. Based on the predicted outcomes, an optimization layer determines a suitable allocation of the available marketing budget while satisfying predefined budget constraints. The system also incorporates a what-if analysis feature that enables users to modify budget allocations or campaign parameters and examine their potential impact on predicted performance and resource distribution. A web-based dashboard presents prediction results, channel recommendations, optimized budget allocations, and simulation outcomes in an accessible format. By integrating predictive analytics, channel recommendation, constrained optimization, and scenario analysis, the proposed system aims to improve advertising resource utilization, minimize inefficient spending, and support informed digital advertising campaign planning.

Keywords : Machine Learning; Digital Advertising; Campaign Performance Prediction; Budget Optimization; Advertising Channel Recommendation; What-if Analysis; Marketing Budget Allocation.

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FACIAL OCCLUSION DETECTION AND CLASSIFICATION IN CONTROLLED LABORATORY ENVIRONMENTS

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Abstract

Laboratory environments require appropriate personal protective equipment (PPE) and reliable monitoring of safety practices. Masks, goggles, and other facial coverings can complicate automated visual assessment. This paper proposes a computer-vision framework for facial occlusion detection and classification in controlled laboratory environments. OpenCV is used to acquire video frames, and YuNet is used to detect faces. The detected facial regions are then processed by a convolutional neural network (CNN) to classify the observed condition into three categories: Fully Visible, Safety PPE Compliant, and Unauthorized Occlusion. Safety PPE Compliant refers to a visible facial condition in which a face mask and protective eyewear are present, while Unauthorized Occlusion refers to facial obstruction that does not meet the project's predefined PPE criteria. Facial key-point and covered-region analysis are intended to support the identification of occlusion around the eyes, nose, and mouth. The system is designed to compare predictions with predefined laboratory safety rules and support alerts and event logging, including timestamps, predicted classes, and snapshots, for review. The planned dataset is intended to include variation in lighting, face angle, distance, skin tone, and degree of occlusion. The framework aims to support more consistent visual monitoring; however, classification performance and processing latency must be measured on a suitable held-out test set before claims about accuracy or real-time performance can be established.

Keywords— Facial Occlusion; PPE Detection; Deep Learning; Convolutional Neural Network; Computer Vision; Laboratory Safety; OpenCV; YuNet

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AI-POWERED MULTI-BRAND E-BIKE RECOMMENDATION AND SMART MONITORING SYSTEM

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ABSTRACT

The AI-Powered Multi-Brand E-Bike Recommendation and Smart Monitoring System is an intelligent solution that helps users choose the most suitable electric bicycle while continuously monitoring battery health and vehicle performance. The system uses Artificial Intelligence (AI) to recommend e-bikes based on user preferences such as budget, travel distance, speed, and battery capacity across multiple brands.It also integrates an ESP32 microcontroller, smart Battery Management System (BMS), current and temperature sensors, and an LCD/mobile interface to monitor real-time parameters including battery percentage, voltage, current, temperature, and estimated range. The collected data enables predictive battery health analysis and provides alerts for overheating, low battery, and abnormal operating conditions.This project improves decision-making, enhances battery safety, reduces maintenance costs, and promotes efficient adoption of electric mobility through a smart, user-friendly platform.

Keywords: Artificial Intelligence, E-Bike, ESP32, Smart BMS, Battery Monitoring, Multi-Brand Recommendation.

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AUTOMATED COMMERCIAL SOLID WASTE SEGREGATION OPTIMIZATION DASHBOARD FOR MUNICIPAL RECYCLING PLANTS

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Abstract

Municipal solid waste management requires coordinated waste identification, collection, transportation, processing and operational monitoring, yet fragmented records can limit visibility into waste composition and plant-level activities. This paper presents WasteDash, an integrated dashboard prototype for supporting commercial solid-waste segregation and municipal recycling operations. The system combines a web-based dashboard with AI-assisted image analysis and structured operational records. Its computer-vision pathway incorporates ResNet50-based image classification and YOLO-based object detection through a FastAPI service. The dashboard brings together waste-category summaries, ward and processing-plant information, collection-vehicle records, alerts and contribution workflows, with role-based access for different user groups. The prototype uses locally seeded development data covering 15 zones, 200 wards, 12 processing plants and 24 collection vehicles to demonstrate its operational views and workflows. These records are seeded for development and do not constitute a live municipal data feed; live GPS/IoT integration and verified real-world performance have not yet been established. Therefore, the current work demonstrates the feasibility of integrating AI-assisted waste analysis with municipal operational monitoring, rather than a measured improvement in segregation accuracy or recycling efficiency. Future work will focus on validation using labelled waste images, reporting per-class detection performance and inference latency, and integrating authenticated live operational data.

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I-Powered Tamil Nadu Welfare Schemes: A Mobile-First Platform for Scheme Discovery and Eligibility Guidance

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Abstract

Government welfare schemes support citizens in areas such as education, health, agriculture, livelihood and social protection, but information can be distributed across multiple websites and departments. Complex wording, language barriers, uncertainty about eligibility and confusion over required documents can make it difficult for people to identify suitable programmes and prepare applications. This paper presents the concept of an AI-powered, mobile-first Tamil Nadu Welfare Schemes platform designed to bring scheme discovery and guidance into one interface. The proposed platform allows users to browse schemes by category or keyword, review likely eligibility by comparing user-provided details with published criteria, and access document checklists based on official guidance. A conversational assistant is intended to explain information in simple Tamil or English and direct users to relevant government sources and application channels. Source links and review dates are included to support verification and reduce reliance on outdated or unofficial information. The platform is designed to improve discoverability, understanding and application preparation; it does not make official eligibility decisions or replace government approval. Future work includes validating scheme details against official sources, developing the mobile interface, and testing usability, clarity and navigation with target users.

Keywords : Tamil Nadu welfare schemes, artificial intelligence, mobile application, eligibility guidance, government services, Tamil-English accessibility, scheme discovery 1

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Decision-Aware Evaluation of ARX, ARMAX and State-Space Models for Soil Moisture Forecasting and Irrigation Scheduling

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Abstract

Efficient irrigation scheduling requires accurate soil moisture forecasting to optimize water application and minimize crop water stress. This study presents a decision-aware framework for evaluating AutoRegressive with eXogenous inputs (ARX), AutoRegressive Moving Average with eXogenous inputs (ARMAX), and State-Space models for soil moisture forecasting and simulated tomato irrigation scheduling. The framework uses 17,544 hourly ERA5-Land reanalysis records from 2024–2025, incorporating soil moisture, air temperature, precipitation, solar radiation, and wind speed. The models are evaluated at 1-hour, 6-hour, and 24-hour prediction horizons using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), coefficient of determination (R^2), and residual diagnostics. ARX achieves the lowest RMSE at 6 hours (0.014647 m^3/m^3) and 24 hours (0.030177 m^3/m^3), while State-Space achieves the lowest 1-hour RMSE (0.005030 m^3/m^3). An illustrative FAO-56-based root-zone water-balance simulation evaluates irrigation timing, water application, and soil-water stress. State-Space achieves the lowest average irrigation timing error (4.33 hours), compared with 5.00 hours for ARX and 7.38 hours for persistence-based scheduling. ARX records a simulated water application of 321.80 mm against 341.35 mm for the reference strategy; however, lower water application does not necessarily indicate better crop water management. Residual diagnostics reveal unexplained temporal dependencies, highlighting limitations in model adequacy. The findings demonstrate that superior forecasting accuracy does not necessarily translate into optimal irrigation decisions, emphasizing the need to evaluate prediction accuracy alongside irrigation timing, water application, and soil-water stress. Since ERA5-Land provides surface soil moisture estimates rather than direct root-zone measurements, the irrigation results are simulation-based and require field validation.

Keywords: Soil Moisture Forecasting, System Identification, ARX, ARMAX, State-Space Models, Irrigation Scheduling, Precision Agriculture.

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Automated Commercial Solid Waste Segregation Optimization Dashboard for Municipal Recycling Plants

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Abstract

Commercial and municipal solid-waste facilities must handle mixed waste streams containing materials such as plastic, paper, metal, organic waste, and non-recyclable items. Reviewing waste records manually can make it difficult to understand the composition of incoming waste, identify contamination, track recycling performance, and decide which recorded bins or waste batches need attention first. This project presents a software-based dashboard designed to support waste monitoring and decision-making in municipal recycling plants. The dashboard provides a centralized interface for managing waste records, including adding, updating, searching, and importing data from CSV files. It organizes records by waste category and presents visual summaries and trends to help users review waste quantities, contamination indicators, and recycling-related information. Configurable alerts can highlight records that meet specified conditions, while a prioritization feature can rank bins or records using the indicators available in the dataset and show the reasons for each priority. An optional image-classification feature can be used to estimate the visible category of an item from an uploaded image or camera input; it does not determine the item's weight and should not be treated as a substitute for physical inspection. The system is intended to work with existing, imported, or manually entered data and does not perform physical waste separation or require dedicated hardware. By combining record management, visual analytics, alerts, and explainable prioritization in one interface, the proposed dashboard aims to help users identify patterns, review operational issues, and make more informed waste-handling and recycling decisions. The reliability of the insights depends on the quality and completeness of the data provided.

Keywords: Solid waste management, waste segregation, recycling analytics, contamination monitoring, decision-support dashboard, CSV data import, image classification.

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Dual-Wavelength Near-Infrared Imaging for Enhanced Vein Visualization

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Abstract:
Vein detection is an important part of medical procedures such as blood collection, IV injection, venipuncture, and cannulation. However, veins are not always clearly visible, especially in patients with difficult vein access. Poor vein visibility can lead to repeated needle attempts, patient discomfort, and increased procedure time. Therefore, a simple and effective method is needed to improve vein visualization. This project proposes a non-invasive vein visualization system using dual-wavelength Near-Infrared (NIR) imaging. The system uses 850 nm and 940 nm NIR light to illuminate the hand and a NoIR camera to capture the vein images. A Raspberry Pi is used to control the system and process the captured images. The images are processed using Python and OpenCV to improve contrast, reduce unwanted noise, and enhance the visibility of the veins. The images captured using 850 nm and 940 nm are compared to identify which wavelength provides better vein visibility. The better image is selected and displayed in real time, allowing healthcare professionals to view the vein pattern more clearly. This can assist in selecting a suitable vein during blood collection and IV procedures. The proposed system is designed to be portable, low-cost, non-invasive, and easy to use. It provides a practical approach for improving vein identification and can be useful in hospitals, clinics, and other healthcare settings. The system can also be further improved by integrating advanced image processing and AI-based vein detection in the future.

Keywords: Near-Infrared Imaging, Dual-Wavelength, Vein Detection, Vein Visualization, Image Processing, OpenCV.

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AI-Driven Road Damage Assessment and Intelligent Maintenance Planning

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Abstract

Road damage, such as potholes and cracks, affects road safety, vehicle performance, and transportation efficiency. Traditional road inspection methods are time-consuming and may not provide consistent information for effective maintenance planning. This work presents an AI-driven framework for road damage assessment and intelligent maintenance planning. The system uses computer vision techniques to detect visible road defects from road images and assess their severity based on the detected damage. Additional contextual information, including road age, traffic intensity, and maintenance history, is considered to support road-condition evaluation and identify segments requiring timely attention. A web-based dashboard is designed to display detected defects, assessment details, inspection information, and recommended maintenance priorities. The framework can also incorporate geographic location data to associate road inspections with real-world locations and support comparisons of road images captured before and after maintenance, where suitable data are available. The proposed system aims to improve road inspection efficiency, support data-driven maintenance decisions, and help authorities identify damaged road segments that require timely intervention. Its performance will be evaluated using annotated road-damage images and appropriate detection metrics.

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A Comprehensive Survey of Recent Phishing Email Attack Detection Techniques

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Abstract

Email remains the principal delivery channel for phishing, and generative artificial intelligence has made phishing messages more fluent, more personalised and cheaper to produce at scale. This paper presents a structured survey of phishing email attack detection techniques reported in peer-reviewed literature published between 2020 and 2026. Using a documented search-and-screening protocol, the selected studies are organised into a taxonomy of eight technique families: feature-engineered machine learning, deep neural networks, transformer-based language models, large language models (LLMs), graph and multimodal hybrids, federated (privacy-preserving) learning, explainable detection, and adaptive or continual learning. For each family, the survey examines the email representation used, the reported performance, the datasets involved and the characteristic failure modes, and it consolidates the evidence in comparative tables of representative studies and benchmark corpora. The synthesis shows that fine-tuned transformer encoders such as BERT, RoBERTa and DistilBERT routinely exceed 98% accuracy on public corpora, and that prompted or fine-tuned LLMs approach this level while also producing natural-language rationales. These figures, however, are obtained largely on ageing, static and English-only datasets, and recent studies report marked degradation under cross-dataset transfer, adversarial rewriting, multilingual input and temporal drift. The paper therefore identifies the open challenges of dataset currency, concept drift and catastrophic forgetting, robustness to LLM-generated phishing, explainability, privacy and computational cost, and it outlines a research agenda centred on adaptive, transformer-based continual learning evaluated under realistic, time-ordered protocols.

Keywords: Phishing email detection, Transformers, Large language models, Continual learning, Concept drift, Explainable AI, Cybersecurity

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Legal Mate AI: A Domain-Specific Indian Legal AI Model for Legal Guidance and Complaint Assistance

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Abstract

The Legal Mate AI is an AI-powered system designed to help users find relevant legal information and better understand Indian laws. The system uses reliable legal sources related to Indian legislation to provide information based on users' queries. For many citizens, understanding legal rights, laws, Acts, sections, and complex legal terminology can be challenging. To address this issue, Legal Mate AI combines Artificial Intelligence (AI), Natural Language Processing (NLP), and Retrieval-Augmented Generation (RAG) to understand user queries and retrieve relevant information from an Indian legal knowledge base. A major part of the project involves developing a customized Indian legal AI model capable of handling legal problem-solving, intent classification, and question-answering tasks. The system aims to simplify legal concepts, explain relevant Acts and sections, assist in analyzing legal documents, and generate legal complaints and evidence checklists. Through these features, Legal Mate AI seeks to make legal information more accessible, understandable, and useful to the general public.

Keywords: Artificial Intelligence, Legal Mate AI, Natural Language Processing, RetrievalAugmented Generation, Indian Legal AI, Legal Information Retrieval, Legal Complaint Generation.

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MRV: Interpreting Medical Reports into Voice in Regional Languages

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Abstract

Understanding medical reports is difficult because of complex terminologies used in medical reports. This study proposes the creation of a healthcare application using AI for interpreting the medical reports and making the reports more user-friendly. Medical reports such as CBC, thyroid, and Pulse Reports can be uploaded by the users into the app and using Optical Character Recognition (OCR), data in the report can be extracted. The Large Language Model (LLM) uses this information to interpret the medical report in simple language. Explanation of this interpretation is given in the form of both text and audio in regional languages using Text-to-Speech (TTS) Technology. In addition, the app provides pictorial explanation of body parts involved in the medical condition, important terms in the medical field, and tutorial videos for the first-time users. In addition to these features, Report History allows the user to view past reports while Health Trend Analysis allows the user to track health parameters.

Keywords: AI, OCR, LLM, TTS, Medical Report Interpretation, Regional Languages, Report History, Health Trend Analysis.

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Drilling Anti-collision Analysis

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ABSTRACT

Well-to-well collision is a significant risk during directional drilling, where maintaining adequate separation between neighbouring well trajectories is essential for safe operations. Conventional anti-collision analysis relies on trajectory calculations, Centre-to-Centre distance measurement, Ellipse of Uncertainty (EOU) analysis and Separation Factor (SF)-based risk assessment. However, these methods may involve time-consuming calculations and have limited predictive and automated anomaly detection capabilities. To address these challenges, this project proposes an intelligent Drilling Anti-Collision Analysis framework that integrates classical anti-collision techniques with Artificial Intelligence and Machine Learning (AI/ML) models. The framework uses a synthetic dataset of 200 well trajectories across five fields for analysis and evaluation. Six classical methods are implemented: Minimum Curvature Method, Centre-to-Centre 3-D Distance, Travelling Cylinder Method, Pedal Curve Method, ISCWSA MWD Rev.4 Ellipse of Uncertainty and Separation Factor calculation. These methods evaluate wellbore trajectories, closest approach, positional uncertainty and separation between wells. Additionally, five AI/ML models, namely Random Forest Classifier, Gradient Boosting Regressor, Isolation Forest, LSTM-based trajectory prediction and Physics-Informed Neural Network (PINN), are employed for risk classification, Separation Factor prediction, trajectory anomaly detection, Dog-Leg Severity prediction and curvature-aware collision analysis. An MCP Server is integrated to provide structured access to analytical functions for Large Language Model (LLM) agents. Based on the combined analytical results, the framework generates risk-based recommendations such as STOP, SLOW, MONITOR and OK. The proposed framework supports intelligent trajectory analysis, collision risk assessment and decision-making in directional drilling.

KEYWORDS: Drilling Anti-Collision, Directional Drilling, Wellbore Trajectory, Separation Factor, Ellipse of Uncertainty, Artificial Intelligence, Machine Learning, MCP Server, Collision Risk Prediction.

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GenVoice Care 2.0: Production-Ready AI-Powered Voice Assistant for Elderly Wellness Monitoring with Real-Time Caregiver Coordination

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Abstract:

A large number of people are getting older around the world.In India, the number of older adults is expected to reach 319 million by 2050, making up about 20% of the total population.At the same time, there are many jobs related to elder care that remain unfilled across the country.Traditional care systems often fail to meet all the needs of the elderly at once. They lack effective monitoring and fail to assist in proper family care management. In order to solve these problems, we created GenVoice Care 2.0. We have also put in place.This is a mobile application system designed for real-world use.It offers voice-based AI chat, live wellness assessments, and support for caregiver coordination.The system operates on the Railway cloud platform.It consists of three interconnected components.First, there is an Android app specifically tailored for seniors.It uses natural voice input through Android's SpeechRecognizer API.Second, there is a Backend server that runs on cloud using FastAPI framework with total 14 ready-to-use REST API end-points.Third, there is an Android caregiver's dashboard with color-coding and real time monitoring system.Forth, for voice communication part, we have used the Groq'sexternal GPT-OSS-120B model.During testing, it provided responses within 2 to 3 seconds, enabling a more natural and smooth interaction for older adults.For detecting health issues, we implemented a rule-based keyword check.It matches spoken words against six medical terms: pain, hurt, ache, bad, terrible, and awful.It then assigns a wellness score on an 8-point scale, with 8 for Good, 6 for Fair, and 4 for Poor.The backend showed 99.6% API response reliability.During production testing, average response times remained between 600 to 700 milliseconds across all endpoints.We also tested keyword detection.We conducted seven check-in scenarios.Each one resulted in the correct classification.This yielded 100% accuracy.Of these, five scenarios involved health concerns, and they were correctly identified.The other two scenarios did not involve health concerns, and they were also correctly identified.This led to zero false positives and zero false negatives.The caregiver dashboard displays alerts in three priority levels: red for high priority, orange for medium, and green for normal.Data updates after a manual refresh and retrieval take about three seconds.This system is currently in production and is accessible at devoted-miracle-production-23ef.up.railway.app.It is available 24 hours a day, seven days a week.

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Edge-AI Framework for Industrial Safety Monitoring Using IoT and Machine Learning

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Abstract

Industrial environments face safety risks from gas leakage, elevated temperature, and abnormal vibration. This paper presents an Internet of Things (IoT)-enabled edge-AI framework for real-time industrial hazard monitoring using DHT11, MQ-3, and vibration sensors interfaced with a Raspberry Pi 3 Model B+. A Random Forest classifier categorizes operating conditions into Safe, Normal, Moderate, and Hazardous states. The system integrates a Flask monitoring dashboard, MQTT-based communication, and buzzer- and relay-based emergency responses. Experimental evaluation reported 98% classification accuracy on benchmark data and 97% on live sensor data. The real-time monitoring dashboard further reported an accuracy of 97.91%, precision of 97.30%, recall of 97.91%, F1-score of 97.60%, response time of 102.76 ms, and false alarm rate of 2.17%. In a recorded hazardous scenario, temperature reached 46.3 °C, humidity 84%, gas reading 3.8, and vibration status 1, triggering the alarm and machine shutdown. These results demonstrate the potential of local multi-sensor classification for timely industrial safety monitoring with reduced dependence on continuous cloud processing. Keywords: Industrial IoT, Random Forest, Edge Computing, Raspberry Pi, Hazard Detection, Real-Time Monitoring, MQTT.

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Dataset-Size and Data-Generation Effects on Machine Learning Prediction of Tensile and Flexural Strength in PEEK Composites

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Abstract

The prediction of mechanical properties of poly (ether ether ketone) (PEEK) composites based on composition and process-related parameters could help to avoid excessive expenses and time required for traditional experimentation by trial and error. Still, the lack of data and features of the datasets themselves might significantly affect the machine learning (ML) results. This research aims to provide a stable and leak-proof framework for the ML-based prediction of tensile and flexural strength of PEEK composites on the basis of synthetic datasets derived from literature data. Ten regression models were compared through cross-validation and independent testing. For the 25,000 rows literature dataset, Hist Gradient Boosting showed the best performance in terms of the highest value of test R^2 (0.9516 and 0.9544 for tensile and flexural strength), while its RMSE value was equal to 5.43 and 8.26 MPa, correspondingly. The six-seed analysis proved its high stability (the mean value of R^2 was 0.9491 ± 0.0025 and 0.9542 ± 0.0023). On the contrary, the much smaller 5,200 rows literature-based datasets led to the significantly lower performance (test R^2 value was around 0.72 and 0.66) and confirmed the impact of the dataset size and its generation features on ML performance.

Keywords: PEEK composites, Machine learning, Tensile strength, Flexural strength, Predictive modeling.

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When Do Sparse and Contextual Representations Transfer? Evidence from a Controlled Zero-Shot and Few-Shot Text Classification Case Study

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Abstract

Sparse lexical features, frozen contextual embeddings, and hybrid combinations of the two encode different, only partly overlapping properties of a document, yet most empirical text-classification studies still look for one representation that wins everywhere. This paper compares three representations under one shared LinearSVC classifier [1]: a word- and character-level TF-IDF representation (B10), a frozen DistilBERT embedding (P1) [2], and a hybrid formed by standardizing and concatenating the two (AUG5). All three are evaluated on a controlled transfer setup from IMDb [3] to SST-2 [4], under strict zero-shot transfer and under classifier-only few-shot adaptation at five target-label budgets per class (10, 50, 100, 500, and 1000). Averaged over five seeds, the hybrid representation reaches the best zero-shot macro-F1 (0.7907) and the best macro-F1 at 1,000 target labels (0.8335); its point-estimate lead over the sparse baseline is positive at every budget tested, though at 10 labels the lead cannot be distinguished from zero given the observed variance. The frozen contextual representation is nominally ahead at the intermediate budgets of 10, 50, 100, and 500 labels, but an approximate Welch's t-test computed from the reported per-seed statistics shows this lead reaching conventional significance ($p < 0.05$) only at 100 labels. We also report accuracy, weighted-F1, the Matthews correlation coefficient, AUROC, and observed runtime for all three representations. Because these results come from a single source-target pair of closely related English-language movie-review corpora, we present them as an initial, appropriately scoped case study rather than a general rule, and set out the additional domain pairs, encoders, and ablations that a broader test would require.

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Lightweight Post Quantum Cryptography in Federated Learning

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Abstract:

Federated Learning enables devices such as wearable devices, smart sensors and connected cars to learn a model while preserving its privacy. This only works if the channels carrying updates to the model are made secure. Today's protocols for those channels rely on RSA or elliptic-curve cryptography, both of which will be broken once a quantum computer powerful enough to run Shor's algorithm exists. Until that happens, attackers can interrupt encrypted traffic and store it, intending to decrypt it when such a machine becomes available.This kind of attack will especially have a long-term impact on FL applications in which data confidentiality must be maintained over a long period of time (such as healthcare). Moreover, quantum-capable attackers could forge digital signatures in order to compromise the integrity of the FL model. Post-quantum cryptography is a promising area of study that aims at defining cryptographic algorithms secure against quantum attacks. PQC algorithms usually require significantly more data to be transmitted as compared to their classical alternates. The impact is greatest on less resource devices, where every byte of payload transmitted over MQTT, CoAP, or 6LoWPAN consumes precious energy and increases the latency of a federated learning round. FL is especially sensitive to this issue as these devices are used in scenarios to communicate the model updates to the server and to synchronize with others in the case of decentralized FL. This critical analysis presents a survey of published research on lightweight PQC and aims at answering the following questions:

• How to reduce the overhead induced by the authentication layer?
• How to enhance the implementations to resist side-channel analysis?
• How to secure FL communication on constrained devices?

This work neither introduces a new protocol nor runs original benchmarks. Instead, it brings together results from several studies to expose patterns that are hard to see in any one paper. The review shows that, in practice, no single operation is the most computationally expensive. Most of the algorithms in the reviewed studies would clearly be too demanding for a lesser resources device, because storing and processing them takes a lot of memory, yet how they are implemented still makes a large difference. The implementation flaws can lead to information leakage, which is a major concern for physically accessible constrained devices. We also find that such devices have enough processing power to perform the necessary mathematical operations, but their limited bandwidth and the need to process many messages in an FL round makes the key exchange and message authentication code a bottleneck.

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With this insight, we propose a set of recommendations that would allow us to secure FL applications against leakage of model information and eavesdropping, while minimizing the performance impact. To this end, we advocate using a stack with different layers, each contributing according to its strengths. Using physical unclonable function (PUF)-based authentication allows to authenticate the device without exposing any long-term secrets to be stored on the device itself. The ML-KEM-512 key encapsulation mechanism is used to establish a shared session key only once, which then allows it to perform the steps in the protocol using AES-256-GCM with its lower overhead. The Falcon-512 signature algorithm is applied to the occasional message that requires confidentiality and authenticity, instead of using it for all messages. Masking guards against side-channel attacks, and the network is built around gateways that take on the heaviest computation. The core idea is not to change costly PQC algorithms for classical cryptography methods. However, it is to spend the available resources where each layer gains the most security.

We also encourage adding security for the data that comes from model updates like gradients or weights. Encrypting the channel cannot stop attacks such as membership inference or gradient inversion. Protecting the updates is a problem from protecting the channel and methods such, as secure aggregation or differential privacy are the solutions.

We keep our claims modest. Our conclusions come from a set of research and those studies ran on various hardware and used different performance metrics. Thus, the comparison is only approximately showing trends and trade-offs and not the exact differences. To improve this we suggest that researchers agree on hardware and common performance metrics when testing lightweight PQC for FL so that the results can be compared equally and the field can move forward in more advanced and in-depth research.

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Non-invasive Behavioral Health Monitoring of Software Developers via Source-Code Analytics and Psychometric Mapping

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Abstract :

In the modern digital era, software developers spend a significant portion of their lives immersed in coding environments, making their code not only a product of logic but also a subtle reflection of their mental and physiological states. CodeSavior AI introduces a groundbreaking perspective that extends the frontier of AI-driven health diagnostics into the behavioral domain of software development patterns. This research aims to identify early indicators of mental strain, cognitive overload, or physiological decline through the analysis of developers' coding behaviors, providing a non-invasive and data-driven health monitoring system. Unlike traditional health assessment techniques that rely on biomedical signals, wearable sensors, or clinical data, CodeSavior AI explores the latent health indicators hidden within the source code itself. Developers often exhibit unconscious variations in their coding styles under stress or fatigue—ranging from inconsistent naming conventions and increased syntax errors to erratic commit patterns and abrupt changes in structural logic. These micro-level irregularities, when tracked over time, can serve as digital biomarkers that reflect shifts in cognitive and emotional well-being. The proposed system integrates machine learning, natural language processing (NLP), and deep learning architectures to model and interpret these behavioral patterns. A hybrid framework combining Transformer-based sequence modeling with Auto- encoder-based anomaly detection enables the system to detect subtle deviations in temporal coding behavior. Furthermore, psychometric mapping links the detected anomalies with validated psychological indicators, allowing for the prediction of potential stress, burnout, or early neurological disorders. Beyond academic contribution, CodeSavior AI offers two major real-world applications. At the organizational level, it can function as an enterprise well-being tool, helping companies monitor developer burnout and sustain productivity. In the healthcare domain, it serves as a supplementary diagnostic instrument, offering clinicians an additional layer of behavioral insight derived from coding activity. By merging computational behavior analysis with AI-driven health inference, CodeSavior AI pioneers a new paradigm in digital mental health research—one that transforms everyday coding patterns into meaningful signals for preventive healthcare. This innovative approach has the potential to reshape both developer wellness management and AI-based behavioral health diagnostics, marking a significant leap toward a future where software itself becomes a mirror of the human condition.

Keywords: Code Behavior Analysis, Developer Health, Digital Biomarkers, Anomaly Detection, AI in Healthcare

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IoT-Based Smart Obstacle Avoiding Robot Using Arduino UNO and ESP32 for Real-Time Remote Monitoring

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Abstract

The autonomous small-scale robot may cost too much, be too difficult to make and lack any remote realtime control. The presented work offers the IoT based smart obstacle avoiding robot, which uses inexpensive materials. For the robot movement control the local microcontroller Arduino UNO board with the Adafruit Motor Shield V1 (L293D) was chosen and four TT geared DC motors are used. The robot is equipped by HC-SR04 ultrasonic sensor with SG90 servo mounted on it for measuring the distance ahead. When the obstacle appears near enough within the 30 cm safe distance the robot stops and moves backwards slightly to check left and right obstacles and then moves towards free paths. The ESP32 Wi-Fi module takes the data from the UNO concerning the distance and status of the robot and transfers it to the ThingSpeak cloud dashboard to provide remote monitoring of the robot status. There are two independent 6 V and 5 V buck voltage converters for motors and logic boards correspondingly to increase power stability. The robot is reactive, lightweight and suitable for educational and inspection route at the warehouse aisles, but it is not a safety critical industrial vehicle.

Keywords: Arduino UNO; ESP32; Internet of Things (IoT); Obstacle Avoiding Robot; HC-SR04 Ultrasonic Sensor; L293D Motor Driver; ThingSpeak

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GenVoice Care 2.0: Production-Ready AI-Powered Voice Assistant for Elderly Wellness Monitoring with Real-Time Caregiver Coordination

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Abstract :

A large number of people are getting older around the world.In India, the number of older adults is expected to reach 319 million by 2050, making up about 20% of the total population.At the same time, there are many jobs related to elder care that remain unfilled across the country.Traditional care systems often fail to meet all the needs of the elderly at once. They lack effective monitoring and fail to assist in proper family care management. In order to solve these problems, we created GenVoice Care 2.0. We have also put in place.This is a mobile application system designed for real-world use.It offers voice-based AI chat, live wellness assessments, and support for caregiver coordination.The system operates on the Railway cloud platform.It consists of three interconnected components.First, there is an Android app specifically tailored for seniors.It uses natural voice input through Android's SpeechRecognizer API.Second, there is a Backend server that runs on cloud using FastAPI framework with total 14 ready-to-use REST API end-points.Third, there is an Android caregiver's dashboard with color-coding and real time monitoring system.Forth, for voice communication part, we have used the Groq'sexternal GPT-OSS-120B model.During testing, it provided responses within 2 to 3 seconds, enabling a more natural and smooth interaction for older adults.For detecting health issues, we implemented a rule-based keyword check.It matches spoken words against six medical terms: pain, hurt, ache, bad, terrible, and awful.It then assigns a wellness score on an 8-point scale, with 8 for Good, 6 for Fair, and 4 for Poor.The backend showed 99.6% API response reliability.During production testing, average response times remained between 600 to 700 milliseconds across all endpoints.We also tested keyword detection.We conducted seven check-in scenarios.Each one resulted in the correct classification.This yielded 100% accuracy.Of these, five scenarios involved health concerns, and they were correctly identified.The other two scenarios did not involve health concerns, and they were also correctly identified.This led to zero false positives and zero false negatives.The caregiver dashboard displays alerts in three priority levels: red for high priority, orange for medium, and green for normal.Data updates after a manual refresh and retrieval take about three seconds.This system is currently in production and is accessible at devoted-miracle-production-23ef.up.railway.app.It is available 24 hours a day, seven days a week.This supports the notion that voice-based monitoring for elderly users can function effectively on a larger scale.The current version is a working prototype in a cloud environment.

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A Computer Aided System for Cervical Cancer Detection Using Image Processing: A Review

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Abstract

Cervical cancer is among the major health problems affecting women across the world. Early diagnosis of cervical cancer is an important factor in improving prognosis for successful therapy. The authors of this work have focused on optimizing the diagnosis of cervical cancer using standardized methods. The suggested method is used in this study to classify and analyze colposcopy images into normal, precancerous and cancerous tissues at different phases which includes image segmentation, feature extraction and classification. The paper also demonstrates that the fusion of clinical data with image-based characteristics could reduce screening mistakes and improve global diagnostic performance. In the present work, an attempt is taken to help in accurate and reliable diagnosis by investigating several machine learning algorithms and sophisticated image processing methods. The results demonstrate that computer-aided diagnostic techniques may be advantageous to doctors, particularly when combined with large screening programs, to provide faster and more accurate results. This study shows the immense potential of the combination of image processing and machine learning for enhancing detection of cervical cancer, reduce the burden of diagnosis and help patients in better treatment planning.

Keywords: Pap smear, Segmentation, Feature Extraction, Classification, ComputerAided Diagnosis

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When Does Hybridization Help? A Multi-Dataset Empirical Analysis of Individual, Lexicon-Based, and Hybrid Classifiers for Social Media Text Classification

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Abstract

Hybrid machine-learning models that combine several base classifiers are routinely presented as a default improvement over single classifiers in social-media text classification, yet this assumption is rarely tested across heterogeneous datasets or benchmarked against the specific component it is meant to improve upon. We report a controlled study of five public social-media text-classification datasets differing in task, size, class cardinality, and imbalance: a short-text sentiment corpus (584 instances, 3 classes), an 11-class Twitter affect corpus (39,711 instances), a hate-speech/offensive-language corpus (24,783 instances, 3 classes), a 5-class review-sentiment corpus (61,594 instances), and a 6-class emotion corpus (20,000 instances). For every dataset we compare up to five individual classifiers, a lexicon-based classifier, and every two- and three-component decision-level hybrid built from them, each evaluated over five seeds and benchmarked against the strongest individual classifier available for that dataset. The best hybrid improved on the best individual classifier by up to +0.068 macro-F1 on the smallest dataset but degraded performance by up to −0.018 macro-F1 on the largest. Of 40 evaluated two-component pairs, only six (15.0%) produced a practically meaningful gain (≥0.005 macro-F1), one (2.5%) was neutral, and thirty-three (82.5%) degraded performance. Paired t-tests, Wilcoxon signed-rank tests, Holm and Benjamini–Hochberg corrections across sixty comparisons, Cohen's d, and learning curves converge on one picture: several nominally significant gains did not survive correction for multiplicity, whereas several large degradations did. A component-level ablation shows that the lexicon component responsible for the largest positive contribution on one dataset (+0.071 macro-F1) was the most damaging component on another (−0.137 macro-F1) when

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added to the same base classifier. A 40-pair diversity analysis shows prediction agreement ($r = 0.327$, $p = 0.039$) and Cohen's κ ($r = 0.502$, $p = 0.001$) correlate positively, not negatively, with hybrid gain, while a complementarity index correlates negatively ($r = -0.500$, $p = 0.001$) — contradicting the assumption that maximal component diversity is what makes hybrids succeed. Hybridization in social-media text classification is therefore a conditional design choice, not a universally beneficial one; we propose a diagnostic checklist for deciding when to hybridize.

Keywords: hybrid classifiers, ensemble diversity, social media text classification, lexicon-based sentiment analysis, hate speech detection, effect size, statistical significance, macro-F1

AI-Driven Clinical Decision Support for Autism Severity Grading: An Interpretable and Uncertainty-Aware Neural Additive Ordinal Model

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Abstract:

Assigning a Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) severity level to a child with Autism Spectrum Disorder (ASD) determines the intensity of support the child receives, yet the judgement is made by clinicians integrating heterogeneous and frequently incomplete information. Machine learning models that support this decision must therefore be accurate, transparent, honest about their uncertainty, and evaluated without data leakage. This paper proposes MA-ONAM, a Missingness-Aware Ordinal Neural Additive Model that learns one small neural network per clinical feature, assigns a learned contribution to missing values instead of imputing them, and produces rankconsistent ordinal predictions of Level 1, Level 2 and Level 3 through a CORAL output layer. Split-conformal prediction converts the model output into prediction sets with a guaranteed 90% coverage. The model is evaluated on the public ASD-SL dataset of 141 clinically assessed children. An audit of the dataset revealed ten pairs of records belonging to the same child or to twins and 47 censored developmental scores; all experiments therefore use childgrouped, stratified 5-fold cross-validation repeated ten times, and nine models are compared with and without the DSM-5 specifier variables. MA-ONAM reached an accuracy of 0.620, a macro-F1 of 0.614 and a quadratic weighted kappa

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of 0.511, statistically comparable in macro-F1 to Logistic Regression (0.633) and Random Forest (0.623), and below XGBoost (0.643). It achieved the highest recall for the minority Level 1 class (0.681), and its learned shape functions recovered clinically coherent relationships: severity rises monotonically with the ADOS comparison score and falls with developmental quotient and adaptive behaviour. A conventional multilayer perceptron failed on this small dataset (macro-F1 0.257). The results show that, on small clinical datasets, an inherently interpretable deep model can match black-box accuracy while exposing its reasoning and its uncertainty, and they establish a leakage-controlled benchmark for ASD-SL. Keywords: Autism spectrum disorder, DSM-5 severity level, neural additive model, ordinal regression, conformal prediction, interpretable machine learning, ASD-SL

Consequence-Aware Closed-Loop Control for Underwater Image Enhancement

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Abstract

Underwater images often suffer from color distortion, reduced contrast, and loss of visual details due to light absorption and scattering. Conventional enhancement methods typically apply a fixed transformation, which may not adapt effectively to varying image degradation conditions. This paper proposes a consequence-aware closed-loop control framework for underwater image enhancement that iteratively adjusts enhancement strength based on the observed consequences of the current enhancement. The framework integrates a continuous convolutional neural network-based enhancer with a lightweight controller that analyzes image descriptors and enhancement-related changes to select actions for decreasing, maintaining, or increasing enhancement strength. This feedback-driven strategy enables adaptive refinement while retaining a frozen enhancement backbone. The proposed approach was evaluated on 1,716 paired test images from the EUVP dataset and compared with a fixed-strength CNN baseline and alternative enhancement backbones. The closed-loop controller achieved a mean peak signal-to-noise ratio (PSNR) of 22.7551 dB and a structural similarity index (SSIM) of 0.8620, compared with 20.5418 dB and 0.8469, respectively, for the fixed-strength baseline. The results indicate that consequence-aware closed-loop control can improve reference-based reconstruction quality over

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fixed-strength enhancement, demonstrating the potential of feedback-driven enhancement for underwater imagery.

Keywords: Underwater Image Enhancement, Closed-Loop Control, Consequence-Aware Learning, Convolutional Neural Networks, Adaptive Enhancement.

CNN-BASED SMART IC IDENTIFICATION, INSPECTION AND FUNCTIONAL TESTING SYSTEM

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Abstract

The project titled "CNN-Based Smart IC Identification, Inspection and Functional Testing System" aims to develop an intelligent and automated system for identifying, inspecting, and testing integrated circuits (ICs). Traditional IC inspection and testing methods require manual effort, consume more time, and may lead to human errors. To overcome these limitations, the proposed system uses Convolutional Neural Networks (CNN) and computer vision techniques to identify ICs automatically from captured images. The system processes the images, recognizes the IC model number using Optical Character Recognition (OCR), and detects and counts the pins to verify their arrangement. It also inspects the IC for physical defects such as damaged pins, cracks, and surface abnormalities. After identification and inspection, a functional testing circuit checks whether the IC performs its intended operation correctly. The test results are processed and displayed through a graphical user interface (GUI), indicating whether the IC has passed or failed the testing process. The system can also store inspection results and generate reports for

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future reference. By integrating image processing, deep learning, OCR, and functional testing, this project provides a systematic approach to IC quality assessment. The proposed system reduces manual inspection, improves testing speed and consistency, minimizes human errors, and supports more reliable electronic component testing in laboratories and electronic manufacturing industries.

AI-ENABLED ZERO TRUST INTRUSION DETECTION AND POST- QUANTUM CRYPTOGRAPHIC SECURITY FOR INDUSTRIAL INTERNET OF THINGS (IIoT)

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Abstract:
The rapid growth of the Industrial Internet of Things (IIoT) has transformed industrial automation by enabling real-time monitoring, intelligent control, and interconnected communication among industrial devices. However, the increasing connectivity of industrial systems has introduced significant cybersecurity challenges, including sophisticated intrusion attacks, unauthorised access, identity spoofing, and vulnerabilities associated with emerging quantum computing technologies. Conventional security mechanisms often lack the adaptability, continuous verification, and quantum resistance required to protect modern industrial environments. To address these challenges, this paper proposes an integrated Artificial Intelligence (AI)-enabled Zero Trust security and Post-Quantum Cryptography (PQC) architecture for secure IIoT environments. The proposed architecture incorporates machine learning-based intrusion detection to identify malicious network traffic and anomalous device behaviour, while the Zero Trust framework continuously evaluates device identity, behavioural

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patterns, and contextual risk before granting access to critical industrial resources. Furthermore, standardised post-quantum cryptographic algorithms, including Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM) and Module-Lattice-Based Digital Signature Algorithm (ML-DSA), are integrated to strengthen secure key establishment and device authentication against potential quantum-enabled attacks. A risk-adaptive security mechanism is designed to support dynamic access control, suspicious-device isolation, and intelligent threat mitigation while considering the latency and resource constraints of industrial systems. The proposed framework can be evaluated using industrial network datasets and a simulated IIoT environment, with performance assessed through intrusion detection accuracy, precision, recall, false-positive rate, authentication latency, computational overhead, and attack mitigation time. The anticipated contribution is a unified and scalable security architecture that combines intelligent threat detection, continuous trust verification, and quantum-resistant cryptographic protection to improve the confidentiality, integrity and availability of industrial IoT systems.

RURAL SELF-HELP GROUP (SHG) MICRO-LOGISTICS MANAGEMENT AND LOCAL E-COMMERCE ORDER AGGREGATION NETWORK

Abstract

Rural women-led Self-Help Groups (SHGs), farmers and micro-entrepreneurs often produce vegetables, pickles, candles, handicrafts, snacks and other locally made products, but their market reach can remain limited by digital literacy, fragmented demand, transportation difficulties and dependence on conventional local selling channels. This paper proposes a physical-digital Rural SHG Micro-Logistics Management and Local E-Commerce Order Aggregation Network to connect rural producers with nearby customers via locally operated hubs. The proposed model does not treat e-commerce as only a website; instead, each hub acts as a collection, digital-assistance, inventory, aggregation and local distribution point. Producers who are unable to independently use digital platforms can bring their products to a nearby hub, where trained local staff create seller accounts, photograph and list products, update stock and assist with order fulfilment. Customers can view currently available products and place orders through the online order interface. Orders from nearby customers can be aggregated to reduce unnecessary delivery trips. The model also introduces freshness-based inventory management for perishable products and promotional pricing for suitable near-expiry stock. The proposed system

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aims to improve local market access, create rural employment, support women entrepreneurship, improve inventory movement and provide an additional revenue channel for rural producers.

CareerCompass AI: Intelligent Skill Demand Aggregation and Career Path Recommendation

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Abstract

Students often face challenges in identifying suitable career paths based on their existing skills and understanding the competencies required to achieve their career goals. This paper presents CareerCompass AI, an intelligent career guidance and skill development system that combines machine-learning-based career prediction with an AI-assisted personalized learning roadmap. The career recommendation module processes student-provided skill keywords using Term FrequencyInverse Document Frequency (TF-IDF) feature extraction and classifies skill profiles into 11 job-role categories using machine learning. Experimental evaluation on a dataset of 440 records showed that Logistic Regression achieved a mean accuracy of 78.86% (±1.54%) under stratified five-fold crossvalidation. The module also provides prediction confidence scores, up to

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three alternative career recommendations when confidence falls below 40%, and skill-gap analysis through matched skills, missing skills, and skill-coverage estimates. To support continuous skill development, the learningroadmap module incorporates AI-generated prerequisite-based knowledge graphs, learning resources, practice questions, answer evaluation, and progress-based next-step recommendations. By connecting career-role prediction with structured learning guidance, CareerCompass AI aims to help students make informed career decisions and systematically address their skill gaps. However, the current prediction model is limited to the job roles represented in its dataset and does not incorporate real-time labour-market demand. Unified end-to-end integration and validation of the complete system remain ongoing work.

keywords: Career Recommendation, Machine Learning, TF-IDF, Skill-Gap Analysis, Personalized Learning, Knowledge Graph.

Sustainable Product Design Assistant with Real-Time Recyclability Analysis and Material Recommendation

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Abstract :
The growing use of diverse materials in manufactured products has increased the need for sustainable product design approaches that consider recyclability at an early stage. Selecting recyclable materials during product development can help reduce waste and support environmentally responsible design decisions. However, assessing recyclability can be challenging when products contain multiple materials with different recycling characteristics. This paper presents a Sustainable Product Design Assistant with Real-Time Recyclability Analysis and Material Recommendation, a web-based application developed to evaluate product recyclability based on material composition and component weights. The proposed system calculates a recyclability score using predefined material recyclability indices, material weight percentages, and penalties for contamination and mixed-material composition. It provides immediate feedback on the assessed product and supports what-if simulation to compare alternative material choices and identify potential improvements. The application also enables

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product information management and the generation of assessment reports, helping users review and refine product designs systematically. The system is implemented using React.js for the frontend, Node.js and Express.js for backend processing, and MongoDB for data storage. Its rule-based scoring approach provides consistent assessments based on configured material parameters and user-provided product data. By integrating recyclability scoring, material comparison, and recommendation support into a single platform, the proposed application aims to assist designers in considering recyclability during early-stage product development. The resulting score is intended as an indicative design-support measure rather than a substitute for a comprehensive life-cycle assessment or formal environmental certification.

Keywords: Sustainable Product Design, Recyclability Assessment, Material Selection, Material Recommendation, Rule-Based Scoring, Web Application.

VIKSHANA: An Evidence-Provenance Framework for Traceable AI-Assisted Crime Investigation

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Abstract

Investigators often work with records stored separately across cases, entities, locations, and investigation notes. Search tools and AI-generated summaries can reduce manual comparison, but they may blur the difference between a fact recorded in a source and a relationship inferred from shared details. VIKSHANA is an AI-assisted crime-intelligence platform that combines natural-language querying and retrieval-augmented generation (RAG) with similar-case retrieval, relational-record analysis, DBSCAN-based spatial clustering, trend analysis, and investigation summaries. This work proposes an investigation-specific Evidence Provenance Ledger that connects each reported lead to its source records and the analytical operations that produced it. The ledger will classify outputs as sourcerecorded facts, system-derived associations, or unverified leads; attach record-level citations; flag missing or conflicting support; and withhold definitive summaries when the available evidence is insufficient. The aim is to preserve the path

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from an investigator's question to retrieved records and downstream analytical results, rather than treating all generated statements as equally established. The proposed layer will be evaluated against the existing RAG workflow using a controlled set of synthetic or appropriately authorized, de-identified records. Measures will include claim-level evidence support, citation correctness, correct classification of facts and associations, unsupported-lead rate, and abstention quality. Any performance improvement will be reported only after testing. VIKSHANA is intended to assist professional review, not determine guilt or make operational decisions. Keywords: crime intelligence; retrieval-augmented generation; evidence provenance; analytical leads; responsible AI

An Explainable Multilingual Framework for Grievance Categorization and Automated Departmental Dispatch in Government Portals

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Abstract

Government grievance portals receive complaints covering a wide range of public services, making manual classification and departmental assignment time-consuming and difficult to manage consistently. These challenges become more pronounced when citizens submit grievances in different languages or use informal expressions. This paper presents an intelligent multilingual grievance management framework that combines machine learning-based categorization, priority assessment, explainable predictions, and automated departmental routing within a unified workflow. The system applies language and script detection to incoming text, followed by text preprocessing and feature extraction using Term Frequency–Inverse Document Frequency (TF-IDF). A Logistic Regression classifier predicts one of nine grievance categories, while a separate priority assessment component considers category severity and urgency-related terms. Based on the predicted category, the framework recommends the responsible department and provides feature-based explanations to support transparency. The application also incorporates role-based access control for citizens, officers, and administrators, enabling controlled grievance tracking, department-specific handling, and authorized correction of

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classification results. The prototype has been evaluated using 675 synthetically generated grievance samples across nine categories, alongside automated software tests. Since the dataset is synthetic, the evaluation provides preliminary evidence of implementation feasibility rather than proof of generalization to real government complaints. The framework demonstrates an integrated approach to explainable grievance triage and dispatch, with further validation on authentic, multilingual grievance data required before operational deployment.

Keywords: Multilingual NLP, Grievance Management, Text Classification, TF-IDF, Logistic Regression, Explainable AI, Priority Assessment, Automated Departmental Routing.

Edge AI-Based Energy Monitoring and Intelligent Load Optimization for Small Industries

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Abstract

The process of smart energy management is gaining importance since small scale industries are looking for ways to cut costs and maximize energy efficiency. Traditional energy monitoring systems mainly rely on cloud-based computing, which causes delays in communication and a decrease in the level of reliability at times when there are problems with connection. The study provides a solution in the form of the Edge AI-Based Smart Energy Monitoring and Intelligent Load Optimization that is using Raspberry Pi or ESP32 for edge data processing in real-time. Energy consumption data is taken from voltage, current and power sensors so that the system can analyze the information and identify abnormal energy consumption patterns and optimize load scheduling at the same time. Only the summarized information and alerts about problems

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are sent to the cloud so that the system would have a minimum need in bandwidth and dependency on the internet connection in order to achieve faster response. As a result, the proposed system leads to increased energy efficiency, all the while minimizing the latency and monitoring.

Keywords—Edge AI, Internet of Things (IoT), Smart Energy Monitoring, Intelligent Load Optimization, Raspberry Pi, ESP32, Small Industries, Energy Management.

Design and Implementation of a Microcontroller-Based Women's Safety System Utilizing Vital Sign Sensing and Local Alerting

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Abstract

Women’s personal safety is an important concern, particularly in public and isolated environments where immediate assistance may not be readily available. This paper presents a Smart Device for Women Safety based on the Arduino UNO microcontroller. The proposed system integrates a TTP223 touch sensor, heartbeat sensor, and respiration sensor to provide both manual and automatic emergency detection. The touch sensor enables the user to manually activate an emergency alert, while the heartbeat and respiration sensors continuously monitor vital parameters. The Arduino UNO processes the sensor inputs and compares the readings with predefined conditions to identify emergency or abnormal situations. When an emergency

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condition is detected, the system activates a buzzer and displays the corresponding warning or sensor status on a 16×2 I2C LCD. The proposed device provides a compact, cost-effective, user-friendly, and real-time safety solution that can be applied during night travel, in workplaces and colleges, and for basic health emergency monitoring. The system can be further enhanced by integrating GPS, GSM, mobile applications, and IoT-based communication for remote emergency notification.

AN ATTENTION-ENHANCED HYENA NETWORK FOR AUTOMATED FETAL ULTRASOUND IMAGE CLASSIFICATION

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Abstract

Fetal ultrasound imaging is an essential diagnostic tool for monitoring fetal growth and detecting abnormalities during pregnancy. However, manual interpretation of ultrasound images is time-consuming and may be affected by operator expertise. This project proposes a deep learning-based framework for the classification of fetal ultrasound images. Initially, the input ultrasound images are preprocessed using the Contrast Limited Adaptive Histogram Equalization (CLAHE) technique to enhance image contrast and improve visibility. The enhanced images are then segmented using the Spatial Fuzzy C-Means (SFCM) algorithm to accurately identify the fetal region of interest. Next, the Convolutional Block Attention Module (CBAM) is employed to extract significant spatial and channel-wise features from the segmented images. These optimized features are then classified using the Hyena Network to effectively capture important feature relationships and improve classification performance. The proposed system predicts the anatomical class of a fetal ultrasound image among six categories: Fetal Abdomen, Fetal Brain, Fetal Femur, Fetal Thorax, Maternal Cervix, and Other. Finally, the proposed system predicts the fetal image category with improved accuracy, sensitivity, and specificity. The framework aims to support radiologists by providing fast, reliable, and automated fetal ultrasound image analysis for early diagnosis and clinical decision-making.

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Keywords: Fetal Ultrasound Imaging, Deep Learning, CLAHE, Spatial Fuzzy C-Means (SFCM), Convolutional Block Attention Module (CBAM), Hyena Network, Image Segmentation, Feature Extraction, Medical Image Classification.

SECURE INTELLIGENT PORTABLE NOHONKING SYSTEM WITH ADAPTIVE SILENT ZONES AND CENTRALIZED MANAGEMENT

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Abstract

Unnecessary vehicle honking in sensitive areas such as hospitals, schools, residential areas, and wildlife zones contributes to noise pollution and disturbance. Existing approaches mainly depend on driver awareness, GPS based geofencing, which may have limitations in privacy, coverage, and infrastructure requirements. This project proposes a Portable RF-Based Intelligent Horn Management System for Silent Zones in Smart Vehicles. The system establishes communication between roadside infrastructure and vehicles using RF signals to automatically manage horn operation within designated zones. An ESP32-based vehicle unit processes and validates the received command and controls the horn, while an emergency override allows necessary horn usage. The roadside unit can be remotely monitored and configured through a centralized platform using 4G/LTE, MQTT, and secure communication mechanisms. The proposed system provides a portable, scalable, and practical approach to support silent-zone management and improve the commuting environment.

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Biomedical Signal Processing and Deep Learning Architectures for Cardiovascular Disease Diagnosis

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Abstract

Cardiovascular diseases (CVDs) are one of the leading causes of mortality in India and all over the globe. Non-invasive and real-time monitoring and diagnosis for quick medical intervention are vital in saving the patient's life. Among the non-invasive diagnostic processes for CVDs, medical practitioners and researchers mostly rely on electrocardiograms (ECGs). However, other biomedical signals such as photoplethysmograms (PPG), phonocardiograms (PCG), and seismocardiograms (SCG) also provide vital information about the cardiovascular health of the patient. Researchers face many challenges during signal acquisition from patients, including signal inference, artefacts, and noise. Signal variation due to diverse demographics, unique patient characteristics, and class imbalance due to rare disorders also pose significant challenges in signal processing and in accurate diagnosis.

Recent studies have demonstrated the capability of Deep Learning architectures for accurate binary and multiclass classification of ECG signals from patients. Pre-processed multimodal biomedical signals with wavelet transform, along with emerging deep learning models such as convolutional neural networks (CNNs), long short-term memory networks (LSTMs), and vision transformers, gave promising results in CVD diagnosis. Despite a lot of advancement in processing algorithms, bottlenecks exist for real-world application in CVD diagnosis and prognosis. Some of the challenges are explainability, trust in the classification results, privacy preservation, self-supervised learning, and federated training of models.

Keeping these in perspective, it is vital to review and analyse the emerging state-of-the-art architectures for biomedical signal classification. So the present study identifies and systematically reviews the major challenges and the emerging architectures in the field of signal processing and classification for CVD diagnosis. This paper will help regulators, researchers,

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and physicians compare and identify different proposed deep learning models for diagnosis and further improvement.

Keywords: CVDs, Biomedical Signal Processing, ECG, Deep Learning, CVD Diagnosis.

JabarAgent: An Explainable Agentic Clinical Decision-Support System for Patient-Centered Risk Stratification and Evidence-Grounded Clinical Reasoning

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ABSTRACT

Decision-support tools driven by artificial intelligence (AI) are proliferating, but many lack three key qualities: their reasoning is not transparent, their utility is difficult to integrate into the hospital workflow, and rarely provide explanations of how a particular finding led to a particular clinical action. This paper describes JabarAgent, an agentive decision-support prototype for individual patients, which transforms a single description of a heterogeneous case into a structured and interpretable clinical assessment. The input includes presenting symptoms and their duration, demographics, history, current medications, allergies, vital signs, laboratory values, and environmental or occupational exposure. A ten-layer pipeline performs data validation, red-flag screening, risk and severity scoring, condition assessment, a three-item differential, feature attribution, medication-safety review and plain-language follow-up guidance. Safety-critical checks are deliberately left out of the generative model so that a fluent answer cannot override an emergency rule. Explainability is treated as a design requirement: each suspected condition is linked to the symptoms, physiological abnormalities, laboratory findings and risk factors that support it, with a short explanation of how each contributes, and the language is deliberately tentative ("most consistent with", "possible", "needs clinical confirmation"). Phase I is a functional validation based on five test cases (scrub typhus, severe dengue and warning signs, preeclampsia with severe features, community-acquired pneumonia with hypoxia, and acute viral gastroenteritis) and nine predefined evaluation metrics. The paper

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describes the architecture and the behavior of an evidence-based, explainable and stage-gated clinical decision support system that does not claim accuracy or effectiveness until prospective evaluation. It follows established principles about evaluation of AI-based decision support.

Keywords: *Artificial intelligence, clinical decision support system, explainable artificial intelligence, large language models, risk stratification, clinical reasoning, patient safety, health informatics, agentic artificial intelligence.*

MaintenanceIQ: A Deterministic-First Hybrid AI Architecture for Evidence-Grounded Industrial Maintenance Decision Support

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Abstract

Industrial maintenance systems integrate equipment records, work orders, reliability analytics, and spare-parts inventory. However, conventional query interfaces limit natural-language accessibility, while unrestricted large language models (LLMs) introduce risks involving analytical reliability, database security, latency, and computational cost. This paper presents MaintenanceIQ, a deterministic-first hybrid AI architecture for secure, evidence-grounded industrial maintenance decision support. The system routes 50 predefined maintenance questions to verified backend routines and processes novel or compound queries through an LLM restricted to 17 logical, read-only analytical tools. Tenant-scoped authorization, customer-specific database routing, encrypted AI credentials, parameterized queries, and local fallback mechanisms strengthen security and operational resilience.

The system was evaluated using a frozen 240-trial benchmark and a separate 60-question validation of the packaged customer desktop application. Following documented remediation, all 80 trials in each benchmark condition completed successfully. The deterministic baseline achieved a mean latency of 0.21 ms, while hybrid routing recorded 3,124.68 ms and reduced token consumption by 73.9% compared with the live-LLM condition. In customer application validation, 55 queries returned analytical answers and five produced valid empty-data responses, satisfying predefined acceptance criteria for all 60 queries.

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These results demonstrate the feasibility of combining deterministic analytics with controlled LLM-based interaction to support accessible and reliable maintenance decision-making. However, hybrid-processing latency and the finite evaluation scope highlight the need for further optimization and validation across diverse industrial environments.

AI-Assisted Performance Prediction and Optimization of a Piezoelectric Footstep Energy Harvesting System Under Different Piezoelectric Disc Configurations

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Abstract

This study investigates a piezoelectric footstep energy harvesting prototype designed to convert repeated mechanical loading into electrical voltage. Four piezoelectric disc configurations are examined: a single disc, two discs, three discs connected in series, and three discs connected in parallel. Each configuration is evaluated under Light Press, High Press, and Frequent Press conditions. Five measured readings are retained for every configuration–condition combination, forming a primary dataset of 60 voltage observations. Descriptive statistics are used to summarize the response, while condition-wise one-way analysis of variance and Tukey post-hoc comparisons examine differences between configurations. A two-way analysis is also used to investigate the combined influence of configuration and pressing condition. To support prediction and configuration screening, a supervised machine-learning workflow is developed using configuration, pressing condition, and piezoelectric disc count as input features, with measured voltage as the target. Linear Regression provides a baseline, while Random Forest Regression and Gradient Boosting Regression are evaluated as ensemble alternatives. Leave-one-out cross-validation is used to compare predictive performance through the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE). The experimental analysis identifies the configuration with the strongest voltage response within the tested conditions, while the machine-learning comparison identifies the most effective of the evaluated prediction models. The study combines physical measurement,

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statistical validation, and data-driven prediction in one workflow. The findings are intended to support early-stage configuration selection; further validation with controlled mechanical input, electrical load measurements, and independent experimental data is needed before broader practical application.

BUYSURE AI: An AI-Based Platform for Smarter Online Shopping Decisions

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Abstract

Online shopping has become an essential part of modern life, providing users with a wide range of products and convenient purchasing options. However, selecting the right product remains challenging due to fake reviews, misleading product information, fluctuating prices, unreliable sellers, and the large number of similar products available online. Users often depend on ratings and reviews to make purchasing decisions, but these factors alone may not provide an accurate understanding of a product's actual quality and value. BUYSURE AI is an AI-based shopping assistant developed to help users make safer, smarter, and more informed purchasing decisions. The proposed system uses Machine Learning (ML) and Natural Language Processing (NLP) techniques to analyze product details, customer reviews, ratings, prices, features, and comparable products. The system identifies suspicious or potentially misleading reviews and analyzes genuine customer feedback to extract important positive and negative aspects of a product. It also compares multiple products based on price, features, quality, ratings, and user-defined requirements. Based on the collected and analyzed information, BUYSURE AI generates a simple and understandable recommendation such as "Buy," "Wait," or "Avoid," along with the key reasons supporting the decision. Users can provide their budget, preferences, and specific requirements to receive personalized recommendations. Future enhancements may include product image and screenshot analysis, price tracking, seller risk detection, and automatic identification of better alternatives. The primary objective of BUYSURE AI is to reduce the complexity of online shopping by transforming large amounts of product and review data into meaningful insights, enabling users to make reliable and informed purchasing decisions.

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Keywords— BUYSURE AI, online shopping, machine learning, natural language processing, review analysis, product comparison, personalized recommendations.

SmartRoute AI: An Explainable Risk-Aware Deep Reinforcement Learning Framework for Dynamic Last-Mile Delivery Assignment

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Abstract

Last-mile delivery is a critical stage of logistics where assigning delivery orders to suitable delivery partners efficiently is challenging due to changing traffic conditions, delivery deadlines, partner availability, distance, and operational risks. This project proposes SmartRoute AI, an explainable and risk-aware deep reinforcement learning framework for dynamic last-mile delivery assignment. The system continuously analyzes delivery requests, delivery-partner availability, traffic conditions, distance, estimated travel time, and delivery priorities. A Deep Reinforcement Learning (DRL) agent learns to select suitable delivery partners for incoming orders based on the current logistics environment. The framework incorporates risk factors such as traffic congestion, delayed deliveries, excessive travel distance, and partner workload into the decision-making process. Unlike conventional assignment methods, the proposed system dynamically adapts to changes in real-time delivery conditions. The Hungarian Algorithm can be integrated as a baseline or optimization component for comparing assignment performance. An explainable AI layer provides understandable reasons for each assignment decision, such as lower travel time, reduced risk, or better workload balance. The system aims to minimize delivery delays, travel distance, and operational risk while improving partner utilization. A simulation-based environment can be used to train and evaluate the DRL model without requiring physical hardware. Performance can be measured using delivery time, assignment accuracy, total distance, workload distribution, and risk reduction. The proposed framework provides a scalable software-based solution for intelligent, adaptive, and explainable last-mile delivery management.

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Keywords— Deep Reinforcement Learning, Last-Mile Delivery, Dynamic Assignment, Explainable AI, Risk-Aware Optimization

Deep Learning-Based Detection and Classification of Brain Tumors Using Medical Imaging

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Abstract:

Cancer is one of the most life-threatening diseases and can have a fatal impact on patients. Medical treatment during the advanced stages of cancer is often critical, expensive, and does not always result in full recovery. Therefore, early detection of cancer is essential. With the advancement of modern deep learning techniques such as YOLO (You Only Look Once) neural architecture and powerful feature extraction algorithms like SIFT (Scale-Invariant Feature Transform), there is a growing potential for accurate and timely cancer diagnosis using medical imaging. Index Terms: Brain Tumor, Deep Learning, Medical Imaging, YOLOv8, Glioblastoma, Image Processing.

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TINYGUARD-MACHINE MONITORING SYSTEM

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Abstract

Industrial machines require continuous monitoring to ensure reliable operation and reduce unexpected failures. This work proposes a TinyML-based multi-machine fault detection system for real-time industrial monitoring. The system uses voltage, current, and frequency sensors to collect operating parameters from multiple machines. The acquired data is processed locally using an embedded controller integrated with a TinyML model to identify normal and abnormal operating patterns. Each machine is assigned a unique identification, allowing the system to determine the specific machine experiencing an abnormal condition. The detected machine status is displayed through a centralized touch-screen interface, enabling operators to monitor multiple machines from a single unit. By performing data analysis at the edge, the proposed system reduces dependence on continuous cloud processing and supports faster fault identification. The system provides an efficient and intelligent approach for real-time industrial machine monitoring and fault detection.

Keywords: TinyML, Industrial Machine Monitoring, Fault Detection, Embedded System, Machine Learning, IoT, Sensors, Real-Time Monitoring

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Deepfake Image Detection: A Comprehensive Systematic Review of Traditional, CNN, Vision Transformer, and Vision-Language Approaches

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Abstract

The rapid development of AI-generated synthetic imagery, broadly termed deepfake, represents one of the most consequential challenges confronting contemporary digital media ecosystems. Advances in generative architectures, including generative adversarial networks (GANs), variational autoencoders (VAEs), and latent diffusion models, have substantially lowered the barrier to producing photorealistic synthetic content, thereby threatening the integrity of forensic evidence, journalistic verification, and public trust in visual information. This systematic review surveys the deepfake image detection literature spanning 2019 to 2025, synthesizing findings from 85 peer-reviewed studies retrieved from Scopus and Elsevier. Adopting a structured methodology aligned with established systematic-review protocols, the study defines explicit inclusion and exclusion criteria, extracts standardized performance and dataset attributes from each primary source, and organizes the retrieved corpus according to detection paradigm, backbone architecture, and evaluation strategy. The review traces the methodological evolution from handcrafted feature extraction and frequency-domain forensics, through deep convolutional networks, to Vision Transformer architectures and Vision-Language Models (CLIP, SigLIP, and SigLIP 2). The principal findings indicate that Vision Transformer-based approaches attain 95 to 99.2 percent accuracy, whereas Vision-Language Models offer superior cross-generator generalization by leveraging semantically grounded, spatially aware representations. Critical research gaps include insufficient cross-dataset robustness evaluation, limited explainability in transformer-based detectors and the absence of lightweight, deployment-ready architectures. The review concludes that future progress will depend on multimodal detection, explainable artificial intelligence, federated learning, and real-time detection systems capable of operating reliably in adversarial, resource-constrained environments.

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Keywords: *Vision Transformer, GAN, diffusion models, CLIP, SigLIP, convolutional neural network, synthetic media, systematic review.*

AI-Assisted Performance Prediction and Optimization of a Thermoelectric Waste Heat Energy Harvesting

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Abstract

Efficient recovery of low-grade thermal waste remains a critical challenge in sustainable energy systems. This paper presents an AI-assisted performance prediction and optimization framework for a thermoelectric waste heat energy harvesting system. The experimental setup employs a Peltier module operating in reverse as a thermoelectric generator (TEG) to convert thermal gradients into electrical energy. Due to the inherently low output voltage of the thermoelectric module, a DC-DC step-up (boost) converter is integrated to efficiently elevate the voltage levels required for battery charging.An Arduino Uno micro-controller serves as the core processing unit, interfaced with a DS18B20 digital temperature sensor for precise thermal monitoring of the module and an INA219 high-precision sensor module for real-time voltage and current measurements. Telemetry data is wirelessly transmitted and monitored via a mobile interface. Machine learning models are integrated to analyze the acquired experimental dataset, enabling accurate prediction of power output and dynamic optimization of energy conversion efficiency under varying thermal loads. The proposed smart harvesting system demonstrates a reliable, low-cost, and scalable approach for autonomous IoT nodes and micro-power generation applications.

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Prediction of Genetic Disorders Using Chromosome Images with a Convolutional Neural Network (CNN)

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Abstract

Genetic disorders caused by chromosomal abnormalities can lead to severe developmental and inherited diseases. Early and accurate detection is essential for effective diagnosis, treatment planning, and genetic counselling. Conventional chromosome analysis relies on manual examination of karyotype images by cytogenetic experts, which is time-consuming, labour-intensive, and prone to human error. To overcome these limitations, this project proposes an automated system for predicting genetic disorders using chromosome images and a Convolutional Neural Network (CNN). The proposed system uses chromosome karyotype images as input and applies preprocessing techniques such as resizing, normalisation, and data augmentation to improve image quality and increase model robustness. The pre-processed images are then fed into a CNN model, which automatically extracts salient features and classifies chromosomes as normal or abnormal. The model is trained to identify chromosomal abnormalities associated with genetic disorders such as Down Syndrome (Trisomy 21), Edwards Syndrome (Trisomy 18), Patau Syndrome (Trisomy 13), Turner Syndrome, and Klinefelter Syndrome. The performance of the CNN model is evaluated using metrics including accuracy, precision, recall, F1-score, and the confusion matrix. The experimental results demonstrate that the proposed approach provides high classification accuracy while significantly reducing diagnosis time and minimising human intervention. This AI-based system serves as an efficient decision-support tool for healthcare professionals and has the potential to improve the reliability of genetic disorder diagnosis. Future work includes expanding the dataset and integrating advanced deep learning models to further enhance prediction performance.

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Keywords: Genetic Disorders, Chromosome Images, CNN, Deep Learning, Karyotype Analysis, Medical Image Classification, Artificial Intelligence.

Detection of Skin Cancer Using Machine Learning: A CNN-Based Preliminary Screening System

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Abstract

Early identification of skin cancer can improve treatment outcomes, but access to expert assessment is not always available. This paper presents a machine-learning-based preliminary screening system that analyses skin-lesion images using a convolutional neural network (CNN). The proposed workflow includes image collection, preprocessing, CNN training, testing, prediction and result display through a Flask-based web interface. Input images are converted to RGB, resized to a fixed size and normalised before being passed to the CNN, which learns visual features such as shape, colour and texture. To reduce the effect of class imbalance, class weighting and data augmentation are incorporated during training. The system produces a preliminary benign or malignant prediction and displays the result with a clear disclaimer that it is not a medical diagnosis. The proposed prototype is implemented using Python and is designed to provide a simple and accessible screening-support tool. Performance is intended to be evaluated using accuracy, precision, recall, F1-score and a confusion matrix, with future work focusing on diverse skin tones, explainable AI and independent clinical validation.

Keywords— Skin cancer detection, Convolutional Neural Network (CNN), Machine Learning, Skin-lesion Classification, HAM10000, ISIC, Preliminary Screening, Flask.

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Vivasayi AI: A Multilingual, Voice-Enabled Artificial Intelligence Platform for Intelligent Farm Management and Agricultural Automation

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Abstract
Agriculture has many activities like crop management, irrigation, soil monitoring, equipment booking, labor management, market analysis, government schemes and document preparation. Farmers usually have to rely on several applications and sources to manage these activities, which makes the whole process tiresome and time- consuming. To tackle this challenge, Vivasayi Agentic AI is presented as an intelligent AI-powered agriculture assistant that aggregates multiple farming services on a single platform. The system is based on the multi-agent system where different AI agents do different agricultural works. A Supervisor Agent manages the requests of the users and assigns them to agents such as the Farm Manager Agent, Booking. A Supervisor Agent coordinates user requests and assigns them to agents, for example the Farm Manager Agent, Booking Agent, Document Agent, Market Agent and Policy Agent. The system leverages Google Gemini AI to comprehend farmer requests made through natural-language and voice, and to deliver suitable assistance. It also includes multi-lingual interaction, support for agricultural decisions, simulated IoT sensor data, weather-based recommendations, document generation and human approval for sensitive operations. The proposed system is aimed at reducing the complexity of agricultural management by providing farmers with a comprehensive intelligent platform for information, decision support and task-oriented automation. Vivasayi Agentic AI demonstrates the potential of agentic AI for supporting modern farming and paves the way for future integration with real-time IoT devices, agricultural data sources, government services, and field-level applications.

Keywords— Vivasayi Agentic AI, Agentic Artificial Intelligence, Smart Agriculture, Precision Agriculture, Multi Agent Systems, Agricultural Automation, Farm Management, Artificial

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Intelligence, Gemini AI, IoT, Agricultural Decision Support, Human-in-the-Loop, Multilingual AI

An Agentic AI Framework for Context-Aware Pharmacovigilance Document Extraction and CIOMS Form Automation

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Abstract
Pharmacovigilance specialists extract information from unstructured clinical safety documents and enter it into reporting forms such as CIOMS Form I and sponsor safety trackers. This process can be time-consuming and error-prone, particularly when stakeholders provide different causality assessments for the same event. This paper presents a prototype of a controlled, supervised agentic AI workflow for pharmacovigilance document extraction and form automation. The system accepts PDF, DOC, DOCX, TXT and JSON documents, along with voice commands, and checks whether a document contains relevant pharmacovigilance information before proceeding. A Gemini-based extraction agent generates structured JSON with supporting evidence snippets, while a deterministic regular-expression fallback is available when the language model is unavailable. A mapping agent preserves Investigator, Sponsor Pharmacovigilance Officer and Medical Reviewer causality assessments as separate contextual fields in the internal schema. A validation agent applies mandatory-field rules and heuristic confidence thresholds to route extracted fields for automatic approval, human review or flagging. Human reviewers can approve, edit or reject fields, followed by explicit supervisor authorization before browser-based robotic process automation populates CIOMS Form I and the 36-field CREST Safety Tracker. The system supports JSON, Word and Excel exports and records workflow actions in an audit log. A real execution on one synthetic case demonstrates the workflow. However, accuracy, recall and time-saving performance have not been measured using a labelled dataset. A plan for future experimental evaluation is presented.

Keywords
pharmacovigilance; agentic AI; clinical information extraction; causality assessment; human-in-the-loop; robotic process automation; CIOMS

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VEHICLE TRACKING AND ACCIDENT ALERT USING IOT PROVIDES GPS TRACKING AND AUTOMATIC ACCIDENT ALERTS TO EMERGENCY SERVICES.

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Abstract

Road accidents are a major concern, and delays in emergency response can lead to serious consequences. This project proposes a Vehicle Tracking and Accident Alert System using the Internet of Things (IoT) to detect possible vehicle accidents and provide immediate alerts to registered emergency contacts. The system uses an ESP32 DevKit V1 as the main controller, an MPU6050 accelerometer and gyroscope to monitor sudden changes in vehicle movement and orientation, and a vibration sensor to detect sudden impacts. When the sensor readings indicate a possible accident, the ESP32 processes the data and activates the alert mechanism. The NEO-6M GPS module obtains the vehicle's geographical coordinates, while the SIM800L GSM module sends an emergency SMS containing the accident location to a predefined mobile number. An OLED display shows the current system status, and a DFPlayer Mini with a speaker provides an audio alert. A push button allows the user to cancel a false alarm before the emergency message is sent. This system aims to reduce the time required to report accidents, improve vehicle location tracking, and support faster emergency assistance. The proposed system offers an affordable IoT-based solution for enhancing road safety and accident response.

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Smart Life: An ESP32-Based IoT Platform for Integrated Home Automation

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Abstract

Controlling home appliances using multiple independent devices is complicated, and such conditions rarely provide the inhabitants with complete knowledge about the state of the environment or allow timely response to dangerous situations. Smart Life is a prototype developed using the ESP32 DevKit platform that consolidates several methods of controlling connected devices into a single dashboard. The dashboard is hosted on the local Wi-Fi server and allows interacting with relays controlled by a four-channel relay board. The prototype also features infrared emitters programmed to operate televisions and air conditioners. Three sensors are used to monitor the environment: PIR motion sensor, photoresistor, and gas sensor MQ-2. The gas sensor output voltage is used to trigger relays, blink an onboard LED, and send a signal to the dashboard, notifying the user of the gas level threshold being exceeded. The additional functions include learning and re-transmitting infrared signals from consumer electronics remote controllers, user authentication using CAPTCHA, account recovery mechanism based on one-time passwords, and text-to-speech command recognition. Since the dashboard is hosted on the local server, no subscription to external services is necessary to use the prototype. In general, the project demonstrates that microcontroller-based solutions, when combined with appropriate sensors and remote communication methods, can be used to design complete home automation systems. More practical experiments are necessary to evaluate how the prototype will behave under different environmental conditions and ensure that all components operate reliably and securely.

Keywords: IoT; ESP32; home appliances; web dashboard; PIR sensor; LDR; MQ-2; infrared control; embedded systems; wireless communication

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ZcannerReality: An AI-Assisted 3D Virtual Try-On and Augmented Reality Framework for Personalized Fashion Visualization

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Abstract

Online fashion shopping offers customers many clothing options, but it is often difficult to imagine how an outfit will look on their own body before purchasing it. ZcannerReality is a proposed AI-powered virtual try-on system designed to make online shopping more interactive by helping users visualize clothing on a digital representation of their body. The system takes a user's image and processes it using computer vision techniques to identify body parts, estimate posture, and reconstruct body shape. This information helps position the selected garment more naturally on the digital model. By combining Artificial Intelligence (AI) with Augmented Reality (AR), the system aims to provide a more realistic preview than simply placing a clothing image over a photograph. Technologies such as OpenCV, PyTorch, MediaPipe, and suitable 3D and AR tools can support its development. The framework also considers practical challenges, including varying lighting conditions, overlapping body parts, changes in posture, and the processing limitations of mobile devices. Its effectiveness will be evaluated through future experiments that examine garment alignment, response time, performance under different conditions, and perceived realism. Ultimately, ZcannerReality aims to simplify online clothing selection and help customers make more informed purchasing decisions.

Keywords: Virtual Try-On, Augmented Reality, 3D Body Reconstruction, Human Pose Estimation, Digital Twin, Garment Fitting, Fashion Technology

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SMART DATA ANALYTICS ASSISTANT

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Abstract

The Smart Data Analytics Assistant is an Artificial Intelligence (AI)-powered conversational system designed to simplify business data analysis through natural language interaction. Traditional data analysis methods often require technical knowledge of database queries, manual data processing, and separate tools for analysing structured and unstructured information. These challenges can increase the time and effort required to obtain meaningful business insights. The proposed system addresses these limitations by enabling users to retrieve, analyse, and interpret information from multiple data sources through a unified interface.

The system integrates MySQL, PostgreSQL, and MongoDB databases through the Model Context Protocol (MCP), allowing users to access business information by asking questions in natural language. The Gemini Large Language Model (LLM) interprets user requests, generates suitable database queries, and converts retrieved information into understandable responses. Additionally, the system supports document analysis using Retrieval-Augmented Generation (RAG), text embeddings, and vector similarity search. Uploaded files, including PDF, Word, Excel, CSV, and PowerPoint documents, are processed to retrieve relevant information and generate context-based answers.

The application is developed using React.js with Vite for the frontend and Python with FastAPI for backend operations. Qdrant is used for storing and retrieving document embeddings. The system also supports conversational context, chat history, and graphical visualization of analytical results. The implemented modules demonstrate database querying, document summarization, and automated information retrieval through a user-friendly interface. Future development focuses on secure live-server database integration, read-only access, real-time analytics, and improved query performance. The system aims to reduce manual analytical effort and make business information more accessible to users without advanced technical expertise.

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An Attention-Enhanced Tabtransformer-GRU Framework Optimized By Ant Colony Algorithm for Battery SOH and RUL Prediction

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Abstract

A battery is a rechargeable energy storage device that stores electrical energy and supplies power to electronic systems. The battery's current health is determined by evaluating its remaining capacity and overall degradation level. The remaining operating cycles are estimated before the battery reaches its end-of-life condition. Accurate prediction improves battery reliability, supports predictive maintenance, enhances safety, and reduces unexpected battery failures. Traditional methods cannot accurately predict battery degradation under varying operating conditions. To overcome this issue, an Attention-Enhanced TabTransformer-GRU Framework Optimized by the Ant Colony Algorithm for Battery SOH and RUL Prediction is proposed. The proposed methodology begins with collecting the battery dataset as input. Initially, data preprocessing is performed to handle missing values, check string values, and clean the dataset for analysis. The preprocessed data is then visualized to understand its distribution and relationships among battery parameters. Next, the dataset is divided into training and testing sets for model development and evaluation. The training data is processed using the proposed Hybrid TabTransformer–Attention–GRU model optimized with the Ant Colony Optimization (TAG-ACO) algorithm. In this framework, the TabTransformer extracts meaningful representations from tabular battery data, the attention mechanism emphasizes the most important features, the GRU learns battery degradation patterns over charging cycles, and the Ant Colony Optimization algorithm optimizes the model for improved prediction accuracy. Support Vector Machine (SVM), Recurrent Neural Network (RNN), and Gated Recurrent Unit (GRU) are compared with the proposed TAG-ACO model. Finally, the trained model predicts the State of Health (SOH) and Remaining Useful Life (RUL) of the battery, and its performance is

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evaluated using error metrics such as MAE, RMSE, and R^2 score. This project is implemented in Python.

Keywords: Hybrid TabTransformer–Attention–GRU model optimized, Support Vector Machine (SVM), Recurrent Neural Network (RNN), and Gated Recurrent Unit (GRU)

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AI-ENHANCED SMART GESTURE COMMUNICATION GLOVE FOR ASSISTIVE COMMUNICATION

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ABSTRACT:

Communication barriers can affect individuals with speech and hearing impairments, particularly when communicating with people who are unfamiliar with sign language. This paper proposes an AI-enhanced smart gesture communication glove intended to support accessible communication by translating predefined hand gestures into text and, potentially, speech. The proposed wearable system integrates five flex sensors to measure finger bending and an MPU6050 inertial measurement unit (IMU) to capture hand orientation and motion. An ESP32 microcontroller is intended to collect and process the sensor readings. These readings will be used to develop a gesture-recognition pipeline based on a Random Forest classification model, covering selected Indian Sign Language (ISL) signs as well as general hand gestures. The recognized gesture can be mapped to a corresponding text label and displayed on a 0.96-inch OLED screen, while voice output is considered as an additional accessibility feature. By combining finger-flexion measurements with motion and orientation data, the design aims to distinguish a broader range of gestures in a portable wearable form. The project is currently at the proposed design stage; dataset collection, model training, hardware integration, and systematic evaluation will be required to assess recognition performance and usability. The proposed system seeks to explore an affordable embedded-AI approach to assistive communication and provide a foundation for future expansion of the gesture vocabulary and real-world testing.

Keywords: Assistive Communication; Gesture Recognition; Indian Sign Language; Flex Sensors; ESP32; Random Forest; Wearable Computing.

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MULTI-HAZARD ROAD SURVEILLANCE AND EMERGENCY ROUTING SYSTEM

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Abstract

Multiple environmental hazards such as rain, waterlogging, animal crossing, fallen trees, landslide, poor visibility, etc., often affect the accessibility of roads especially in forest areas, hills, and low lying areas. These can result in road obstruction and congestion as well as delays in the operation of emergency services like ambulance, firefighting etc. The paper presents a multi-hazard intelligent road surveillance and emergency routing system that uses existing CCTV cameras, traffic radar, rainfall sensor, water-level sensors, weather data and road condition monitoring. This system will keep track of the road conditions to detect changes in visibility, traffic flow, water accumulation, animal movement and any physical obstructions. The information will be analysed to detect the road accessibility as well as its risk and generate warnings on a central dashboard. Depending upon the detected road conditions as well as the prediction, alternative routes may be advised to the emergency vehicles prior to reaching the affected area.

Keywords: Multi-Hazard Monitoring, Road Accessibility, Emergency Routing, Waterlogging, Wildlife Detection, Traffic Monitoring, CCTV, Radar, IoT.

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SkillLoop: A Reciprocal Skill-Exchange Platform with Rule-Based Match Scoring, Token Incentives, and Live Collaboration Rooms

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Abstract

Technical skills are often learned outside the classroom, where many learners lack access to an instructor or a paid mentor. Peer learning offers an alternative, but existing platforms rarely model the two-sided condition that keeps an exchange sustainable, namely that each participant can teach what the other wishes to learn. This paper presents SkillLoop, a web-based platform for reciprocal peer-to-peer skill exchange. Users declare the skills they can teach and the skills they want to learn, and the system pairs them with members whose profiles complement their own. Matching compares skill sets in both directions and assigns an interpretable rule-based score from 1 to 100, constructed so that every reciprocal pair ranks above every one-way pair. A recommendation layer adds an activity term and generates a textual reason for each suggestion. Matched users collaborate through exchange requests, live rooms with text chat and browser-based video calls (WebRTC), and mentor bookings. A token wallet with an append-only ledger rewards confirmed teaching within anti-abuse limits. The platform is implemented in Python with Django and SQLite, and it includes a self-hosted captcha, a honeypot field, and login throttling. The evidence reported consists of worked scoring examples, a verified module inventory, and an automated test suite covering security and room workflows. Skills are currently matched as exact strings, the score constants are uncalibrated, and no user study has yet been conducted. Planned work includes semantic skill matching, project recommendation, and a formal evaluation with real users.

Keywords— Skill Exchange, Peer Learning, Reciprocal Matching, Recommender System, Learning by Teaching, Django, WebRTC

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Deadline-Aware Adaptive-Parallelism VLSI Accelerator for Energy-Efficient Real-Time Edge-AI Inference

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Abstract

Real-time Edge-AI applications need hardware accelerators that can process data quickly while using less power. Conventional VLSI accelerators usually operate with a fixed number of Processing Elements (PEs) for all types of workloads. This can lead to unnecessary energy consumption during low workloads and increased processing time during high workloads. To address this issue, this paper proposes a Deadline-Aware Adaptive-Parallelism VLSI Accelerator that changes the number of active PEs according to the workload and available processing time. The proposed architecture consists of a workload monitor, deadline monitor, adaptive PE controller, parallel MAC-based processing elements, and an accumulator. When the workload is low and sufficient time is available, fewer PEs are activated. When the workload is high or the deadline is close, more PEs are activated to complete the computation faster. The design is implemented using Verilog/System Verilog and evaluated through RTL simulation, synthesis, timing analysis, and power estimation using AMD Vivado. The proposed approach aims to reduce unnecessary hardware activity and energy consumption while maintaining the required performance for real-time Edge-AI applications.

Keywords: Edge-AI, VLSI Accelerator, Adaptive Parallelism, Processing Element, MAC, Low Power, Real-Time Processing.

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A Centralized Patient-Centric Application for Digital Prescriptions and Healthcare Management

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Abstract

In an era of digital transformation, the healthcare sector demands intelligent solutions that transcend the limitations of traditional paper-based medical record management. Physical prescriptions and clinical documents are frequently misplaced, damaged, or forgotten, creating obstacles to uninterrupted treatment, accurate diagnosis, and informed medical decision-making. This project introduces an innovative, cloud-powered healthcare management application that revolutionizes the way patients and healthcare professionals interact with medical information. The proposed platform functions as a secure digital ecosystem, consolidating prescriptions, and easily laboratory records, and healthcare documents into a unified accessible easily repository. Through seamless integration, authorized healthcare providers can instantly cand instantly upload medical records, while patients can retrieve their complete healthcare history anytime, anywhere. Furthermore, the application incorporates an intelligent appointment orchestration module that enables real-time consultation scheduling, availability tracking, and personalized healthcare planning. consultation scheduling, availability tracking, and personalized healthcare planning. State-of-the-art encryption, multi-layered authentication, and privacy-centric security mechanisms ensure the confidentiality, integrity, and resilience of sensitive patient data. By replacing fragmented hospital portals and vulnerable paper records with a scalable, interoperable, and patient-centric digital platform, the proposed system fosters continuity of care, enhances operational efficiency, and strengthen clinical records

Keywords: Digital Health Records; Electronic Prescriptions; Healthcare Management; Appointment Scheduling; Patient-Centric Application.

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